

AP EAMCET-2019-Engineering-23 Apr-Shift-1 Mathematics

1. Match the following

List-I

(A) $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that
 $f(x) = px + q, (p \neq 0), \forall x \in \mathbb{R}$

(B) $f : \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$ is such that
 $f(x) = x^2, \forall x \in \mathbb{R}$

(C) $f : \mathbb{N} \rightarrow \mathbb{N}$ is such that
 $f(n) = n^2 + 2n + 3, \forall n \in \mathbb{N}$

(D) $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that
 $f(x) = 2(\cos^2 5x + \sin^2 5x), \forall x \in \mathbb{R}$

List-II

(I) f is neither one-one nor onto

(II) f is both one-one and onto

(III) f is one-one but not onto

(IV) f is onto but not one-one

(V) f is a constant function and also a bijection

The correct answer is

	A	B	C	D
(1)	II	IV	III	I

	A	B	C	D
(2)	II	IV	V	I

	A	B	C	D
(3)	II	I	III	IV

	A	B	C	D
(4)	III	II	I	IV

2. The range of $f(x) = \sqrt{\frac{a-|x|}{(a+1)-|x|}}$, ($a > 0$) is

(1) $[0, a]$

(2) $[0, \infty) - \left[-\sqrt{\frac{a}{a+1}}, \sqrt{\frac{a}{a+1}}\right]$

(3) $\left[0, \sqrt{\frac{a}{a+1}}\right] \cup (1, \infty)$

(4) $\left[0, \sqrt{\frac{a}{a+1}} + 1\right]$

3. If $2 \cdot 4^{2k+1} + 3^{3k+1} = 11t$ and $2 \cdot 4^{2k+3} + 3^{3k+4} = 11(pt + 3^q)$, where $k, t \in \mathbb{Z}^+$ then $(p, q) =$

$$(1) (16, 3k + 1)$$

$$(2) (16, 3k + 4)$$

$$(3) (32, 3k + 1)$$

$$(4) (32, 3k + 4)$$

4. The equation obtained by eliminating a, b, c from the equation

$$x = \frac{a}{b-c}, y = \frac{b}{c-a}, z = \frac{c}{a-b} \text{ is}$$

$$(1) \begin{vmatrix} 1 & -x & x \\ 1 & -y & y \\ 1 & -z & z \end{vmatrix} = 0$$

$$(2) \begin{vmatrix} 1 & -x & x \\ 1 & 1 & -y \\ 1 & z & 1 \end{vmatrix} = 0$$

$$(3) \begin{vmatrix} 1 & -x & x \\ y & 1 & -y \\ -z & z & -1 \end{vmatrix} = 0$$

$$(4) \begin{vmatrix} x & y & 1 \\ y & x & 1 \\ 1 & x & y \end{vmatrix} = 0$$

5. If A , is a 3×3 matrix and $|A| = 2$ then $(\text{Adj } A)|\text{Adj}(\text{Adj } A)| =$
- (1) $32A$
 - (2) $64A$
 - (3) $16A$
 - (4) $8A$
6. If $x = \alpha, y = \beta, z = \gamma$ is the solution for the system of equation,
- $$2x - y + 8z = 13$$
- $$3x + 4y + 5z = 18$$
- $$5z - 2y + 7z = 20$$
- Then $\alpha\beta + \beta\gamma + \gamma\alpha =$
- (1) 1
 - (2) 0
 - (3) 7
 - (4) -3
7. For complex number $Z = a + ib$, let $\hat{Z} = b + ia$. If Z_1, Z_2 are such complex number then $\widehat{Z_1 Z_2} =$
- (1) $\hat{Z}_1 \hat{Z}_2$

$$(2) \hat{Z}_1 \widehat{\bar{Z}}_2$$

$$(3) \bar{Z}_1 \hat{Z}_2$$

$$(4) \hat{Z}_1 Z_2$$

8. The points in the Argand plane represented by the complex conjugate of $1 + 2i$, $2 - 3i$, $3 - 4i$

(1) are collinear

(2) form an equilateral triangle

(3) form an obtuse angled triangle

(4) form an acute angle triangle

9. If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the n^{th} root of unity and n is an even natural number then $(1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1}) =$

(1) 1

(2) -1

(3) 0

(4) 2

10. If $x + \frac{1}{x} = 2\sin\alpha$ and $y + \frac{1}{y} = 2\cos\beta$ then $x^3 y^3 + \frac{1}{x^3 y^3} =$

(1) $2\cos 3(\beta - \alpha)$

(2) $2\cos 3(\beta + \alpha)$

(3) $2\sin 3(\beta - \alpha)$

(4) $2\sin 3(\beta + \alpha)$

11. If the product of the roots of the equation $x^2 + 4kx + 12e^{3\log k} - 1 = 0$, $k > 0$ is 323, then the sum of its roots is

(1) $9k$

(2) 12

(3) -12

(4) $-16k$

12. If a and b are the maximum and minimum values of quadratic expression $1 - 2x - 5x^2$ and $x^2 - 2x + 5$ respectively then the set of values of x for which the expression $5ax^2 + bx + 7$ is positive is

(1) (a, b)

(2) $(-\infty, 7)$

(3) $(5, \infty)$

(4) $(-\infty, \infty)$

13. If $\left| \frac{x^2 + kx + 1}{x^2 + x + 1} \right| < 3$ for all real numbers x , then the range of parameter k is

(1) $(0, 4)$

(2) $(-1, 5)$

(3) $(-4, 0)$

(4) $(-5, 1)$

14. let $a, b, c, d \in \mathbb{R}$ if the equations $2bx^2 + 3cx - d = 0$ and $2ax^2 + 3bx + 4c = 0$ have a common root and

$$\frac{4bc + ad}{k(b^2 - ac)} = \frac{bd + 4c^2}{4bc + ad} \text{ then } k =$$

(1) $\frac{9}{2}$

(2) $\frac{2}{9}$

(3) $\frac{1}{9}$

(4) $\frac{1}{3}$

15. If all the letters of the word REPEAT are permuted in all possible ways and if the six letter permutation thus formed are arranged in dictionary order, then the rank of the word REPEAT is

(1) 133

(2) 267

(3) 266

(4) 132

16. 15 persons are sitting around a circular table, the number of ways selecting three persons at a time from them, such that the selected three did not sit together at one place is

(1) 455

(2) 15

(3) 45

(4) 440

17. If the coefficient of r^{th} and $(r+1)^{\text{th}}$ terms in the expansion of $(1+x)^{24}$ are in the ratio of 12:13, the r is the root of quadratic equation

(1) $x^2 - 5x + 6 = 0$

(2) $x^2 - 11x + 30 = 0$

(3) $x^2 - 14x + 13 = 0$

(4) $x^2 - 14x + 24 = 0$

18. If $x = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{2!} \left(\frac{2}{5}\right)^3 + \dots$ then $x + \frac{1}{x} =$

(1) $\frac{1+\sqrt{5}}{4}$

(2) 3

(3) $\frac{5\sqrt{5}+3}{4}$

(4) $\frac{5\sqrt{5}-3}{4}$

19. The coefficient of x^4 in the expansion of $\frac{1}{(1-x)(1-2x)(1-3x)}$

is

(1) 602

(2) 301

(3) $\frac{601}{2}$

(4) 302

20. If $\tan A - \tan B = x$ and $\cot A - \cot B = y$ then $\cot(A - B) =$

(1) $\frac{xy}{x+y}$

(2) $\frac{xy}{x-y}$

(3) $\frac{x-y}{xy}$

(4) $\frac{y-x}{xy}$

21. $\tan \frac{\pi}{5} + 2 \tan \frac{2\pi}{5} + 4 \cot \frac{4\pi}{5} =$

(1) $\cot \frac{\pi}{5}$

(2) $\cot \frac{2\pi}{5}$

(3) $\cot \frac{3\pi}{5}$

(4) $\cot \frac{4\pi}{5}$

22. If $\sin x + \sin y = \frac{\sqrt{3} + 1}{2}$ and $\cos x + \cos y = \frac{\sqrt{3} - 1}{2}$ then

$$\tan^2 \left(\frac{x-y}{2} \right) + \tan^2 \left(\frac{x+y}{2} \right) =$$

(1) $8 + 4\sqrt{3}$

(2) $6 + 4\sqrt{3}$

(3) $3 + \sqrt{3}$

(4) $12 + 6\sqrt{3}$

23. In ΔPQR let $\angle P > \angle Q$. If radian measures of $\angle P$ and $\angle Q$ satisfy the equation $4\sin^3 x - 3\sin x + a = 0$ then the radian measure of $\angle R$ is

(1) $\frac{\pi}{3}$

(2) $\frac{\pi}{2}$

(3) $\frac{2\pi}{3}$

(4) $\frac{5\pi}{6}$

24. The solution of the equation $\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}$ =

(1) $-\frac{1}{2}\sqrt{\frac{3}{7}}$

(2) $\frac{1}{2}\sqrt{\frac{3}{7}}$

(3) $\frac{1}{2}\sqrt{\frac{2}{7}}$

(4) $-\frac{1}{3}\sqrt{\frac{2}{7}}$

25. If $-\frac{\pi}{4} < \theta < \frac{\pi}{2}$ then $\log\left(\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)\right) =$

$$(1) \tanh^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$(2) 2 \tanh^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$(3) \coth^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$(4) 2 \coth^{-1}\left(\tan\frac{\theta}{2}\right)$$

26. if ABC is a right angled triangle with 90° at C and $a > b$, then

$$\frac{a^2 + b^2}{a^2 - b^2} \sin(A - B) =$$

$$(1) \frac{3}{2}$$

$$(2) 1$$

$$(3) \frac{1}{2}$$

$$(4) 0$$

27. a, b, c are the sides of scalene triangle ABC , if angles α, β, γ

lies between 0 and π such that $\cos \alpha = \frac{a}{b+c}, \cos \beta = \frac{b}{c+a},$

$\cos \gamma = \frac{c}{a+b}$, then $\tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2} + \tan^2 \frac{\gamma}{2} =$

(1) $\frac{1}{3}$

(2) 2

(3) 1

(4) $\frac{3}{2}$

28. In triangle ABC , if $r_1 = 36, r_2 = 18$ and $r_3 = 12$ then $a + b =$

(1) 36

(2) 24

(3) 30

(4) 54

29. If α, β, γ are the distinct real numbers are $\alpha + \beta + \gamma \neq 0$, then the

points with position vectors $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k},$

$\beta \hat{i} + \gamma \hat{j} + \alpha \hat{k}, \gamma \hat{i} + \alpha \hat{j} + \beta \hat{k}$ are

- (1) collinear
- (2) vertices of scalene triangle
- (3) vertices of isosceles triangle
- (4) vertices of equilateral triangle

30. Given three vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$, a vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is

- (1) $-2\hat{i} - \hat{j} + 5\hat{k}$
- (2) $2\hat{i} + 3\hat{j} + 3\hat{k}$
- (3) $2\hat{i} + \hat{j} + 5\hat{k}$
- (4) $2\hat{i} - 3\hat{j} + 3\hat{k}$

31. $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$, $\vec{c} = 4\hat{i} + 5\hat{j} - \hat{k}$ are three vectors and $\vec{r}$ is perpendicular to both $\vec{b}$ and $\vec{c}$. If $\vec{r} \cdot \vec{a} = 9$ then $\vec{r} =$

- (1) $3(\hat{i} - \hat{j} - \hat{k})$
- (2) $3(\hat{i} - \hat{j} + \hat{k})$

$$(3) 9(\hat{i} - \hat{j} - \hat{k})$$

$$(4) 9(\hat{i} - \hat{j} + \hat{k})$$

32. $\overline{AB} = \overline{a}$ and $\overline{AC} = \overline{b}$ are the sides of triangle ABC. P is a point on $\overline{AB}$ and Q is a point on $\overline{BC}$ such that $\frac{AP}{PB} = \frac{1}{2}$ and $\frac{BQ}{QC} = \frac{1}{2}$.

If the point of intersection of $\overline{AQ}$ and $\overline{CP}$ is D and the area of ΔBCD is 7 square units, then the area of triangle ABC (in the same sq units) is

$$(1) \frac{49}{4}$$

$$(2) \frac{49}{2}$$

$$(3) \frac{7}{2}$$

$$(4) \frac{7}{4}$$

33. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, then the magnitude of projection on $\vec{c}$ of a unit vector that is perpendicular to both $\vec{a}$ and $\vec{b}$ is

(1) $\frac{1}{\sqrt{29}\sqrt{3}}$

(2) $\frac{1}{\sqrt{6}}$

(3) $\frac{1}{\sqrt{58}}$

(4) $\frac{3}{\sqrt{29}}$

34. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\sqrt{3}}{2}\vec{b} + \frac{1}{2}\vec{c}$, then

the angle between $\vec{a}, \vec{b}$ and $\vec{a}, \vec{c}$ respectively are,

(1) $60^\circ, 30^\circ$

(2) $120^\circ, 30^\circ$

(3) $60^\circ, 150^\circ$

(4) $120^\circ, 150^\circ$

35. The variance of the following data is,

x_i	1	2	3	4	5	6	7	8	9	10
f_i	1	2	3	4	5	6	7	8	9	10

(1) 10

(2) 9

(3) 8

(4) 6

36. The coefficient of variation of following distribution is,

Class interval	0-5	5-10	10-15	15-20	20-25
Frequency	4	1	10	3	2

(1) $\frac{25\sqrt{139}}{12}$

(2) $\frac{25\sqrt{139}}{6}$

(3) $\frac{39}{6}$

(4) $\frac{25 \times 139}{12}$

37. From a group of 50 students, two sections comprising of 20 and 30 students are formed. If Ram and Rahim are two particular

students among the 50 students, then the probability that the belong to same section is

(1) $\frac{25}{49}$

(2) $\frac{12}{23}$

(3) $\frac{13}{23}$

(4) $\frac{24}{49}$

38. If E_1 and E_2 are two events of random experiment such that

$$P(E_1) = \frac{1}{8}, P(E_1 | E_1) = \frac{1}{3}, P(E_2 | E_1) = \frac{1}{4}, \text{ then match the items}$$

of List-I with the items of List-II.

List-I

List-II

(A). $P(E_2)$

$\therefore \frac{3}{16}$

(B). $P(E_1 \cup E_2)$

$\therefore \frac{3}{29}$

(C). $P(\overline{E_1} | \overline{E_2})$

$\therefore \frac{3}{32}$

$$(D). P(E_1 | \overline{E_2}) = \frac{26}{29}$$

$$= \frac{13}{32}$$

The correct match is

$$\begin{array}{cccc} & A & B & C & D \\ (1) & I & III & IV & II \end{array}$$

$$\begin{array}{cccc} & A & B & C & D \\ (2) & III & I & IV & V \end{array}$$

$$\begin{array}{cccc} & A & B & C & D \\ (3) & III & I & IV & II \end{array}$$

$$\begin{array}{cccc} & A & B & C & D \\ (4) & I & II & V & IV \end{array}$$

39. A box B_1 contains 3 blue balls and 6 red balls, another box B_2 contains 8 blue balls and 'n' red balls ($n \in \mathbb{N}$). A ball is selected at random from a box is found to be red. if p is the probability that this red ball drawn is from box B_2 then,

$$(1) \frac{1}{7} \leq p < \frac{3}{5}$$

$$(2) \frac{3}{5} \leq p < 1$$

$$(3) 0 < p \leq \frac{3}{5}$$

$$(4) 0 \leq p < \frac{1}{7}$$

40. From a bag containing 4 white and 5 red balls, if 3 balls are drawn at random, then the mean of the number of red balls among the balls drawn is,

$$(1) \frac{5}{3}$$

$$(2) \frac{20}{7}$$

$$(3) \frac{22}{7}$$

$$(4) \frac{25}{9}$$

41. If the mean of Poisson variate X is 1, then $\sum_{r=0}^{\infty} |r-1| P(X=r) =$

$$(1) 1$$

(2) 0

(3) $\frac{2}{e}$

(4) $\frac{1}{e}$

42. A line moves such that the portion of it intercepted between the coordinate axis is of constant length a , then the locus of the mid point of that line segment is

(1) $\frac{x^2}{4} + \frac{y^2}{4} = a^2$

(2) $x^2 + y^2 = a^2$

(3) $x^2 + y^2 = \frac{a^2}{4}$

(4) $x^2 + y^2 = \frac{a^2}{2}$

43. If (h, k) be the point to which the origin has to be shifted in order to get the transformed equation of $y^2 - 4x + 6y + 17 = 0$ as $y^2 = ax$ then $h^2 + k^2 =$

(1) 11

- (2) 1
- (3) 25
- (4) 13

44. The intercepted form of the equation of the straight line passing through the point $(4, -3)$ and perpendicular to the line passing through the points $(1, 1)$ and $(2, 3)$ is

(1) $\frac{x-4}{1} = \frac{y+3}{1}$

(2) $\frac{x}{-2} + \frac{y}{-1} = 1$

(3) $\frac{x}{8} - \frac{y}{6} = 1$

(4) $\frac{x}{1} + \frac{y}{1} = 1$

45. If $x + 2y - 3 = 0$, $3x + 4y - 7 = 0$, $2x + 3y - 4 = 0$, and $4x + 5y - 6 = 0$ are the equations of the four lines then,

- (1) they are the sides of a square
- (2) they are all concurrent lines
- (3) they are the sides of a parallelogram

(4) not all of them are concurrent

46. Two equal sides of an isosceles triangle are given by $7x - y + 3 = 0$ and $x + y - 3 = 0$. If the slope m of the third side is an integer, then $m =$

(1) 3

(2) 1

(3) -1

(4) -3

47. The pair of lines $lx^2 + 2(1+m)xy + my^2 = 0$ lies along two diameters of a circle and divides the circle into 4 sectors. If the area of the bigger sector is 5 times the area of smaller sector,

then $\frac{lm}{(1+m)^2} =$

(1) $\frac{1}{2}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\frac{11}{12}$

(4) $\frac{13}{12}$

48. The line $3x + 4y - 5 = 0$ cuts the curve $2x^2 + 3y^2 = 5$ at A and B.

If O is the origin, then $\angle AOB =$

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{8}$

49. The centre of the circle which passes through the vertices of the triangle formed by the lines $y = 0$, $y = x$ and $2x + 3y = 10$ is

(1) $\left(-\frac{5}{2}, -\frac{1}{2}\right)$

(2) $\left(\frac{5}{2}, -\frac{1}{2}\right)$

(3) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

(4) $\left(\frac{5}{2}, \frac{1}{2}\right)$

50. The distance between the polar of $P(2, 3)$ with respect to the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ and polar of the inverse point of P with respect to the same circle is

(1) 0

(2) $\frac{4}{\sqrt{5}}$

(3) $\frac{12}{\sqrt{5}}$

(4) 1

51. A circle S of radius 2 units lies in the first quadrant and touches both the quadrant axes. The equation of the circle with centre at $(6, 5)$ and touching the circle S externally is

(1) $x^2 + y^2 - 12x - 10y + 12 = 0$

(2) $x^2 + y^2 - 12x - 10y - 20 = 0$

(3) $x^2 + y^2 - 12x - 10y + 25 = 0$

(4) $x^2 + y^2 - 12x - 10y + 52 = 0$

52. If the circles $(x + a)^2 + (y + b)^2 = a^2$ and $(x + c)^2 + (y + d)^2 = d^2$ cuts orthogonally then $b(b - 2d) =$

(1) $c(c - 2a)$

(2) $c(2a - c)$

(3) $d(2c - a)$

(4) $a(a - 2c)$

53. The equation of the circle having common chord of the circles $x^2 + y^2 - 8x = 0$ and $x^2 + y^2 - 9 = 0$ as its diameter is

(1) $x^2 + y^2 - 72x - 207 = 0$

(2) $x^2 + y^2 + 72x + 207 = 0$

(3) $32x^2 + 32y^2 - 72x - 207 = 0$

(4) $32x^2 + 32y^2 + 72x - 207 = 0$

54. The parametric equations of the parabola $y^2 - 8x - 4y - 12 = 0$ are

(1) $x = 2 + 2t^2, y = -2 + 4t$

$$(2) \ x = 2 + 4t, y = -2 + 2t^2$$

$$(3) \ x = -2 + 2t^2, y = 2 + 4t$$

$$(4) \ x = -2 + 4t, y = 2 + 2t^2$$

55. For any non zero real value of m , the equation of the parabola to which the line $mx - y + 10 + m^2 = 0$ is a tangent is,

$$(1) \ x^2 = y - 10$$

$$(2) \ y^2 = 4(x - 2)$$

$$(3) \ x^2 = -4(y - 10)$$

$$(4) \ x^2 = -4y$$

56. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($b > a$) and the parabola $y^2 = ax$ cut at

right angles, if e is the eccentricity of the ellipse, then $2e^2 =$

$$(1) \ 1$$

$$(2) \ \frac{1}{2}$$

$$(3) \ \frac{1}{8}$$

(4) $\frac{1}{3}$

57. Let a tangent drawn at any point on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ cut the X-axis at Q. Let R be the image of Q with respect to $y = x$. If S is a circle with QR as its diameter, then the fixed point through which the circle S passes is

(1) (5, 4)

(2) (4, 5)

(3) (0, 0)

(4) (0, 5)

58. If the normal drawn to the hyperbola $xy = 4$ at $(\alpha_i, \beta_i) (i = 1, 2, 3, 4)$ are concurrent at the point (a, b) then

$\frac{(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4)}{(\beta_1 + \beta_2 + \beta_3 + \beta_4)} (\alpha_1 \alpha_2 \alpha_3 \alpha_4)$ is

(1) $\frac{-16b}{a}$

$$(2) \frac{-16a}{b}$$

$$(3) \frac{4b}{a}$$

$$(4) \frac{4a}{b}$$

59. The distance between the orthocenter and circumcentre of the triangle formed by the points $(1, 2, 3)$, $(3, -1, 5)$ and $(4, 0, -3)$ is

$$(1) \sqrt{\frac{33}{2}}$$

$$(2) \sqrt{\frac{31}{2}}$$

$$(3) \sqrt{\frac{27}{2}}$$

$$(4) \sqrt{\frac{23}{2}}$$

60. The direction cosines of the normal drawn to the plane passing through the points $(2, -1, 5)$, $(1, -3, 4)$, $(5, 2, 1)$ are

$$(1) \frac{11}{\sqrt{179}}, \frac{-7}{\sqrt{179}}, \frac{3}{\sqrt{179}}$$

$$(2) \frac{9}{\sqrt{134}}, \frac{-7}{\sqrt{134}}, \frac{2}{\sqrt{134}}$$

$$(3) \frac{11}{\sqrt{179}}, \frac{7}{\sqrt{179}}, \frac{-3}{\sqrt{179}}$$

$$(4) \frac{9}{\sqrt{134}}, \frac{7}{\sqrt{134}}, \frac{-2}{\sqrt{134}}$$

61. The equation of the plane π through the line of intersection of the planes $\pi_1 = x + 3y - 6 = 0$ and $\pi_2 = 3x - y + 4z = 0$ is $\pi_1 + \lambda\pi_2 = 0$ if the plane π is at the unit distance from the origin then an equation of the plane π is

$$(1) 2x + y + 2z - 3 = 0$$

$$(2) 2x - y - 2z + 3 = 0$$

$$(3) 2x + y + 2z + 3 = 0$$

$$(4) x + 2y + 2z + 3 = 0$$

62. Let $[x]$ denote the greatest integer not exceeding x . If

$$l_1 = \lim_{x \rightarrow 2^+} (x^2 + [x]), l_2 = \lim_{x \rightarrow 3^-} (2x - [x]) \text{ and } l_3 = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{x - \frac{\pi}{2}} \right), \text{ then}$$

(1) $l_2 < l_3 < l_1$

(2) $l_1 < l_3 < l_2$

(3) $l_1 < l_2 < l_3$

(4) $l_3 < l_2 < l_1$

63. If $\lim_{x \rightarrow 0} \frac{[(a - n)nx - \tan x] \sin nx}{x^2} = 0, n \neq 0$ then the minimum

possible positive value of a is

(1) 0

(2) -2

(3) 2

(4) 1

64. If a function f is defined by;

$$f(x) = \begin{cases} 0, & x = 1 \\ x^3 - 1, & 1 < x < \infty \\ x - 1, & -\infty < x < 1 \end{cases}$$

Then at $x = 1$ f is

- (1) continuous and differentiable
- (2) continuous but not differentiable
- (3) discontinuous and differentiable
- (4) discontinuous and not differentiable

65. If $\cos(f(x)) = \frac{1-x^2}{1+x^2}$ and $\tan(g(x)) = \frac{3x-x^3}{1-3x^2}$ then $\frac{df}{dg} =$

(1) $\frac{3}{2}$

(2) $\frac{1+x^2+2x^3}{(1-3x^2)^2}$

(3) $\frac{2}{3}$

(4) $\frac{x^2+x^3}{(1+x^2)(1-3x^2)}$

66. If $x^2 + y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$, then $\frac{dy}{dx} =$

(1) $\frac{2}{x^3}$

(2) $\frac{2}{x^3 y}$

(3) $\frac{1}{x^3}$

(4) $\frac{1}{x^3 y}$

67. If $x = \sin \theta$ and $y = \cos p\theta$, then $(1 - x^2)y_2 =$

(1) $xy_1 - p^2 y$

(2) $p^2 y - xy_1$

(3) xy_1

(4) $p^2 y$

68. If T is the length of the subtangent drawn at any point on the curve $3y^2 = 4x^3$ and N is the length of the subnormal at the same point, then $(3T)^2 =$

(1) $4N^2$

(2) $4N$

(3) $2N$

(4) $8N^2$

69. The interval in which the function $f(x) = \frac{\log(7+x)}{\log(3+x)} (x > 0)$

decreases is

(1) $\left(0, \frac{7}{3}\right)$

(2) $\left(0, \frac{3}{7}\right)$

(3) $(0, 1)$

(4) $(0, \infty)$

70. Let f be a polynomial function defined on $[2, 7]$. If $f(2) = 3$ and $f'(x) \leq 5$ for all x in $(2, 7)$, then the maximum possible value attained by f at $x = 7$ is

(1) 7

(2) 14

(3) 18

(4) 28

71. In the interval $[-2, 4]$, the absolute maximum of occurs at $x =$

(1) 4

(2) -2

(3) -1

(4) 2

72. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = k \cos 4x + c$, then k is

(1) $-\frac{1}{2}$

(2) $-\frac{1}{4}$

(3) $-\frac{1}{8}$

(4) -1

73. If $\int e^{2x} (\cos(3x + 4) + 5x^2) dx =$ then k is

$$(1) e^{2x} \left[\frac{2}{13} \cos(3x+4) + \frac{3}{13} \sin(3x+4) + \frac{5x^2}{2} - \frac{5x}{2} + \frac{5}{4} \right] + c$$

$$(2) e^{2x} \left[\frac{2}{13} \cos(3x+4) - \frac{3}{13} \sin(3x+4) + \frac{5x^2}{2} + \frac{5x}{2} + \frac{5}{4} \right] + c$$

$$(3) e^{2x} \left[\frac{2}{13} \cos(3x+4) - \frac{3}{13} \sin(3x+4) - \frac{5x^2}{2} - \frac{5x}{2} - \frac{5}{4} \right] + c$$

$$(4) e^{2x} \left[\frac{2}{13} \cos(3x+4) - \frac{3}{13} \sin(3x+4) + \frac{5x^2}{2} - \frac{5x}{2} + \frac{5}{4} \right] + c$$

74. If $\int \frac{5 \cot x + 1}{(\cot x - 1)(\cot x - 2) \sin^2 x} dx = 6 \log |f(x)| + 11 \log |g(x)| + c$

then $(f(x), g(x)) =$

$$(1) ((\cot x - 1), (\cot x - 2)^{-1})$$

$$(2) ((\cot x - 1)^{-1}, (\cot x - 2))$$

$$(3) ((\cot x - 1)^{-1}, (\cot x - 2)^{-1})$$

$$(4) (\cot x - 1, \cot x + 2)$$

75. If $I_{m,n} = \int e^{mx} \cdot x^n dx$ then $I_{m,n} + \frac{n}{m} I_{m,n-1} =$

$$(1) x^n e^{mx} + c$$

$$(2) \frac{x^n e^{mx}}{n} + c$$

$$(3) \frac{x^n e^{mx}}{m} + c$$

$$(4) \frac{-x^n e^{mx}}{m} + c$$

$$76. \lim_{n \rightarrow \infty} \frac{1}{n} [(n+1)(n+2)\dots(2n)]^{\frac{1}{n}} =$$

$$(1) 1$$

$$(2) 0$$

$$(3) \frac{2}{e}$$

$$(4) \frac{4}{e}$$

$$77. \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx =$$

$$(1) \frac{3}{\sqrt{2}} \log(\sqrt{2} + 1)^{\frac{1}{2}}$$

$$(2) \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1)$$

$$(3) \frac{\sqrt{2}}{3} \log(\sqrt{3} + 1)$$

$$(4) \frac{2}{\sqrt{3}} \log(\sqrt{2} - 1)$$

78. The area (in sq units) enclosed by the loop of the curve $ay^2 = x^2(a - x)$, ($a > 0$) is

$$(1) 2\pi a^2$$

$$(2) \frac{\pi}{3} a^2$$

$$(3) \frac{4}{15} a^2$$

$$(4) \frac{8}{15} a^2$$

79. The differential equation corresponding to the family of curves

$$y = e^x (A \cos x + B \sin x) \text{ is}$$

$$(1) y'' + y' + y = 0$$

$$(2) y'' + 2y' + 2y = 0$$

$$(3) y'' - 2y' + 2y = 0$$

$$(4) y'' - 2y' - 2y = 0$$

80. The solution to the differential equation $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$ is

(Here, k is an arbitrary constant)

$$(1) x = y \sin^{-1}\left(\frac{k}{x}\right)$$

$$(2) y = x \sin^{-1}\left(\frac{k}{x}\right)$$

$$(3) x \sin y + k = 0$$

$$(4) y = x \cos(kx)$$

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81. Assertion (A) : The number 0.00764 has three significant figures.

Reason (R) : If the number is less than 1, the zeros on the right of the decimal point but to the left of the first non-zero digit are not significant.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

82. A car moving with a velocity 6.25 ms^{-1} is decelerated with $2.5\sqrt{v} \text{ ms}^{-2}$ ('v' is instantaneous velocity). Time taken by the car to come to rest is

- (1) 2 s
- (2) 3 s

(3) 2.5 s

(4) 4 s

83. A bullet fired from a gun falls at a distance half of its maximum range. The angle of projection of the bullet is

(1) 45°

(2) 60°

(3) 30°

(4) 15°

84. A body is projected at an angle of 45° from a point on the ground at a distance of 30 m from the foot of a vertical pole of height 20 m. The body just crosses the top of the pole and strikes the ground at a distance s from the foot of the pole on the other side of the pole. Then $s =$

(1) 20 m

(2) 30 m

(3) 50 m

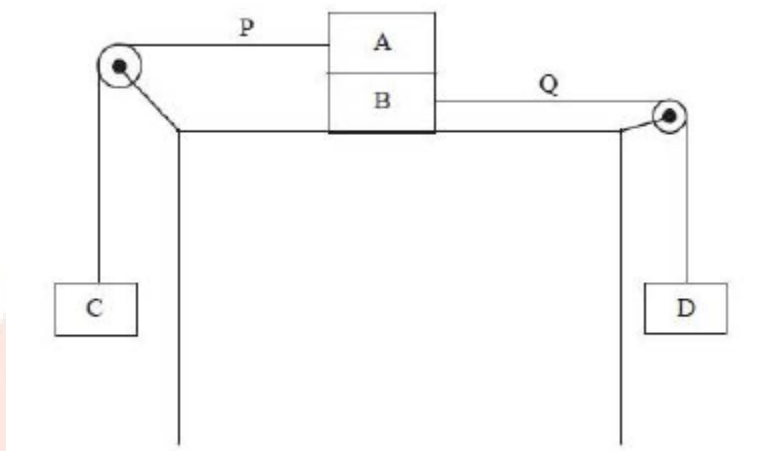
(4) 60 m

85. An explosion blows a stationary rock into three parts. Two parts of masses 1 kg and 2 kg move at right angles to one another with velocities 12 ms^{-1} and 8 ms^{-1} respectively. If the velocity of third part is 4 ms^{-1} , the mass of the rock is

- (1) 8 kg
- (2) 5 kg
- (3) 17 kg
- (4) 3 kg

86. Four blocks A, B, C and D of masses 6 kg, 3 kg, 6 kg and 1 kg respectively are connected by light strings passing over frictionless pulleys as shown in the figure. The strings P and Q are horizontal. The coefficient of friction between the horizontal surface and the block B is 0.2 and the blocks A and B move together. If the system is released from rest then the tension in the string Q is

(Acceleration due to gravity = 10 ms^{-2})



- (1) 48 N
- (2) 24 N
- (3) 12 N
- (4) 6 N

87. A constant power of 7 watt is applied on a toy car of mass 15 kg. The distance travelled by the car when its velocity increases from 3 ms^{-1} to 5 ms^{-1} is.

- (1) 56 m
- (2) 7 m
- (3) 61 m
- (4) 70 m

88. A body A moving with momentum 'P' collides one dimensionally with another stationary body B of same mass.

During impact, A gives impulse 'J' to B. Then which of the following is/are correct?

(a) The total momentum of A and B is 'P' before and after impact and $(P - J)$ during the impact.

(b) During the impact, B gives impulse of magnitude J to A.

(c) The coefficient of restitution is $\left[\frac{2J}{P} - 1 \right]$

(d) The coefficient of restitution is $\left[\frac{2J}{P} + 1 \right]$

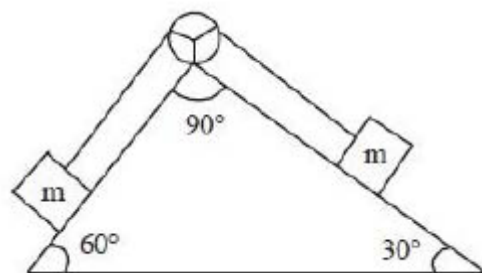
(1) only a is correct

(2) a and c are correct

(3) b and c are correct

(4) only c is correct

89. In the figure shown, the blocks have equal masses. Friction, mass of the string and the mass of the pulley are negligible. The magnitude of the acceleration of the centre of mass of the two blocks is (g – acceleration due to gravity)



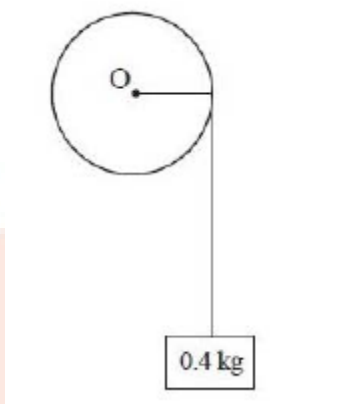
$$(1) \left(\frac{\sqrt{3}-1}{\sqrt{2}} \right) g$$

$$(2) \frac{g}{2}$$

$$(3) (\sqrt{3}-1)g$$

$$(4) \left(\frac{\sqrt{3}-1}{4\sqrt{2}} \right) g$$

90. A wheel of radius 8 cm is attached to a support so as to rotate about a horizontal axis through its centre. A string of negligible mass wrapped around its circumference carries a mass of 0.4 kg attached to its free end. When the mass is released, it descends through 1 m in 10 seconds, then its moment of inertia is (Acceleration due to gravity = 10 ms^{-2})



- (1) 1.277 kg m^2
- (2) 2.177 kg m^2
- (3) 21.77 kg m^2
- (4) 12.77 kg m^2

91. A body of mass 1 kg is suspended from a spring of negligible mass. Another body of mass 500 g moving vertically upwards hits the suspended body with a velocity of 3 ms^{-1} and gets embedded in it. If the frequency of oscillation of the system of the two bodies after collision is $\frac{10}{\pi} \text{ Hz}$, the amplitude of motion and the spring constant are respectively

- (1) 5 cm , 300 Nm^{-1}
- (2) 10 cm , 300 Nm^{-1}
- (3) 10 cm , 600 Nm^{-1}

(4) 5 cm, 600 Nm^{-1}

92. The gravitational field in a region is given by $\vec{E} = (5\hat{i} + 12\hat{j}) \text{ Nkg}^{-1}$. If a particle of mass 2 kg is moved from the origin to the point (12 m, 15 m) in this region, the change in gravitational potential energy is

- (1) -450 J
- (2) -480 J
- (3) -240 J
- (4) -500 J

93. A uniform wire of length 10 m and diameter 0.6 mm is stretched by 6 mm with certain force. If the Poisson's ratio of the material of the wire is 0.3, then the change in diameter of the wire is

- (1) $108 \times 10^{-8} \text{ m}$
- (2) $108 \times 10^{-6} \text{ m}$
- (3) $10.8 \times 10^{-8} \text{ m}$
- (4) $1.08 \times 10^{-8} \text{ m}$

94. Two tubes of same length and diameter 4 mm and 8 mm are joined together to form a U-shaped tube open at both ends. If the U-tube contains water, then the difference between the levels of water in the two limbs of the tube is

(Surface Tension of water at the temperature of experiment is $7.3 \times 10^{-2} \text{ Nm}^{-1}$, angle of contact = 0° , density of water = $1.0 \times 10^3 \text{ kgm}^{-3}$ and acceleration due to gravity = 10 ms^{-2})

- (1) 3.65 mm
- (2) 36.5 mm
- (3) 0.365 mm
- (4) 365 mm

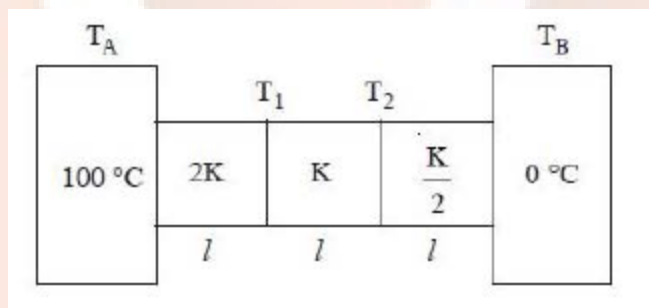
95. A uniform metal bar of length 10 m with a crack at its midpoint is clamped between two rigid supports. The bar buckles upwards due to a temperature rise of 40°C . If the coefficient of linear expansion of the metal is $25 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, the maximum displacement of the midpoint of the bar is

- (1) 11.3 cm
- (2) 22.3 cm
- (3) 33.3 cm

(4) 44.3 cm

96. Three rods each of length ' l ' and cross sectional area ' A ' are joined in series between two heat reservoirs as shown in the figure. Their conductivities are $2K$, K and $\frac{K}{2}$ respectively.

Assuming that the conductors are insulated from surrounding, the temperatures T_1 and T_2 of the junctions in steady state conditions are respectively



(1) $\frac{600}{7}^\circ\text{C}, \frac{400}{7}^\circ\text{C}$

(2) $\frac{600}{7}^\circ\text{C}, \frac{700}{4}^\circ\text{C}$

(3) $\frac{500}{6}^\circ\text{C}, \frac{600}{5}^\circ\text{C}$

(4) $\frac{600}{4}^\circ\text{C}, \frac{400}{7}^\circ\text{C}$

97. Two heat engines X and Y of same efficiency are connected in series in such a way that the sink of X works as source of Y. X receives heat at 900 K and rejects some heat to its sink at T K and in turn Y rejects heat to its sink at 400 K, then the temperature T is

- (1) 550 K
- (2) 600 K
- (3) 650 K
- (4) 700 K

98. The specific heat capacities of an ideal gas at constant pressure and at constant volume are $620 \text{ Jkg}^{-1}\text{K}^{-1}$ and $420 \text{ Jkg}^{-1}\text{K}^{-1}$ respectively. The density of the gas at STP is approximately

- (1) 2.88 kg m^{-3}
- (2) 4.86 kg m^{-3}
- (3) 3.88 kg m^{-3}
- (4) 1.86 kg m^{-3}

99. Three closed vessels A, B and C are at the same temperature 'T' and contain gases. Vessel A contains only O_2 , B contains only N_2 and C contains a mixture of equal quantities of O_2 and N_2 . If

the rms speed of O_2 molecules in vessel 'A' is v_1 and that of N_2 molecules in vessel 'B' is v_2 then the rms speed of O_2 molecules in vessel 'C' is

(1) $\left(\frac{v_1 + v_2}{2} \right)$

(2) v_1

(3) $(v_1 v_2)^{\frac{1}{2}}$

(4) $\frac{v_1}{2}$

100. Match the following List-I with List-II.

<u>List-I</u>		<u>List-II</u>
(A) Transverse wave	(I)	Vibrations parallel to the direction of propagation
(B) Longitudinal wave	(II)	Vibrations perpendicular to the direction of propagation
(C) Beats	(III)	Superposition of waves travelling in opposite direction
(D) Stationary waves	(IV)	Superposition of waves

travelling in same direction

The correct answer is

- | | A | B | C | D |
|-----|-----|----|-----|-----|
| (1) | II | I | III | IV |
| | A | B | C | D |
| (2) | II | I | IV | III |
| | A | B | C | D |
| (3) | III | IV | I | II |
| | A | B | C | D |
| (4) | IV | I | II | III |

101. A police car moving at 22 ms^{-1} chases a motor cyclist. The police man sounds horn at 176 Hz , while both of them move towards a stationary siren of frequency 165 Hz . If the number of beats heard by the motor cyclist per second is zero, then the speed of motorcycle is

(speed of sound in air = 330 ms^{-1})

- (1) 33 ms^{-1}
- (2) 22 ms^{-1}
- (3) 44 ms^{-1}
- (4) 11 ms^{-1}

102. When an object is moved along the principle axis of a concave mirror placed in air, the image coincides with the object if the object is 50 cm from the mirror. If the mirror is placed at a depth of 20 cm in a transparent medium, the image coincides with the object when the object is 40 cm from the mirror. The refractive index of the liquid is

(1) $\frac{5}{4}$

(2) $\frac{4}{3}$

(3) $\frac{3}{2}$

(4) $\frac{5}{3}$

103. In Young's double slit experiment, light of wavelength $5900 \text{ \AA}$ is used. When the slits are 2 mm apart, the fringe width is 1.2 mm. If the slit separation is increased to one and half times the previous value, then the fringe width will be

(1) 0.9 mm

(2) 0.8 mm

(3) 1.8 mm

(4) 1.6 mm

104. Two particles with charges $+3.72 \mu\text{C}$ and $+1.86 \mu\text{C}$ are some distance apart. If 20% of charge is transferred from first particle to second particle then electrostatic force between them is

(1) decreases by 12%

(2) increases by 12%

(3) increases by 4%

(4) decreases by 4%

105. ABC is a right angled triangle in which $AB = 3 \text{ cm}$, $BC = 4 \text{ cm}$ and right angle is at B. Three charges $+15 \mu\text{C}$, $+12 \mu\text{C}$ and $-20 \mu\text{C}$ are placed respectively at A, B and C. The force acting on charge at B is

(1) 1250 N

(2) 3500 N

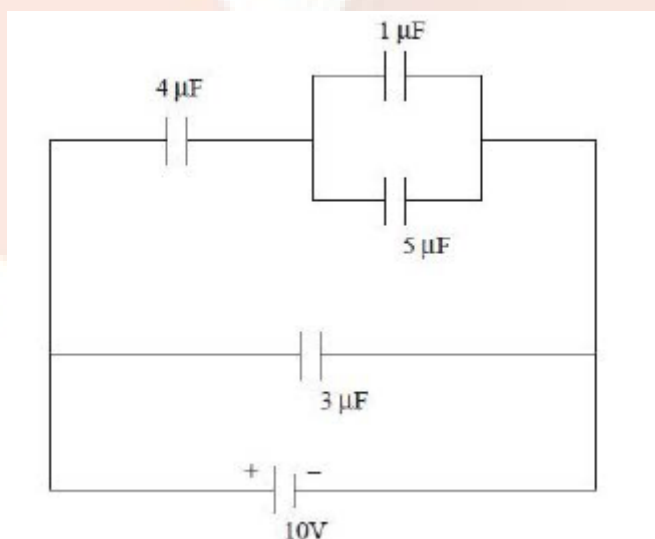
(3) 1200 N

(4) 2250 N

106. A spherical capacitor has outer sphere of radius 5 cm and inner sphere of radius 2 cm. When the inner sphere is earthed, its capacity is C_1 and when the outer sphere is earthed its capacity is C_2 . Then $\frac{C_1}{C_2}$ is

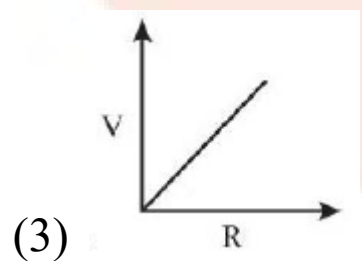
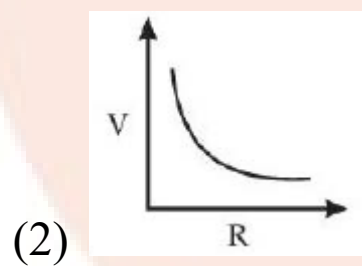
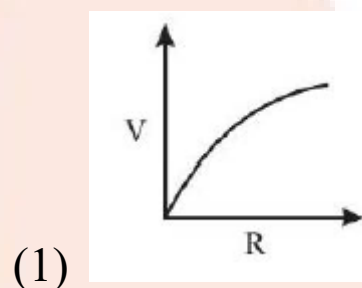
- (1) $\frac{5}{2}$
- (2) $\frac{2}{5}$
- (3) $\frac{7}{3}$
- (4) $\frac{3}{7}$

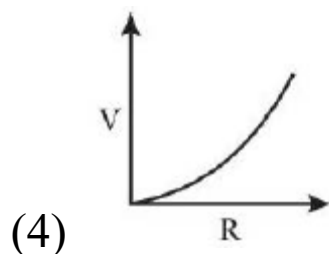
107. The charge on $4\ \mu\text{F}$ capacitor, in the given circuit is



- (1) $24\ \mu\text{C}$
- (2) $100\ \mu\text{C}$
- (3) $2.4\ \mu\text{C}$
- (4) $30\ \mu\text{C}$

108. A cell of emf ' ε ' and internal resistance ' r ' is connected across a variable load resistance R . The graph drawn between its terminal voltage and the resistance R is





109. In a meter bridge if the left and right gaps are connected with $2\ \Omega$ and $3\ \Omega$ resistances respectively then the bridge is balanced. The resistance to be connected with $3\ \Omega$ resistance to get the balancing point at midpoint of the bridge wire is

- (1) $3\ \Omega$ in series
- (2) $3\ \Omega$ in parallel
- (3) $6\ \Omega$ in series
- (4) $6\ \Omega$ in parallel

110. Magnetic field at the centre of a circular loop of area 'A' is 'B'. Then the magnetic moment of the loop is

(μ_0 - permeability of free space)

(1) $\frac{BA^2}{\mu_0\pi}$

(2) $\frac{BA\sqrt{A}}{\mu_0}$

$$(3) \frac{BA\sqrt{A}}{\mu_0\pi}$$

$$(4) \frac{2BA\sqrt{A}}{\mu_0\sqrt{\pi}}$$

111. A circular coil of radius 10 cm with 100 turns carrying a current of 0.5 A lies in a magnetic field of 2 T such that the normal drawn to the plane of the coil makes an angle ' θ ' with the direction of the field. Work done in rotating the coil to change the angle ' θ ' from 0° to 180° is

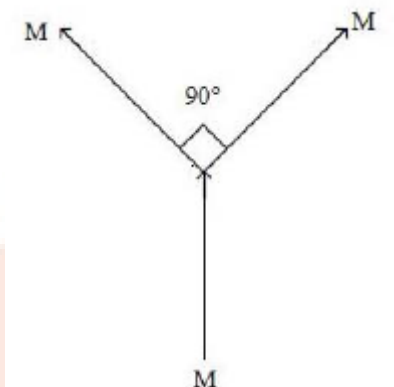
(1) π J

(2) 2π J

(3) 4π J

(4) 8π J

112. The resultant magnetic moment of three magnetic dipoles, each of magnetic moment ' M ' shown in the arrangement is



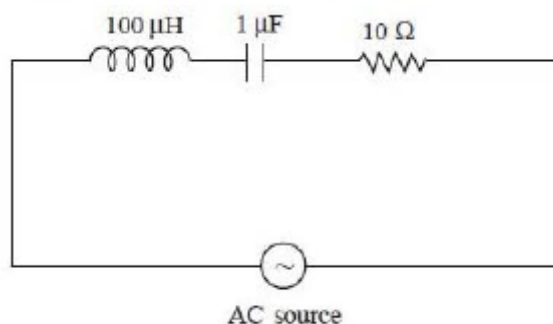
- (1) $\sqrt{2} M$
- (2) $(\sqrt{2} + 1) M$
- (3) $(\sqrt{2} - 1) M$
- (4) M

113. A long solenoid with 2000 turns per metre has a small loop of radius 3 cm placed inside the solenoid normal to its axis. If the current through the solenoid increases steadily from 1.5 A to 5.5

A in $\frac{\pi^2}{100}$ s, the induced emf in the loop is

- (1) 0.144 mV
- (2) 0.288 mV
- (3) 0.072 mV
- (4) 0.316 mV

114. In the given circuit, the angular frequency of the voltage sources is $70 \times 10^3 \text{ rads}^{-1}$. The circuit effectively behaves like



- (1) Purely resistive circuit
- (2) Series RL circuit
- (3) Series RC circuit
- (4) Series LC circuit with $R = 0$

115. A parallel plate capacitor consists of two circular plates each of radius 2 cm, separated by a distance of 0.1 mm. If the potential difference across the plates is varying at the rate of $5 \times 10^6 \text{ Vs}^{-1}$, then the value of displacement current is

- (1) 5.56 A
- (2) 5.56 mA
- (3) 0.556 mA
- (4) 2.28 mA

116. Light of wavelength 488 nm produced by an Argon laser is used in the photoelectric effect. When light from this spectral line is incident on the cathode, the stopping potential of photoelectrons is 0.38 V. The work function of the cathode material is

- (1) 2.16 eV
- (2) 216 eV
- (3) 21.6 eV
- (4) 0.216 eV

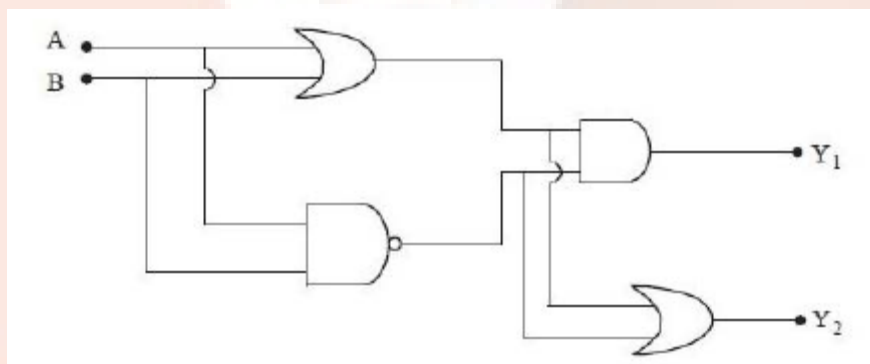
117. If the first excitation potential of a hypothetical hydrogen like atom is 15 V, then the third excitation potential of the atom is

- (1) 13.6 V
- (2) $\frac{4}{75}$ V
- (3) $\frac{15}{16}$ V
- (4) $\frac{75}{4}$ V

118. The energy released when one nucleus of ${}_{92}\text{U}^{235}$ undergoes fission is 188 MeV. The energy released when 100 g of ${}_{92}\text{U}^{235}$ undergoes fission is

- (1) 3.55×10^{12} J
- (2) 7.71×10^{12} J
- (3) 3.55×10^{13} J
- (4) 7.71×10^{13} J

119. The values of Y_1 and Y_2 respectively in the following logic circuit if both A and B are 1 are



- (1) 1, 1
- (2) 1, 0
- (3) 0, 1
- (4) 0, 0

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120. If E_c and E_m are peak values of carrier and modulating signals respectively then for 100% modulation

(1) $E_c = \frac{E_m}{2}$

(2) $\frac{E_c^2}{2} = E_m^2$

(3) $E_c = E_m$

(4) $E_c = 2E_m$

121. The work function (W_0) of Li, K, Mg, Ag and Cu are 2.42, 2.25, 3.70, 4.30 and 4.80 eV respectively. The number of metals which undergo photoelectric effect if a radiation of wavelength 540 nm falls on them is-

(1 eV = 1.602×10^{-19} J)

(1) 4

(2) 2

(3) 1

(4) 3

122. What is the mass of a particle with a wavelength of $3.13 \text{ \AA}$ moving with a speed of $2.0 \times 10^8 \text{ m s}^{-1}$?

(1) $1.0 \times 10^{-28} \text{ kg}$

(2) $2.0 \times 10^{-32} \text{ kg}$

(3) $1.0 \times 10^{-32} \text{ kg}$

(4) $2.0 \times 10^{-28} \text{ kg}$

123. If the electronic configuration of M^{3+} is $[\text{Xe}] 4f^3$, then M^{3+} is

(1) Nd^{3+}

(2) Pr^{3+}

(3) Sm^{3+}

(4) Dy^{3+}

124. Observe the following statements.

- i. According to VSEPR theory ClF_3 and SO_2 are shown as AB_3E and AB_2E type molecules respectively.
- ii. SF_4 has “See-saw” shape.

iii. HgCl_2 and PbCl_2 have same shape.

- (1) i, ii only
- (2) i, iii only
- (3) i, ii, iii
- (4) ii, iii only

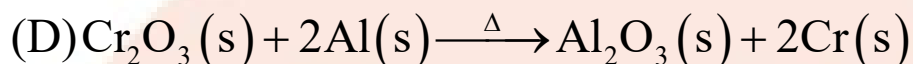
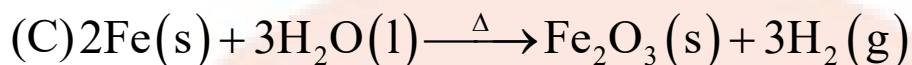
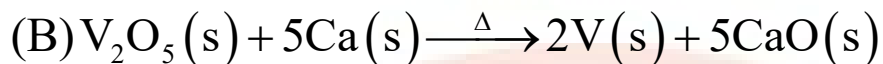
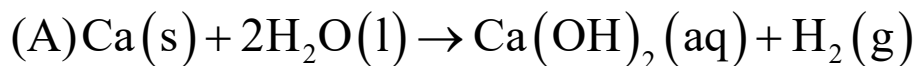
125. Observe the following molecules, C_2 , N_2 , O_2 , F_2 Which one of the following statements for the above molecules ?

- (1) They exhibit same magnetic property
- (2) They have same number of bonding molecular orbitals and same number of antibonding molecular orbitals.
- (3) The sequence of molecular orbitals is as follows.
- (4) They have same bond order.

126. The ratio of rates of diffusion of gases A and B is 1:0.707. If the molecular weight of B is 32, the molecular weight of A is,

- (1) 2
- (2) 64
- (3) 16
- (4) 8

127. Which of the following are non-metal displacement reactions ?



(1) A, B, C, D

(2) B, C only

(3) C, D only

(4) A, C only

128. Match the following.

LIST – I

A) $\Delta U = W_{\text{ad}}$

B) $\Delta U = q - W$

C) $\Delta U = -q$

D) $\Delta U = 0$

LIST – II

I) Isothermal reversible expansion

II) Wall is adiabatic

III) Thermally conducting walls

IV) Isolated system

V) Closed system

A B C D
(1) V I II III

A B C D
(2) I III II IV

A B C D
(3) II V III I

A B C D
(4) II V I III

129. 18.4 g N_2O_4 was placed in 1 L vessel at 400 K and allowed to attain the following equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$

If the total pressure at equilibrium was 10.64 bar, approximate K_p is ($R = 0.083 \text{ L bar K}^{-1}\text{mol}^{-1}$)

(Assume N_2O_4 , NO_2 , as ideal gases)

- (1) 57.20
- (2) 24.24
- (3) 14.30
- (4) 6.64

130. If the pH of a buffer solution containing 0.1 M of monoacidic base and 0.01 M of its salt is 10.5. the pK_a of conjugate acid is

- (1) 9.5
- (2) 4.5

(3) 3.5

(4) 11.5

131. Three vessels (A, B, C) contain H_2O_2 solution. In vessel A, 500 mL of 10 vol H_2O_2 is present. 100 mL of 30 vol H_2O_2 is present in vessel B. Vessel C is filled with 250 mL of 2M H_2O_2 . The weight (in g) of H_2O_2 present in these vessels follows the order

(1) $C > A > B$

(2) $C > B > A$

(3) $B > A > C$

(4) $A > B > C$

132. Identify correct statements from the following:

i. Beryllium oxide is an amphoteric oxide.

ii. Group II elements dissolve in liquid ammonia to form deep blue-black solutions.

iii. The hydration enthalpy of group II ions decreases from Be^{2+} to Ba^{2+} .

(1) i, ii only

(2) ii, iii only

(3) i, iii only

(4) i, ii, iii

133. Identify correct statements from the following.

i. H_3BO_3 , is a monobasic acid.

ii. The correct formula of borax is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_3] \cdot 8\text{H}_2\text{O}$

iii. NaBH_4 is a reducing agent.

(1) i, ii, iii

(2) i, ii only

(3) ii, iii only

(4) i, iii only

134. Observe the followings statements regarding C_{60} .

i. All carbons are sp^2 hybridized.

ii. It contains 12 rings of five carbons each and 20 ring of six carbon each.

iii. It is a non-aromatic compound

iv. It is pure form of carbon

v. C-C bond lengths in it are 143.5 and 138.3 pm

vi. It is prepared by heating graphite in an electric arc in the presence of oxygen

- (1) i, ii, iii, iv, v
- (2) i, ii, iv, v only
- (3) i, ii, iv, vi only
- (4) i, ii, iii only

135. Identify the correct statements from the following:

- i. In the presence of UV light, CF_2Cl_2 (g) gives chlorine free radicals which will react with O_3 (g) to form O_2 (g)
- ii. Drinking water with 10 ppm fluoride is better than drinking water with 1 ppm fluoride
- iii. The maximum permissible concentration of lead in drinking water is 50 ppb

- (1) i, ii, iii
- (2) i, ii only
- (3) ii, iii only
- (4) i, iii only

136. Which of the following conversion represents Fries rearrangement?

- (1) O-acylated phenol to C-acylated phenol
- (2) C-acylated phenol to O-acylated phenol

(3) N-acylated phenol to C-acylated phenol

(4) C-acylated phenol to N-acylated phenol

137. What are X and Y in the following reaction?



X

Y



X

Y



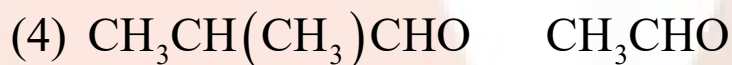
X

Y

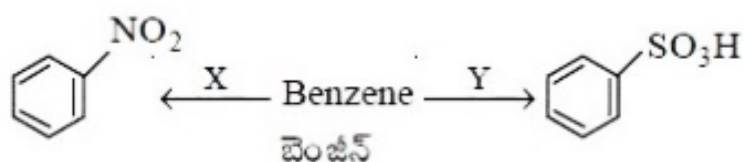


X

Y



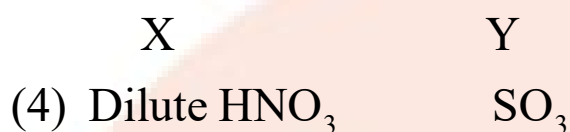
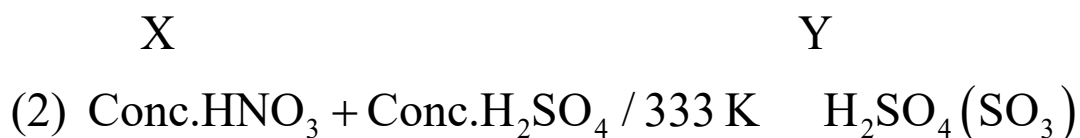
138. What are X and Y in the following reactions?



X

Y





139. Which of the following statements are not correct?

- (A) Diode is a combination of n-type and p-type semiconductors
- (B) Silicon or germanium is electron rich impurity
- (C) Phosphorous and arsenic are electron deficient impurities
- (D) Schottky defect decreases the density of the crystal

- (1) B, C
- (2) A, D
- (3) C, D
- (4) A, C

140. At 298 K, the vapour pressure of a solution of 7.5 g of non-volatile solute in 90 g of water is 2.8 kPa. If 18 g of water is added to this solution the vapour pressure becomes 2.81 kPa at same temperature, the molar mass of solute in g mol⁻¹ is

- (1) 17.5
- (2) 68.2
- (3) 71.5
- (4) 51.8

141. At $T(K)$, the vapour pressures of pure liquids A and B are 100 mm and 160 mm respectively. An ideal solution is formed by mixing 2 moles of A and 3 moles of B at the same temperature. The mole fraction of A and B in the vapour state respectively are

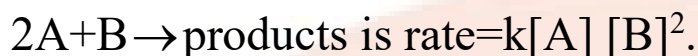
- (1) 0.706, 0.294
- (2) 0.294, 0.706
- (3) 0.40, 0.60
- (4) 0.60, 0.40

142. In which of the following cells, the space between cathode and anode is filled by a moist mixture of ammonium chloride and zinc chloride

- (1) Mercury cell
- (2) Leclanche cell
- (3) Nickel-Cadmium cell

(4) Fuel cell

143. The rate equation for the reaction



If k at $T(K)$ is $5.0 \times 10^{-6} \text{ mol}^{-2} \text{L}^2 \text{s}^{-1}$, the initial rate of the reaction, when $[A] = 0.05 \text{ mol L}^{-1}$ and $[B] = 0.1 \text{ mol L}^{-1}$ is

(1) $1.25 \times 10^{-9} \text{ L mol L}^{-1} \text{s}^{-1}$

(2) $1.25 \times 10^{-9} \text{ mol L}^{-1} \text{s}^{-1}$

(3) $2.50 \times 10^{-9} \text{ mol L}^{-1} \text{s}^{-1}$

(4) $2.50 \times 10^{-9} \text{ L mol L}^{-1} \text{s}^{-1}$

144. Which one of the following statements is not correct?

(1) The process of settling down of colloidal particles is coagulation

(2) The mass in milligrams of lyophilic sol which protects the coagulation of 10 mL of a gold sol on adding 1 mL of 10 % NaCl solution is its gold number

(3) The layer of positive or negative charge acquired by selective adsorption of ion on the surface of a colloidal particle is electrokinetic potential

- (4) The potential difference between the fixed layer on the colloidal particles and the diffused layer of opposite charge is zeta potential

145. Identify the correct statements from the following:

- i. In the extraction of Ag and Au, zinc is used as reducing agent.
- ii. Impure zinc can be refined by distillation method.
- iii. Malachite is an ore of nickel.

- (1) i, ii, iii
- (2) i, iii only
- (3) ii, iii only
- (4) i, ii only

146. White phosphorous reacts with sulphuryl chloride to form PCl_5 and X. Chlorine reacts with X in the presence of wood charcoal to form Y. X and Y are respectively

- (1) SO_2 , SO_2Cl_2
- (2) SO_2 , SCl_4
- (3) SO_3 , SO_2Cl_2
- (4) SO_3 , SCl_4

147. Identify the correct statements from the following:

- i. Sulphuric acid is manufactured by contact process.
- ii. SO_3 dissolves in H_2SO_4 to form pyrosulphuric acid.
- iii. H_2SO_4 is used in the manufacture of fertilizers such as ammonium sulphate and super phosphate.
- iv. In the reaction $\text{S} + 2\text{H}_2\text{SO}_4(\text{Conc.}) \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$, H_2SO_4 is oxidized to SO_2 .

- (1) i, ii, iii, iv
- (2) i, ii, iv only
- (3) i, iii, iv only
- (4) i, ii, iii only

148. Assertion (A): Helium has lowest boiling point (4.2 K)

Reason (R): The forces that exist between Helium atoms are weak dispersion forces.

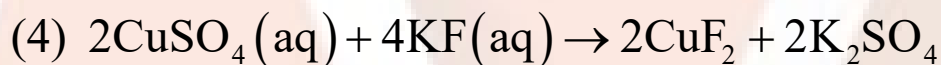
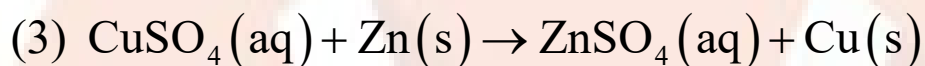
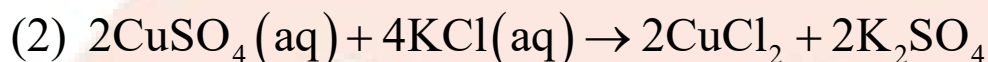
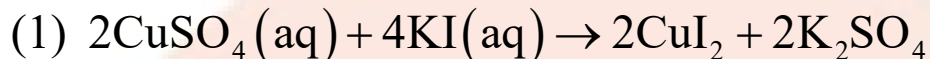
The correct answer is

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

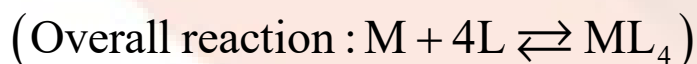
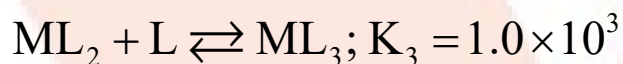
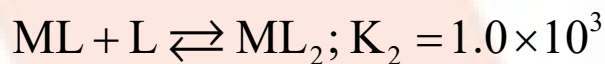
(4) (A) is not correct but (R) is correct.

149. Which one of the following reactions does not take place?



150. The stepwise stability constants of a complex are given below.

What is its overall reaction stability constant (β_4)?



(1) 1.0×10^{12}

(2) 12.1×10^3

(3) 1.0×10^6

(4) 1.0×10^8

151. Number average molecular mass of a polymer that contains 15 molecules with each of mass 8,000 and 15 molecules with each of mass 80,000 is

- (1) 22,000
- (2) 33,000
- (3) 11,000
- (4) 44,000

152. Which of the following statements are correct?

- A. A tripeptide has two peptide bonds
- B. A pentapeptide contains five amino acids
- C. Nucleotide is a product of base and sugar
- D. In cellulose, β -glycosidic linkages are present

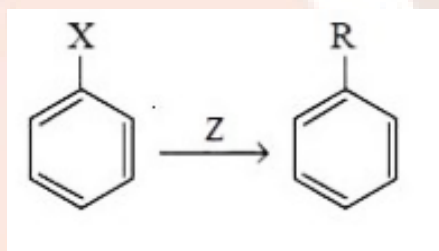
- (1) B, C, D
- (2) C, D only
- (3) A, B, D
- (4) A, C only

153. Identify antihistamines from the following

Serotonin	Dimethane	Phenelzine	Seldane
1	2	3	4

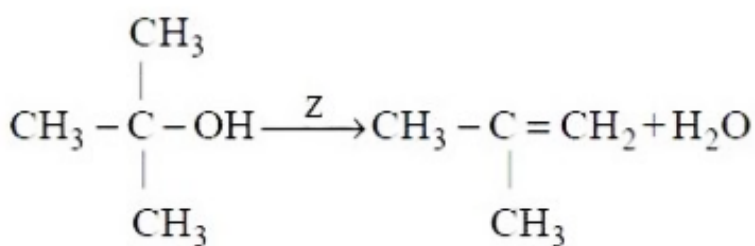
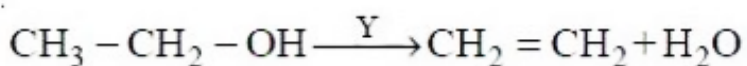
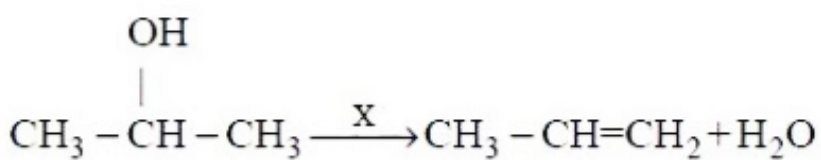
- (1) 1, 3 only
- (2) 1, 3, 4
- (3) 2, 4 only
- (4) 1, 2, 3

154. Identify the Z in the reaction.



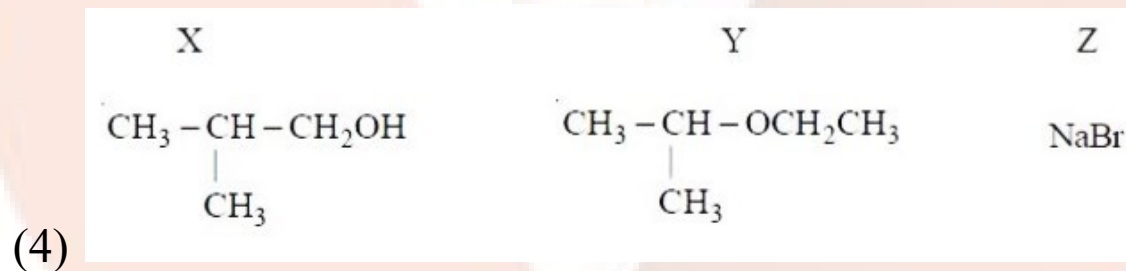
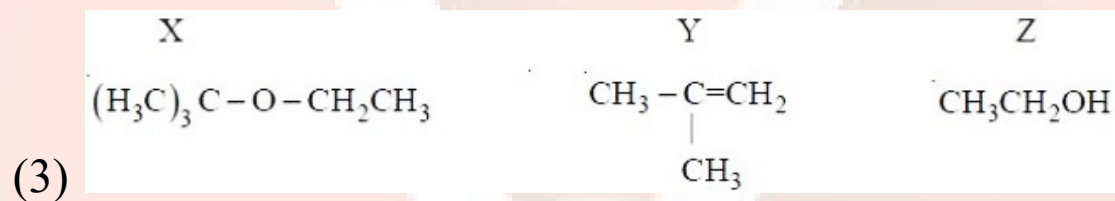
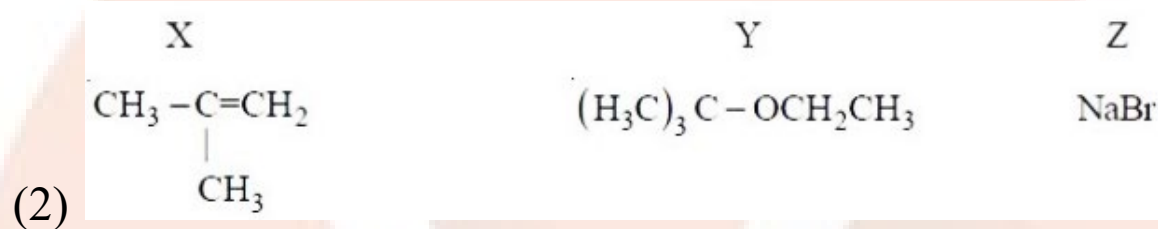
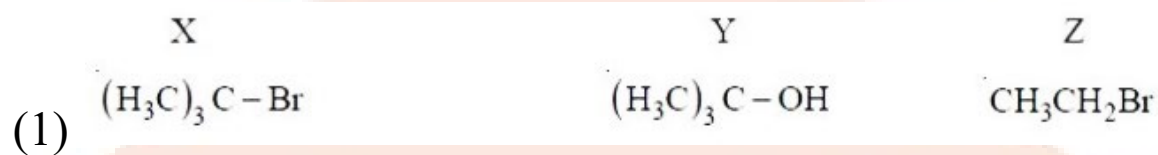
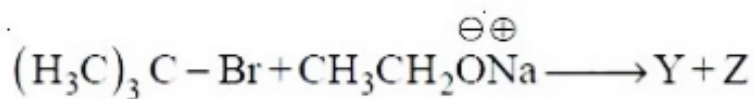
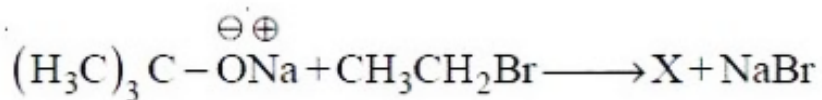
- (1) $\text{RCOX}/\text{Anhydrous AlCl}_3$
- (2) $\text{RX}/\text{Na}/\text{dry (C}_2\text{H}_5)_2\text{O}$
- (3) $\text{Na}/\text{dry (C}_2\text{H}_5)_2\text{O}$
- (4) $\text{RCOX}/\text{FeCl}_3$

155. What are X, Y, Z in the following reactions?

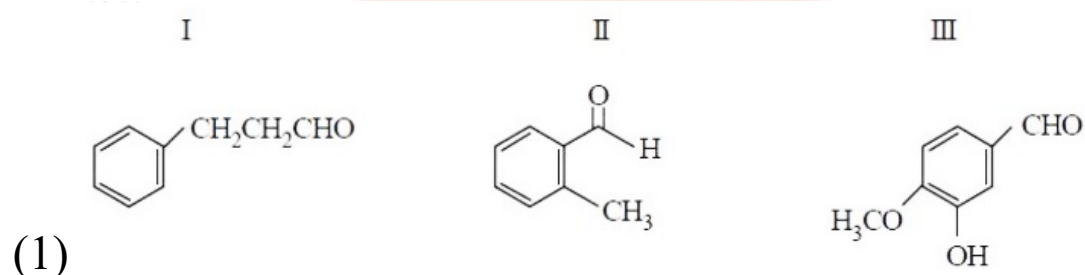


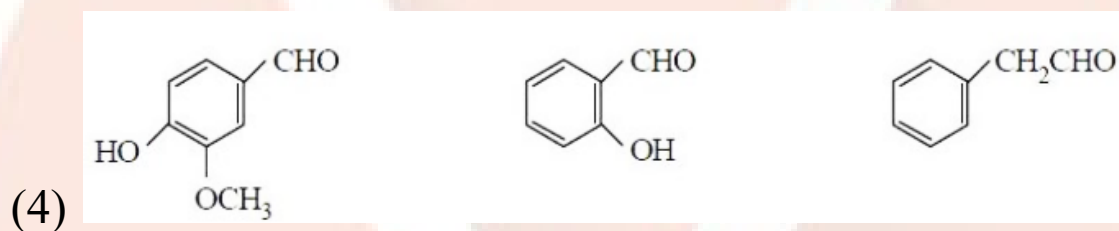
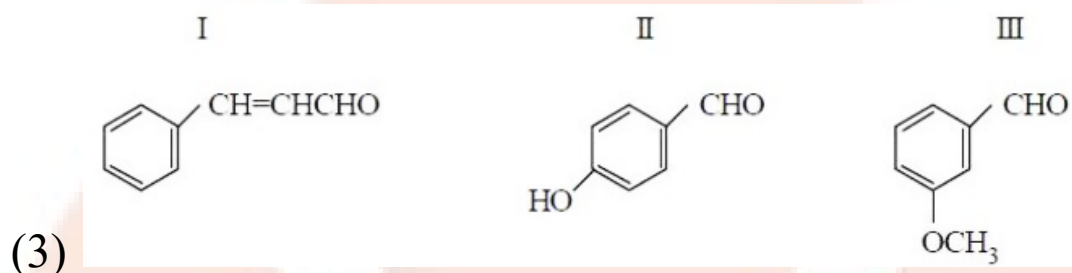
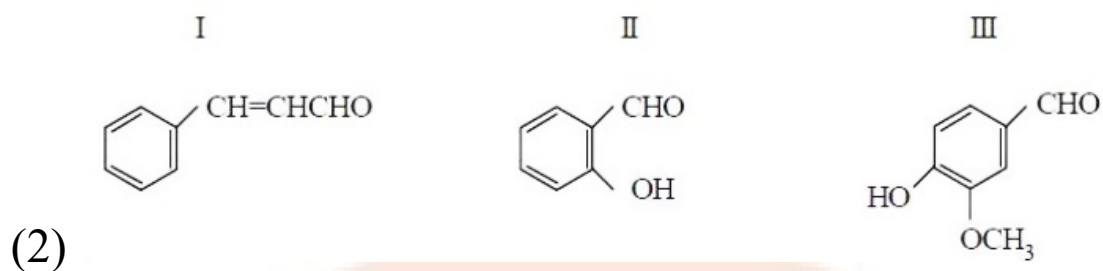
- | X | Y | Z |
|---|---|---|
| (1) $\text{H}_2\text{SO}_4, 443 \text{ K}$ | $85\% \text{H}_3\text{PO}_4, 440 \text{ K}$ | $20\% \text{H}_3\text{PO}_4, 358 \text{ K}$ |
| X | Y | Z |
| (2) $85\% \text{H}_3\text{PO}_4, 440 \text{ K}$ | $\text{H}_2\text{SO}_4, 443 \text{ K}$ | $20\% \text{H}_3\text{PO}_4, 358 \text{ K}$ |
| X | Y | Z |
| (3) $20\% \text{H}_3\text{PO}_4, 358 \text{ K}$ | $\text{H}_2\text{SO}_4, 443 \text{ K}$ | $85\% \text{H}_3\text{PO}_4, 440 \text{ K}$ |
| X | Y | Z |
| (4) $\text{H}_2\text{SO}_4, 443 \text{ K}$ | $20\% \text{H}_3\text{PO}_4, 358 \text{ K}$ | $85\% \text{H}_3\text{PO}_4, 440 \text{ K}$ |

156. What are X, Y and Z in the following reactions ?

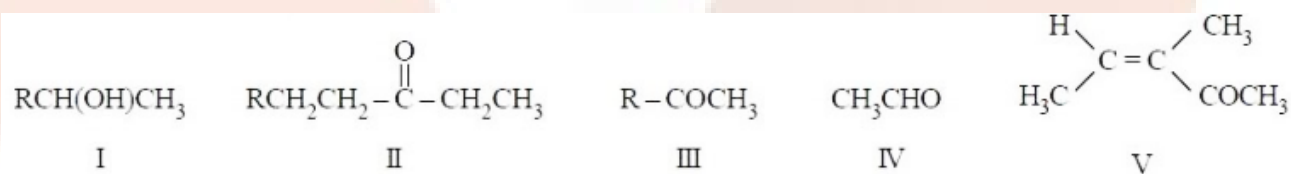


157. Structures of cinnamaldehyde (I), salicylaldehyde (II) and vanillin (III) are



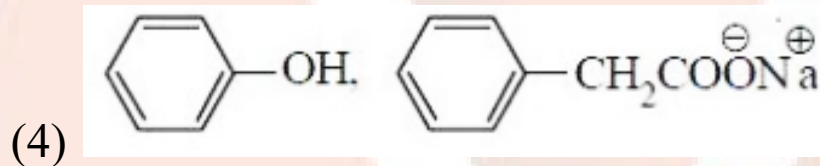
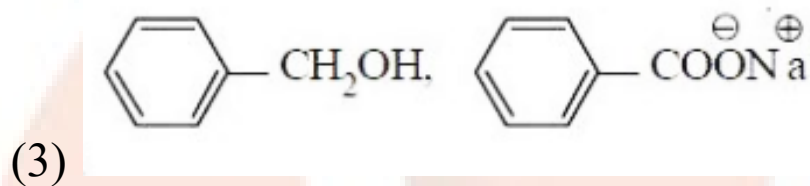
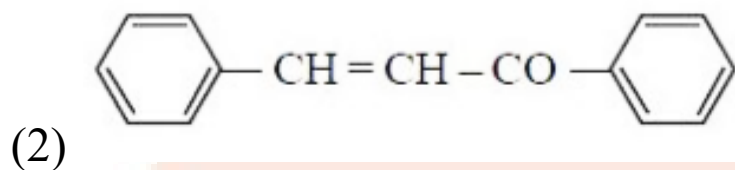
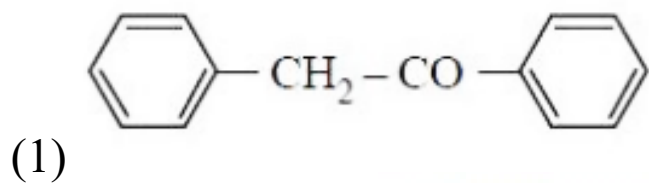


158. Which of the following are oxidized by NaOCl?



- (1) I, III, IV, V
 (2) I, II, III
 (3) II, IV, V
 (4) II, III, IV

159. Benzaldehyde on heating with concentrated NaOH gives



160. The reaction



- (1) Swarts reaction
- (2) Gatterman reaction
- (3) Sandmeyer reaction
- (4) Stephen reaction

AP EAMCET-2019-Engineering-23 Apr-Shift-1
Answer Key

1	1	33	3	65	3	97	2	129	2
2	3	34	2	66	4	98	4	130	4
3	1	35	4	67	1	99	2	131	2
4	2	36	2	68	3	100	2	132	4
5	1	37	1	69	4	101	2	133	4
6	3	38	3	70	4	102	3	134	2
7	3	39	1	71	1	103	2	135	4
8	3	40	1	72	3	104	2	136	1
9	3	41	3	73	1	105	4	137	3
10	3	42	3	74	1	106	1	138	2
11	3	43	4	75	3	107	1	139	1
12	4	44	2	76	4	108	1	140	3
13	2	45	4	77	2	109	4	141	2
14	1	46	4	78	4	110	4	142	2
15	2	47	3	79	3	111	2	143	3
16	4	48	3	80	2	112	2	144	3
17	4	49	2	81	1	113	2	145	4

18	4	50	2	82	1	114	3	146	1
19	2	51	4	83	4	115	3	147	4
20	4	52	2	84	4	116	1	148	1
21	1	53	3	85	1	117	4	149	1
22	1	54	3	86	3	118	2	150	1
23	3	55	3	87	4	119	3	151	4
24	2	56	1	88	3	120	3	152	3
25	2	57	3	89	4	121	3	153	3
26	2	58	2	90	1	122	3	154	2
27	3	59	1	91	4	123	1	155	2
28	4	60	1	92	2	124	2	156	3
29	4	61	1	93	3	125	2	157	2
30	1	62	4	94	1	126	3	158	1
31	1	63	3	95	2	127	4	159	3
32	1	64	2	96	1	128	3	160	2

AP EAMCET-2019-Engineering-23 Apr-Shift-1
Solutions – Mathematics

1. For (A),

$f(x) = px + q, (p \neq 0)$ is a linear function and the function which are linear are one-one or onto in set of real numbers so, $A \rightarrow II$

For (B),

$f: \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$ is defined as,

$$f(x) = x^2$$

So,

$$f(-1) = f(1) = 1$$

So, $f(x)$ is not one-one function but range of function is $[0, \infty)$

Thus,

$$x^2 \geq 0, \forall x \in \mathbb{R}$$

So, f is onto but not one-one. Therefore, $B \rightarrow IV$

For (C),

$f: \mathbb{N} \rightarrow \mathbb{N}$ is defined as,

$f(x) = n^2 + 2n + 3$ is one-one but not onto because there is no value of n for which $f(n) = 3$

So, $C \rightarrow III$

Considering the options the correct option is (1).

2. Consider the function,

$$f(x) = \sqrt{\frac{a - |x|}{(a+1) - |x|}}, (a > 0)$$

So,

$$f(x) \geq 0, \forall x \in \text{domain of } f(x)$$

$$\text{Let, } \frac{a - |x|}{(a+1) - |x|} = y$$

Then,

$$a - |x| = y(a+1) - y|x|$$

$$(y-1)|x| = y(a+1) - a$$

$$|x| = \frac{y(a+1)}{y-1} \geq 0, \forall x \in \text{domain of } f(x)$$

This implies,

$$y \in \left(-\infty, \frac{a}{a+1}\right) \cup (1, \infty)$$

So, range of function is,

$$f(x) = \sqrt{y} \in \left[0, \sqrt{\frac{a}{a+1}} \right] \cup (1, \infty)$$

3. By the method of verification if $k = 0$,

$$2 \cdot 4^1 + 3^1 = 11t$$

$$8 + 3 = 11t$$

$$t = 1$$

Now, from the second given relation on putting the values of $(k, t) = (0, 1)$, we get

$$2 \cdot (64) + 81 = 11(p + 3^q)$$

$$128 + 81 = 11(p + 3^q)$$

$$11(p + 3^q) = 209$$

$$p + 3^q = 19$$

From the option p must be 16 and $q = 1$. So,

$$(3k + 1)_{k=0}$$

4. Consider the equation,

$$x = \frac{a}{b - c}$$

$$a - bx + cx = 0$$

And,

$$y = \frac{b}{c-a}$$

$$ay + b - cy = 0$$

And,

$$z = \frac{c}{a-b}$$

$$az - bz - c = 0$$

Eliminate a, b, c from above equation then,

$$\begin{vmatrix} 1 & -x & x \\ y & 1 & -y \\ z & -z & -1 \end{vmatrix} = 0$$

Apply the operation $C_1 \rightarrow C_1 + C_2 + C_3$ we get,

$$\begin{vmatrix} 1 & -x & x \\ 1 & 1 & -y \\ 1 & z & 1 \end{vmatrix} = 0$$

5. Consider the expression,

$$\text{adj adj}(A) = |A|^{n-2} A$$

And

$$|\text{adj adj}(A)| = |A|^{(n-1)^2}$$

So,

$$\begin{aligned} |\text{adj adj}(A)| &= |A|^{(n-1)^2} |A|^{(n-2)} A \quad [\because |A| = 2, n = 3] \\ &= 2^4 (2A) \\ &= 2^5 A \\ &= 32A \end{aligned}$$

6. By the Cramer's rule,

$$x = \alpha = \frac{\Delta_1}{\Delta}$$

$$y = \beta = \frac{\Delta_2}{\Delta}$$

$$z = \gamma = \frac{\Delta_3}{\Delta}$$

Where,

$$\Delta = \begin{vmatrix} 2 & -1 & -8 \\ 3 & 4 & 5 \\ 5 & -2 & 7 \end{vmatrix}$$

$$= 2(28 + 10) + 1(21 - 25) + 8(-6 - 20)$$

$$= 76 - 4 - 208$$

$$= -136$$

And,

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 13 & -1 & -8 \\ 18 & 4 & 5 \\ 20 & -2 & 7 \end{vmatrix} \\ &= 13(28+10)+1(126-100)+8(-36-80) \\ &= 494+26-928 \\ &= -408\end{aligned}$$

And,

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 2 & 13 & -8 \\ 3 & 18 & 5 \\ 5 & 20 & 7 \end{vmatrix} \\ &= 2(126-100)-13(21-25)+8(60-90) \\ &= 52+52-240 \\ &= -136\end{aligned}$$

And,

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 2 & -1 & 13 \\ 3 & 4 & 18 \\ 5 & -2 & 20 \end{vmatrix} \\ &= 2(80+36)+1(60-90)+13(-6-20) \\ &= 232-30-338 \\ &= -136\end{aligned}$$

So, $\alpha = 3, \beta = 1, \gamma = 1$

This implies,

$$\alpha\beta + \beta\gamma + \gamma\alpha = 3 + 1 + 3 \\ = 7$$

7. Consider the expression,

$$Z = a + ib$$

Let, $Z_1 = a + ib$ and $Z_2 = c + id$ then,

$$Z_1 Z_2 = (ac - bd) + i(ad + cb) \\ = (ad + cb) + i(ac - bd)$$

Then,

$$\overline{Z_1} \hat{Z}_2 = (\overline{a + ib})(d + ci) \\ = (a - ib)(d + ic) \\ = (ad + cb) + i(ac - bd)$$

This implies,

$$\widehat{Z_1 Z_2} = \overline{Z_1} \hat{Z}_2$$

8. The complex conjugate of the given points are in arg and plane

$$A(1 - 2i), B(2 + 3i), C(3 + 4i)$$

So,

$$\begin{aligned}a &= BC \\&= \sqrt{1+1} \\&= \sqrt{2}\end{aligned}$$

And,

$$\begin{aligned}b &= CA \\&= \sqrt{4+36} \\&= \sqrt{40}\end{aligned}$$

And,

$$\begin{aligned}c &= AB \\&= \sqrt{1+25} \\&= \sqrt{26}\end{aligned}$$

This implies,

$$\begin{aligned}\cos B &= \frac{a^2 + c^2 - b^2}{2ac} \\&= \frac{2 + 26 - 40}{2\sqrt{2}\sqrt{26}} \\&= -\frac{12}{4\sqrt{13}} \\&= -\frac{3}{\sqrt{13}} < 0\end{aligned}$$

So, the points A, B, C represent vertices of an obtuse triangle.

9. From given data,

$$x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1})$$

$$(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1}) = \frac{x^n - 1}{x - 1}$$

Put $x = -1$ then,

$$(-1 - \alpha_1)(-1 - \alpha_2) \dots (-1 - \alpha_{n-1}) = 0$$

10. Consider the expression,

$$x + \frac{1}{x} = 2 \sin \alpha$$

This implies,

$$x = \sin \alpha + i \cos \alpha$$

$$= \cos \left(\frac{\pi}{2} - \alpha \right) + i \sin \left(\frac{\pi}{2} - \alpha \right)$$

$$= e^{i \left(\frac{\pi}{2} - \alpha \right)}$$

And,

$$y + \frac{1}{y} = 2 \cos \beta$$

$$y = \cos \beta + i \sin \beta$$

$$= e^{i\beta}$$

Now,

$$\begin{aligned}
 x^3 y^3 &= e^{i\left(\frac{3\pi}{2}-3\alpha\right)} \cdot e^{i3\beta} \\
 &= e^{i\left(\frac{3\pi}{2}+3\beta-3\alpha\right)} \\
 &= \cos\left(\frac{3\pi}{2}+3\beta-3\alpha\right) + i \sin\left(\frac{3\pi}{2}+3\beta-3\alpha\right) \\
 &= \sin(3\beta-3\alpha) - i \cos(3\beta-3\alpha)
 \end{aligned}$$

So,

$$\begin{aligned}
 \frac{1}{x^3 y^3} &= \sin(3\beta-3\alpha) + i \cos(3\beta-3\alpha) \\
 \frac{1}{x^3 y^3} + x^3 y^3 &= 2 \sin(3\beta-3\alpha) \\
 &= 2 \sin 3(\beta-\alpha)
 \end{aligned}$$

11. Consider the equation,

$$x^2 + 4kx + 12e^{3\log k} - 1 = 0$$

By the given information the product of the roots is 323. Then,

$$12e^{3\log k} - 1 = 323$$

$$12k^3 = 324$$

$$k^3 = 27$$

$$k = 3$$

Now the sum of roots is,

$$\begin{aligned}-4k &= -4(3) \\ &= -12\end{aligned}$$

12. Consider the equation,

$$1 - 2x - 5x^2$$

The maximum value of above is given by,

$$\begin{aligned}a &= -\frac{4 + 20}{4(-5)} \\ &= \frac{6}{5}\end{aligned}$$

The minimum value is given by,

$$\begin{aligned}b &= -\frac{4 - 20}{4} \\ &= 4\end{aligned}$$

Now the quadratic expression $5ax^2 + bx + 7$ at $a = \frac{6}{5}$ and $b = 4$

must be positive then,

$$6x^2 + 4x + 7 > 0$$

The discriminant is given by,

$$D = 16 - 4(6)(7)$$

$$D = -152$$

The discriminant $D < 0$ and coefficient of x^2 is positive. So,

$$6x^2 + 4x + 7 > 0, \forall x \in \mathbb{R} = (-\infty, \infty)$$

13. Consider the expression,

$$\left| \frac{x^2 + kx + 1}{x^2 + x + 1} \right| < 3$$

$$-3 < \frac{x^2 + kx + 1}{x^2 + x + 1} < 3$$

$$-3x^2 - 3x - 3 < x^2 + kx + 1 < 3x^2 + 3x + 3$$

This implies,

$$4x^2 + (k+3)x + 4 > 0, \forall x \in \mathbb{R}$$

Then,

$$D < 0$$

$$(k+3)^2 - 4(4)(4) < 0$$

$$(k+3)^2 - 8^2 < 0$$

$$(k+3-8)(k+3+8) < 0$$

$$k \in (-11, 5)$$

And,

$$2x^2 + (3-k)x + 2 > 0, \forall x \in \mathbb{R}$$

Then,

$$D < 0$$

$$(3-k)^2 - 16 < 0$$

$$(k-3-4)(k-3+4) < 0$$

$$k \in (-1, 7)$$

This implies,

$$k \in (-1, 5)$$

14. Consider the expression,

$$2bx^2 + 3cx - d = 0$$

And,

$$2ax^2 + 3bx + 4c = 0$$

Both above equations have common root,

$$(6b^2 - 6ac)(12c^2 + 3bd) = (8bc + 2ad)^2$$

So,

$$\frac{4bc + ad}{\frac{9}{2}(b^2 - ac)} = \frac{bd + 4c^2}{4bc + ad}$$

On comparing,

$$k = \frac{9}{2}$$

15. The alphabetical order of word REPEAT is AEEPRT.

Now the number of words start with R is,

$$\frac{5!}{2!} = 60$$

The number of words start with E is,

$$5! = 120$$

The number of words starts with P is,

$$\frac{5!}{2!} = 60$$

The number of words starts with RA,

$$\frac{4!}{2!} = 12$$

The number of words starts with REA,

$$3! = 6$$

The number of words starts with REE,

$$3! = 6$$

The number of words starts with REPA,

$$2! = 2$$

The number of words starts with REPEAT,

$$1! = 1$$

So, the rank of word REPEAT is,

$$60 + 120 + 60 + 12 + 6 + 6 + 2 + 1 = 267$$

16. The total number of ways selecting 3 persons from 15 persons seat around a circular table is ${}^{15}C_3$.

Now, the number of ways selecting 3 persons who sit together at one place is 15.

So, the required number of ways is,

$$\begin{aligned} {}^{15}C_3 - 15 &= \frac{15 \times 14 \times 13}{3 \times 2} - 15 \\ &= 33 \times 13 - 15 \\ &= 440 \end{aligned}$$

17. From the given information,

$$\frac{{}^{24}C_{r-1}}{{}^{24}C_r} = \frac{12}{13}$$

Simplify the above,

$$\frac{24!}{(r-1)!(25-r)!} = \frac{12}{13}$$

$$\frac{24!}{r!(24-r)!}$$

$$\frac{r}{25-r} = \frac{12}{13}$$

$$r = 12$$

Since, $x^2 - 14x + 24$ so,

$$(x-12)(x-2) = 0$$

$$x = 2, 12$$

18. Let,

$$x = (1+y)^n - 1$$

So,

$$x = ny + \frac{n(n-1)}{2!}y^2 + \frac{n(n-1)(n-2)}{3!}y^3 + \dots$$

$$= \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

Compare first three terms then,

$$ny = \frac{2}{5}, \frac{n(n-1)}{2}y^2 = \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2, \frac{n(n-1)(n-2)}{3!}y^3 = \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3$$

From first two relation,

$$\frac{ny(ny-y)}{2!} = \frac{\frac{2}{5}\left(\frac{2}{5}-y\right)}{2!}$$

$$= \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2$$

$$y = \frac{2}{5} - \frac{6}{5}$$

$$= -\frac{4}{5}$$

And, $n = -\frac{1}{2}$

This implies,

$$\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!} \left(-\frac{4}{5}\right)^3 = \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3$$

= RHS

So,

$$x = \left(1 - \frac{4}{5}\right)^{-\frac{1}{2}} - 1$$

$$= \sqrt{5} - 1$$

$$\frac{1}{x} = \frac{\sqrt{5} + 1}{4}$$

This implies,

$$\begin{aligned}
 x + \frac{1}{x} &= \sqrt{5} - 1 + \frac{\sqrt{5} + 1}{4} \\
 &= \frac{5\sqrt{5} - 3}{4}
 \end{aligned}$$

19. Consider the expression,

$$\begin{aligned}
 \frac{1}{(1-x)(1-2x)(1-3x)} &= (1-x)^{-1} (1-2x)^{-1} (1-3x)^{-1} \\
 &= \begin{pmatrix} 1+x+x^2 \\ +x^3+x^4 \end{pmatrix} \begin{pmatrix} 1+2x+4x^2 \\ +8x^3+16x^4 \end{pmatrix} \begin{pmatrix} 1+3x+9x^2 \\ +27x^3+81x^4 \end{pmatrix}
 \end{aligned}$$

The coefficient of x^4 is required so expand up to x^4 ,

$$(1-ax)^{-n} = 1 + ax + a^2x^2 + a^3x^3 + \dots$$

So, coefficient of x^4 is,

$$81 + 54 + 36 + 24 + 16 + 27 + 18 + 12 + 8 + 9 + 4 + 3 + 2 + 1 = 301$$

20. Consider the expression,

$$\tan A - \tan B = x$$

Then,

$$\frac{1}{\cot A} - \frac{1}{\cot B} = x$$
$$\frac{\cot B - \cot A}{\cot A \cot B} = x$$

It is given that,

$$\cot A - \cot B = y$$

So,

$$\cot A \cot B = -\frac{y}{x}$$

This implies,

$$\begin{aligned}\cot(A - B) &= \frac{\cot A \cot B + 1}{\cot B - \cot A} \\ &= \frac{-\frac{y}{x} + 1}{-y} \\ &= \frac{y - x}{xy}\end{aligned}$$

21. It is given that,

$$\begin{aligned}\theta &= \frac{\pi}{5} \\ &= 36^\circ\end{aligned}$$

This implies,

$$\begin{aligned}\tan\left(\frac{\pi}{5}\right) &= \tan 36^\circ \\ &= \sqrt{5-2\sqrt{5}}\end{aligned}$$

Now,

$$\begin{aligned}\tan 2\theta &= \tan\left(\frac{2\pi}{5}\right) \\ &= \frac{2 \tan 36^\circ}{1 - \tan^2 36^\circ} \\ &= \frac{2\sqrt{5-2\sqrt{5}}}{1-5-2\sqrt{5}} \\ &= \frac{\sqrt{5-2\sqrt{5}}}{\sqrt{5}-2}\end{aligned}$$

Then,

$$\begin{aligned}\tan\left(\frac{4\pi}{5}\right) &= \frac{2\sqrt{5-2\sqrt{5}}}{\sqrt{5}-2} \\ &= \frac{1-\frac{5-2\sqrt{5}}{\sqrt{5}-2}}{2\sqrt{5-2\sqrt{5}}}\end{aligned}$$

$$\begin{aligned}\cot\left(\frac{4\pi}{5}\right) &= \frac{1-\frac{5-2\sqrt{5}}{\sqrt{5}-2}}{2\sqrt{5-2\sqrt{5}}}\end{aligned}$$

$$\begin{aligned}&= \frac{(2-\sqrt{5})}{(\sqrt{5}-2)(\sqrt{5-2\sqrt{5}})} \\ &= \frac{-1}{\sqrt{5-2\sqrt{5}}}\end{aligned}$$

Now,

$$\begin{aligned}\tan\frac{\pi}{5} + 2\tan\frac{2\pi}{5} + 4\cot\frac{4\pi}{5} &= \sqrt{5-2\sqrt{5}} + 2\left(\frac{\sqrt{5-2\sqrt{5}}}{\sqrt{5}-2}\right)\left(-\frac{4}{\sqrt{5-2\sqrt{5}}}\right) \\ &= -1.37 \\ &= \cot\frac{\pi}{5} \\ &= \cot 36^\circ\end{aligned}$$

22. Consider the expression,

$$\sin x + \sin y = \frac{\sqrt{3} + 1}{2}$$

This implies,

$$\begin{aligned} 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} &= \frac{\sqrt{3} + 1}{2} x + \\ \sin \frac{x+y}{2} \cos \frac{x-y}{2} &= \frac{\sqrt{3} + 1}{4} \end{aligned} \quad (I)$$

And,

$$\cos x + \cos y = \frac{\sqrt{3} - 1}{2}$$

Then,

$$\begin{aligned} 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2} &= \frac{\sqrt{3} - 1}{2} \\ \cos \frac{x+y}{2} \cos \frac{x-y}{2} &= \frac{\sqrt{3} - 1}{4} \end{aligned} \quad (II)$$

From equation (I) and (II),

$$\tan \frac{x+y}{2} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

Then,

$$\begin{aligned}\tan^2 \frac{x+y}{2} &= \frac{3+1+2\sqrt{3}}{3+1-2\sqrt{3}} \\&= \frac{2+\sqrt{3}}{2-\sqrt{3}} \\&= \frac{4+3+4\sqrt{3}}{4-3} \\&= 7+4\sqrt{3} \quad \text{(III)}\end{aligned}$$

Square and add equation (III) and (I),

$$\begin{aligned}\cos^2 \frac{x-y}{2} &= \frac{1}{16} \times 2(3+1) \\&= \frac{1}{2}\end{aligned}$$

$$\sec^2 \frac{x-y}{2} = 2$$

This implies,

$$\begin{aligned}\tan^2 \frac{x-y}{2} &= \sec^2 \frac{x-y}{2} - 1 \\&= 1\end{aligned}$$

Then,

$$\begin{aligned}\tan^2 \frac{x-y}{2} + \tan^2 \frac{x+y}{2} &= 1 + (7+4\sqrt{3}) \\&= 8+4\sqrt{3}\end{aligned}$$

23. Consider the expression,

$$4\sin^3 x - 3\sin x + a = 0$$

Then,

$$a = 3\sin x - 4\sin^3 x$$

$$a = \sin 3x$$

This implies,

$$\sin 3P = a = \sin 3Q$$

$$\sin 3P = \sin 3Q$$

$$3P = \pi - 3Q$$

$$3(P + Q) = \pi$$

Further simplify the above,

$$P + Q = \frac{\pi}{3}$$

Then,

$$\angle P + \angle Q + \angle R = \pi$$

$$\begin{aligned}\angle R &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3}\end{aligned}$$

24. Consider the expression,

$$\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}$$

Let, $x = \sin \theta$ then,

$$\theta + \sin^{-1}(2 \sin \theta) = \frac{\pi}{3}$$

$$\sin^{-1}(2 \sin \theta) = \frac{\pi}{3} - \theta$$

$$\begin{aligned} 2 \sin \theta &= \sin\left(\frac{\pi}{3} - \theta\right) \\ &= \frac{\sqrt{3}}{2} \cos \theta - \frac{1}{2} \sin \theta \end{aligned}$$

Then,

$$\frac{5}{2} \sin \theta = \frac{\sqrt{3}}{2} \cos \theta$$

$$\tan \theta = \frac{\sqrt{3}}{5}$$

$$\sin \theta = \sqrt{\frac{3}{28}}$$

$$= \frac{1}{2} \sqrt{\frac{3}{7}} = x$$

25. Consider the expression,

$$x = \log \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$$

Then,

$$\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) = e^x$$

$$\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} = e^x$$

This implies,

$$\frac{e^y - 1}{e^y + 1} = \tan \frac{x}{2}$$

$$\tanh \left(\frac{y}{2} \right) = \tan \left(\frac{x}{2} \right)$$

By the componendo and dividendo rule,

$$\frac{2 \tan \frac{\theta}{2}}{2} = \frac{e^x - 1}{e^x + 1}$$

$$\tan \frac{\theta}{2} = \frac{e^{\frac{x}{2}} - e^{-\frac{x}{2}}}{e^{\frac{x}{2}} + e^{-\frac{x}{2}}}$$

$$= \tanh \left(\frac{x}{2} \right)$$

This implies,

$$\frac{x}{2} = \tanh^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$x = 2 \tanh^{-1} \left(\tan \frac{\theta}{2} \right)$$

26. By the sine rule,

$$\begin{aligned} \frac{\sin^2 A + \sin^2 B}{\sin^2 A - \sin^2 B} &= \frac{\sin^2 A + \sin^2 B}{\sin(A+B)\sin(A-B)} \sin(A-B) \\ &= \frac{\sin^2 A + \sin^2 B}{\sin(\pi - C)} \left[\because A + B + C = \pi \text{ and } C = \frac{\pi}{2} \right] \\ &= \frac{1}{\sin C} \\ &= 1 \end{aligned}$$

27. Consider the expression,

$$\cos \alpha = \frac{a}{b+c}$$

$$\frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{a}{b+c}$$

By compodendo and dividendo rule,

$$\frac{2 \tan \frac{\alpha}{2}}{2} = \frac{b + c - a}{b + c + a}$$

$$\tan \frac{\alpha}{2} = \frac{b + c - a}{a + b + c}$$

Similarly,

$$\tan \frac{\beta}{2} = \frac{c + a - b}{a + b + c}$$

$$\tan \frac{\gamma}{2} = \frac{a + b - c}{a + b + c}$$

So,

$$\begin{aligned} \tan^2 \frac{\alpha}{2} + \tan^2 \frac{\beta}{2} + \tan^2 \frac{\gamma}{2} &= \frac{a + b + c}{a + b + c} \\ &= 1 \end{aligned}$$

28. Consider the expression,

$$r_1 = \frac{\Delta}{s - a} = 36$$

Similarly,

$$r_2 = \frac{\Delta}{s - b} = 18$$

$$r_3 = \frac{\Delta}{s - c} = 12$$

Then,

$$\begin{aligned}\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r^2} &= \frac{1}{36} + \frac{1}{18} + \frac{1}{12} \\ &= \frac{1+2+3}{36} \\ &= \frac{1}{6}\end{aligned}$$

This implies,

$$\begin{aligned}\frac{s-a+s-b+s-c}{\Delta} &= \frac{1}{6} \\ \frac{s}{\Delta} &= \frac{1}{6} \\ \Delta &= 6s\end{aligned}$$

So,

$$\begin{aligned}\frac{6s}{s-a} &= 36 \\ s &= 6s - 6a \\ 6a &= 5s\end{aligned}$$

And,

$$\begin{aligned}\frac{6s}{s-b} &= \\ s &= 3s - 3b \\ 3b &= 2s\end{aligned}$$

And,

$$\frac{6s}{s-c} = 12$$

$$s = 2s - 2c$$

$$2c = s$$

So,

$$\Delta^2 = 36s^2$$

Then,

$$s(s-a)(s-b)(s-c) = 36s^2$$

$$\left(s - \frac{5s}{6}\right)\left(s - \frac{2s}{3}\right)\left(s - \frac{s}{2}\right) = 36s$$

$$s^3\left(\frac{1}{6}\right)\left(\frac{1}{3}\right)\left(\frac{1}{2}\right) = 36s$$

$$s = 36$$

So,

$$a = 30$$

$$b = 4$$

This implies,

$$a + b = 30 + 24$$

$$= 54$$

29. Let,

$$\mathbf{OA} = \alpha \hat{\mathbf{i}} + \beta \hat{\mathbf{j}} + \gamma \hat{\mathbf{k}}$$

$$\mathbf{OB} = \beta \hat{\mathbf{i}} + \gamma \hat{\mathbf{j}} + \alpha \hat{\mathbf{k}}$$

$$\mathbf{OC} = \gamma \hat{\mathbf{i}} + \alpha \hat{\mathbf{j}} + \beta \hat{\mathbf{k}}$$

So,

$$\mathbf{AB} = (\beta - \alpha) \hat{\mathbf{i}} + (\gamma - \beta) \hat{\mathbf{j}} + (\alpha - \gamma) \hat{\mathbf{k}}$$

$$\mathbf{BC} = (\gamma - \beta) \hat{\mathbf{i}} + (\alpha - \gamma) \hat{\mathbf{j}} + (\beta - \alpha) \hat{\mathbf{k}}$$

$$\mathbf{CA} = (\alpha - \gamma) \hat{\mathbf{i}} + (\beta - \alpha) \hat{\mathbf{j}} + (\gamma - \beta) \hat{\mathbf{k}}$$

This implies,

$$|\mathbf{AB}| = \sqrt{(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2}$$

And,

$$|\mathbf{BC}| = \sqrt{(\beta - \gamma)^2 + (\gamma - \alpha)^2 + (\alpha - \beta)^2}$$

And

$$|\mathbf{CA}| = \sqrt{(\gamma - \alpha)^2 + (\alpha - \beta)^2 + (\beta - \gamma)^2}$$

This implies,

$$|\mathbf{AB}| = |\mathbf{BC}| = |\mathbf{CA}| = \sqrt{(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2}$$

So, the given position vectors are the vertices of equilateral triangle.

30. Consider the given vectors, let a vector in plane **b** and **c** is,

$$\mathbf{d} = \mathbf{b} + \lambda \mathbf{c}$$

$$= (1 + \lambda)\hat{\mathbf{i}} + (2 + \lambda)\hat{\mathbf{j}} - (1 + 2\lambda)\hat{\mathbf{k}}$$

Since the projection of **d** on **a** is $\sqrt{\frac{2}{3}}$ then,

$$\frac{\mathbf{d} \cdot \mathbf{a}}{|\mathbf{a}|} = \sqrt{\frac{2}{3}}$$

$$\frac{2(1 + \lambda) - 2(2 + \lambda) - (1 + 2\lambda)}{\sqrt{4 + 1 + 1}} = \sqrt{\frac{2}{3}}$$

$$2 + 2\lambda - 2 - \lambda - 1 - 2\lambda = 2$$

$$\lambda = -3$$

So, the required vector is,

$$\mathbf{d} = -2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$$

31. From the given vectors,

$$\begin{aligned}
 \mathbf{b} \times \mathbf{c} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -4 & 5 \\ 4 & 5 & -1 \end{vmatrix} \\
 &= \hat{\mathbf{i}}(4 - 25) - \hat{\mathbf{j}}(-1 - 20) + \hat{\mathbf{k}}(5 + 16) \\
 &= -21\hat{\mathbf{i}} + 21\hat{\mathbf{j}} + 21\hat{\mathbf{k}} \\
 &= 21(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})
 \end{aligned}$$

$$|\mathbf{b} \times \mathbf{c}| = 21\sqrt{3}$$

The vector $\mathbf{r}$ is perpendicular to $\mathbf{b}$ and $\mathbf{c}$ so,

$$\mathbf{r} = \pm |\mathbf{r}| \frac{\mathbf{b} \times \mathbf{c}}{|\mathbf{b} \times \mathbf{c}|}$$

Now, $\mathbf{r} \cdot \mathbf{a} = 9$ so,

$$\pm \frac{|\mathbf{r}|}{|\mathbf{b} \times \mathbf{c}|} \left[21(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) \right] = 9$$

$$\pm \frac{|\mathbf{r}|}{21\sqrt{3}} \left[21(-3 + 1 - 1) \right] = 9$$

$$|\mathbf{r}| = 3\sqrt{3}$$

This implies,

$$\begin{aligned}
 \mathbf{r} &= \pm 3\sqrt{3} \frac{21(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})}{21\sqrt{3}} \\
 &= \pm 3(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \\
 &= -3(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \text{ or } 3(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})
 \end{aligned}$$

32. From the given data,

$$\mathbf{AP} = \frac{\mathbf{a}}{3} \text{ and } \mathbf{AQ} = \frac{2\mathbf{a} + \mathbf{b}}{3}$$

Let, D divide the line $\mathbf{AQ}$ in ratio $\lambda : 1$ and $\mathbf{CP}$ in $\mu : 1$ then,

$$\frac{\lambda(2\mathbf{a} + \mathbf{b})}{3(\lambda + 1)} = \frac{\mu \frac{\mathbf{a}}{3} + \mathbf{b}}{\mu + 1}$$

On comparison,

$$\frac{\mu}{3(\mu + 1)} = \frac{2\lambda}{3(\lambda + 1)} \text{ and } \frac{1}{(\mu + 1)} = \frac{\lambda}{3(\lambda + 1)}$$

This implies,

$$\begin{aligned}
 \frac{1}{3(\mu + 1)} &= \frac{2}{\mu + 1} \\
 \mu &= 6
 \end{aligned}$$

So,

$$\frac{1}{7} = \frac{\lambda}{3(\lambda+1)}$$

$$3\lambda + 3 = 7\lambda$$

$$\lambda = \frac{3}{4}$$

So,

$$\mathbf{AD} = \frac{2\mathbf{a} + \mathbf{b}}{7}$$

Then the area of triangle BCD is,

$$\begin{aligned}\frac{1}{2}|\mathbf{BC} \times \mathbf{BD}| &= \frac{1}{2}\left|(\mathbf{b} - \mathbf{a}) \times \left(\frac{\mathbf{b} - 5\mathbf{a}}{7}\right)\right| \\ &= \frac{1}{14}|(-5\mathbf{b} \times \mathbf{a}) - (\mathbf{a} \times \mathbf{b})| \\ &= \frac{4}{14}|\mathbf{a} \times \mathbf{b}|\end{aligned}$$

It is given that area is 7 units so,

$$\frac{1}{2}|\mathbf{a} \times \mathbf{b}| = \frac{49}{4}$$

So, the area of triangle ABC is $\frac{49}{4}$.

33. From the given vectors,

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} \\ &= \hat{\mathbf{i}}(2-1) - \hat{\mathbf{j}}(2-1) + \hat{\mathbf{k}}(1-1) \\ &= \hat{\mathbf{i}} - \hat{\mathbf{j}}\end{aligned}$$

So, the unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is,

$$\pm \frac{\hat{\mathbf{i}} - \hat{\mathbf{j}}}{\sqrt{2}}$$

The magnitude of projection of unit vector on $\mathbf{c}$ is,

$$\begin{aligned}\left| \pm \frac{\hat{\mathbf{i}} - \hat{\mathbf{j}}}{\sqrt{2}} \cdot \frac{(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}})}{\sqrt{4+9+16}} \right| &= \left| \frac{2-3}{\sqrt{2}\sqrt{29}} \right| \\ &= \frac{1}{\sqrt{58}}\end{aligned}$$

34. Consider the given data,

$$\begin{aligned}\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \frac{\sqrt{3}}{2} \mathbf{b} + \frac{1}{2} \mathbf{c} \\ (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} &= \frac{\sqrt{3}}{2} \mathbf{b} + \frac{1}{2} \mathbf{c}\end{aligned}$$

Let the angle between **a** and **c** be α and the angle between **a** and **b** be β then,

$$(\cos \alpha)\mathbf{b} - (\cos \beta)\mathbf{c} = \frac{\sqrt{3}}{2}\mathbf{b} + \frac{1}{2}\mathbf{c}$$

From above,

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\alpha = 30^\circ$$

And,

$$\cos \beta = -\frac{1}{2}$$

$$\beta = 120^\circ$$

35. Consider given data,

x_i	(f_i)	$x_i f_i$	$\bar{x} - x_i$	$f_i (\bar{x} - x_i)^2$
1	1	1	6	36
2	2	4	5	50
3	3	9	4	48
4	4	16	3	36
5	5	25	2	20

6	6	36	1	6
7	7	49	0	0
8	8	64	1	8
9	9	81	2	36
10	10	100	3	90

So, $N = \sum f_i = 55$

The mean is given by,

$$\begin{aligned}\bar{x} &= \frac{\sum x_i f_i}{N} \\ &= \frac{10 \times 11 \times 21}{6 \times 55} \\ &= 7\end{aligned}$$

Thus the mean deviation is,

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum f (\bar{x} - x_i)^2 = \frac{36 + 50 + 48 + 36 + 20 + 6 + 8 + 36 + 90}{55} \\ &= 6\end{aligned}$$

36. Consider given data,

Class Interval	(f_i)	x_i	$x_i f_i$	$ \bar{x} - x_i $
0-5	4	2.5	10	9.5

5-10	1	7.5	7.5	4.5
10-15	10	12.5	125	0.5
15-20	3	17.5	52.5	5.5
20-25	2	22.5	45	10.5
	$N = \sum f_i = 20$			$\sum x_i f_i = 240$

The mean is given by,

$$\begin{aligned}\bar{x} &= \frac{\sum x_i f_i}{N} \\ &= \frac{240}{20} \\ &= 12\end{aligned}$$

Thus the mean deviation is,

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum f_i \left| \bar{x} - x_i \right|^2 \\ &= \frac{4(9.5)^2 + 1(4.5)^2 + 10(0.5)^2 + 3(5.5)^2 + 2(10.5)^2}{20} \\ &= \frac{139}{4}\end{aligned}$$

This implies,

$$\begin{aligned} CV &= \frac{\sigma}{x} \times 100 \\ &= \frac{\sqrt{139}}{2 \times 12} \\ &= \frac{25\sqrt{139}}{6} \end{aligned}$$

37. The total number of students are 50.

Number of ways if both the students are in first group is

$${}^{48}C_{18} \times {}^{50}C_{30} = {}^{48}C_{18}$$

Number of ways if both are in second group is,

$${}^{48}C_{27} \times {}^{20}C_{20} = {}^{48}C_{28}$$

The total number of ways is,

$${}^{50}C_{20} \times {}^{30}C_{30} = {}^{50}C_{20}$$

So, the required probability is,

$$\begin{aligned}\frac{{}^{48}C_{18} + {}^{48}C_{28}}{{}^{50}C_{20}} &= \frac{\frac{48!}{18!30!} + \frac{48!}{28!20!}}{\frac{50!}{20!30!}} \\ &= \frac{(20)(19) + (30)(29)}{(50)(49)} \\ &= \frac{25}{49}\end{aligned}$$

38. From the given information,

$$P(E_1) = \frac{1}{8}, P(E_1|E_2) = \frac{1}{3}, P(E_2|E_1) = \frac{1}{4}$$

Then,

$$\frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{1}{4}$$

$$P(E_1 \cap E_2) = \frac{1}{32}$$

And,

$$P(E_2) = \frac{P(E_1 \cap E_2)}{P(E_1|E_2)}$$

$$\begin{aligned} &= \frac{1}{\frac{32}{1}} \\ &= \frac{3}{32} \end{aligned}$$

Then,

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \\ &= \frac{1}{8} + \frac{3}{32} - \frac{1}{32} \\ &= \frac{3}{16} \end{aligned}$$

This implies,

$$P(\overline{E_1}) = \frac{7}{8}$$

$$P(\overline{E_2}) = \frac{29}{32}$$

Now,

$$\begin{aligned}P(\bar{E}_1 \cap \bar{E}_2) &= P(\overline{E_1 \cup E_2}) \\&= 1 - P(E_1 \cup E_2) \\&= \frac{13}{16}\end{aligned}$$

Then,

$$\begin{aligned}P(\bar{E}_1 | \bar{E}_2) &= \frac{P(\bar{E}_1 \cap \bar{E}_2)}{P(\bar{E}_2)} \\&= \frac{\frac{13}{16}}{\frac{29}{32}} \\&= \frac{26}{29}\end{aligned}$$

And,

$$\begin{aligned}P(E_1 | \bar{E}_2) &= 1 - P(\bar{E}_1 | \bar{E}_2) \\&= 1 - \frac{26}{29} \\&= \frac{3}{29}\end{aligned}$$

39. From the given information the required probability is,

$$\frac{\frac{1}{2} \times \frac{n}{n+8}}{\left(\frac{1}{2} \times \frac{n}{n+8}\right) + \left(\frac{1}{2} \times \frac{6}{3+6}\right)}$$

By the Baye's theorem,

$$\begin{aligned} \frac{\frac{n}{n+8}}{\frac{n}{n+8} + \frac{2}{3}} &= \frac{3n}{3n + 2n + 16} \\ &= \frac{3n}{5n + 16} \\ &= p \end{aligned}$$

For minimum value of p , n needs to be 1 so,

$$p \geq \frac{3}{21}$$

$$p \geq \frac{1}{7}$$

For maximum value of p , n should be ∞ so,

$$p < \frac{3}{5}$$

This implies,

$$\frac{1}{7} \leq p < \frac{3}{5}$$

40. Let the random variable be X then,

X	0	1	2	3
P(X)	$\frac{{}^4C_3}{{}^9C_3}$	$\frac{{}^4C_2 \times {}^5C_1}{{}^9C_3}$	$\frac{{}^4C_1 \times {}^5C_2}{{}^9C_3}$	$\frac{{}^4C_0 \times {}^5C_3}{{}^9C_3}$

So, the mean is,

$$\begin{aligned}
 & 0 \times \left(\frac{{}^4C_3}{{}^9C_3} \right) + 1 \times \left(\frac{{}^4C_2 \times {}^5C_1}{{}^9C_3} \right) + 2 \times \left(\frac{{}^4C_1 \times {}^5C_2}{{}^9C_3} \right) + 3 \times \left(\frac{{}^4C_0 \times {}^5C_3}{{}^9C_3} \right) \\
 &= \frac{(6)(5) + 2(2)(10) + (3)(10)}{9(8)(7)} \\
 &= \frac{30 + 80 + 30}{84} \\
 &= \frac{5}{3}
 \end{aligned}$$

41. It is given that mean of Poisson variate X is $\lambda = 1$ then,

$$\begin{aligned}
 P(X=r) &= \frac{e^{-\lambda} \lambda^r}{r!} \\
 &= \frac{e^{-1}}{r!}
 \end{aligned}$$

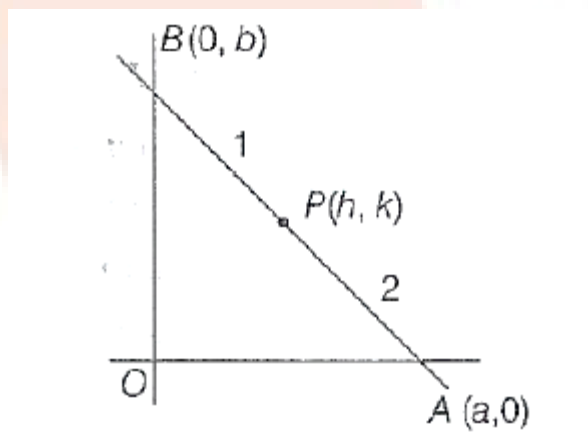
This implies,

$$\begin{aligned}
 \sum_{r=0}^{\infty} |r-1| P(X=r) &= \sum_{r=0}^{\infty} |r-1| \frac{e^{-1}}{r!} \\
 &= e^{-1} \left[\frac{1}{0!} + \frac{0}{1!} + \frac{1}{2!} + \frac{2}{3!} + \dots \right] \\
 &= e^{-1} \left(1 + \frac{1}{2!} + \frac{2}{3!} + \dots \right) \\
 &= e^{-1} (1+1)
 \end{aligned}$$

This implies,

$$\sum_{r=0}^{\infty} |r-1| P(X=r) = \frac{2}{e}$$

42. Consider the diagram,



It is given that $\overline{AB} = a$ so the centre,

$$(h, k) = \left(\frac{p}{2}, \frac{q}{2} \right)$$

This implies

$$p = 2h, q = 2k$$

So,

$$\begin{aligned} A &= (p, 0) \\ &= (2h, 0) \end{aligned}$$

And,

$$\begin{aligned} B &= (0, q) \\ &= (0, 2k) \end{aligned}$$

So, the length of AB is,

$$\sqrt{(2h)^2 + (2k)^2} = a$$

$$\sqrt{4h^2 + 4k^2} = a$$

$$4h^2 + 4k^2 = a^2$$

$$h^2 + k^2 = \frac{a^2}{4}$$

So, the required locus is,

$$x^2 + y^2 = \frac{a^2}{4}$$

43. Consider the expression,

$$y^2 - 4x + 6y + 17 = 0$$

Then,

$$(y + 3)^2 = 4(x - 2)$$

If the origin is shifted to $(2, -3)$ then the equation

$$y^2 - 4x + 5y + 17 = 0 \text{ get transformed as } y^2 = 4ax$$

So,

$$(h, k) = (2, -3)$$

This implies,

$$\begin{aligned} h^2 + k^2 &= 4 + 9 \\ &= 13 \end{aligned}$$

44. Let,

$$A = (4, -3)$$

$$B = (1, 1)$$

$$C = (2, 3)$$

The slope of BC is,

$$\frac{3-1}{2-1} = 2$$

So, the slope of the required line is $-\frac{1}{2}$.

So, the equation of the line passing through $(4, -3)$ and have

slope $-\frac{1}{2}$ is,

$$y - y_1 = m(x - x_1)$$

$$y + 3 = -\frac{1}{2}(x - 4)$$

$$x + 2y = -2$$

$$\frac{x}{-2} + \frac{y}{-1} = 1$$

45. The given lines are,

$$x + 2y - 3 = 0$$

$$3x + 4y - 7 = 0$$

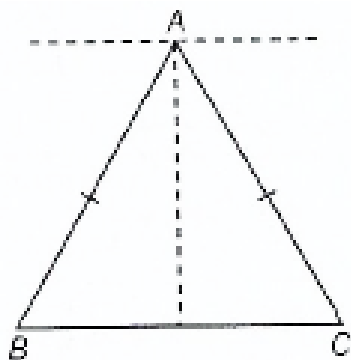
$$2x + 3y - 4 = 0$$

$$4x + 5y - 6 = 0$$

From first two equations the intersection point is $P(1,1)$.

Here, $P(1,1)$ does not satisfy the last two equations therefore, all the lines given are not concurrent.

46. Consider the diagram,



The equation of sides is,

$$7x - y + 3 = 0$$

$$x + y - 3 = 0$$

The equation for the angle bisector of above two is,

$$\frac{7x - y + 3}{\sqrt{49 + 1}} = \pm \frac{x + y - 3}{\sqrt{1 + 1}}$$

$$\frac{7x - y + 3}{\sqrt{50}} = \pm \frac{x + y - 3}{\sqrt{2}}$$

$$7x - y + 3 = 5x + 5y - 15 \text{ or } 7x - y + 3 = -5x - 5y + 15$$

$$2x - 6y + 12 = 0 \quad \text{or} \quad 12x + 4y - 12 = 0$$

This implies,

$$x - 3y + 6 = 0 \text{ or } 3x + y - 3 = 0$$

So,

$$\text{slope} = \frac{1}{3}, \text{slope} = -3 (\text{integer})$$

Therefore, the required slope of third side is -3 .

47. From given information,

$$A_2 = A_1$$

$$\frac{1}{2}(\pi - \theta)r^2 = \left(\frac{1}{2}\theta r^2\right)5$$

$$\pi - \theta = 5\theta$$

$$\theta = \frac{\pi}{6}$$

Then,

$$\cos \frac{\pi}{6} = \frac{1+m}{\sqrt{(1-m)^2 + 4(1+m)^2}}$$

$$\frac{\sqrt{3}}{2} = \frac{1+m}{\sqrt{5l^2 + 5m^2 + 6lm}}$$

$$\frac{3}{4} = \frac{1+m}{5l^2 + 5m^2 + 6lm}$$

$$15l^2 + 15m^2 + 18lm = 4l^2 + 4m^2 + 8lm$$

This implies,

$$11l^2 + 11m^2 + 10lm = 0$$

$$11(l^2 + m^2 + 2lm) - 22lm + 10lm = 0$$

$$11(1+m)^2 - 12lm = 0$$

$$\frac{lm}{(1+m)^2} = \frac{11}{12}$$

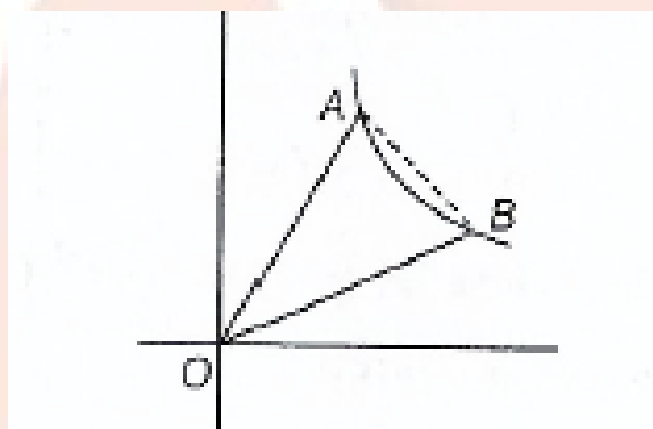
48. The given equation of line is,

$$3x + 4y - 5 = 0$$

$$\frac{3x + 4y}{5} = 1$$

The equation of curve is $2x^2 + 3y^2 = 5$

The graph for above equation is,



the equation of curve and line are homogeneous so,

$$2x^2 + 3y^2 = 5(1)^2$$

$$2x^2 + 3y^2 = 5\left(\frac{3x + 4y}{5}\right)^2$$

$$2x^2 + 3y^2 = 5\left(\frac{9x^2 + 16y^2 + 24xy}{25}\right)$$

$$x^2 - y^2 + 24xy = 0$$

Now,

$$\begin{aligned}(\text{coefficient of } x^2) + (\text{coefficient of } y^2) &= 1 + (-1) \\ &= 0\end{aligned}$$

This implies,

$$\angle AOB = 90^\circ \text{ or } \frac{\pi}{2}$$

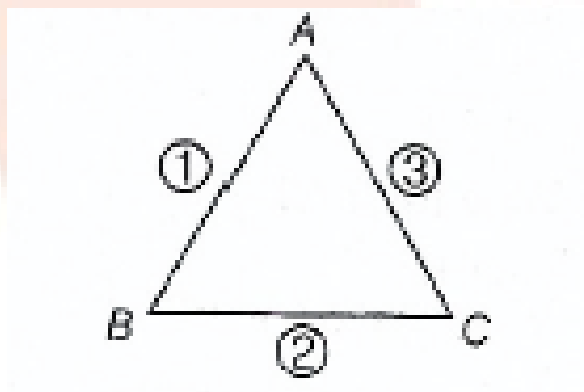
49. Consider the given equations,

$$y = 0 \quad \text{(I)}$$

$$y = x \quad \text{(II)}$$

$$2x + 3y = 10 \quad \text{(III)}$$

Consider the diagram,



The vertex A from equation (I) and (III) is $(5, 0)$.

The vertex B from equation (I) and (II) is $(0, 0)$.

The vertex C from equation (II) and (III) is $(2, 2)$.

Let the equation of circle be,

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (IV)$$

The equation (IV) passes through point B(0,0) so, $C = 0$.

Since, equation (IV) passes through A(5,0) so,

$$25 + 0 + 10g + 0 + 0 = 0$$

$$g = -\frac{5}{2}$$

The equation (IV) passes through C(2,2) so,

$$4 + 4 + 4g + 4f = 0$$

$$g + f + 2 = 0$$

$$-\frac{5}{2} + f + 2 = 0$$

$$f = \frac{1}{2}$$

Therefore, the center of circle is $(-g, -f) = \left(\frac{5}{2}, \frac{-1}{2}\right)$

50. The equation of the polar of point P(2,3) with respect to equation of circle $x^2 + y^2 - 2x - 2y + 1 = 0$ is,

$$T = 0$$

This implies,

$$2x + 3y - (x + 2) - (y + 3) + 1 = 0$$

$$x + 2y - 4 = 0 \quad (I)$$

The equation of the line joining point $P(2, 3)$ and center of circle $C(1, 1)$ is,

$$y - 1 = \frac{2}{1}(x - 1)$$

$$2x - y - 1 = 0$$

The equation of polar of point $Q\left(\frac{6}{5}, \frac{7}{5}\right)$ with respect to circle is,

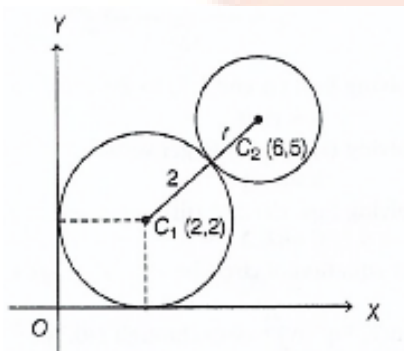
$$\frac{6}{5}x + \frac{7}{5}y - \left(x + \frac{6}{5}\right) - \left(y + \frac{7}{5}\right) + 1 = 0$$

$$x + 2y - 8 = 0 \quad (II)$$

So, the required distance from equation (I) and (II) is,

$$\frac{4}{\sqrt{1+4}} = \frac{4}{\sqrt{5}}$$

51. Consider the diagram,



The center of first circle is $C_1(2, 2)$ and radius is 2 units.

The center of second circle is $C_2(6, 5)$

So,

$$C_1C_2 = r + 2$$

$$5 = r + 2$$

$$r = 3$$

The required equation of circle with radius 3 units and center $C_2(6, 5)$ is,

$$(x - 6)^2 + (y - 5)^2 = 3^2$$

$$x^2 + y^2 - 12x - 10y + 52 = 0$$

52. The given equation of circle is,

$$(x + a)^2 + (y + b)^2 = a^2$$

$$x^2 + y^2 + 2ax + 2by + b^2 = 0 \quad (I)$$

And,

$$(x + c)^2 + (y + d)^2 = a^2$$

$$x^2 + y^2 + 2cx + 2dy + c^2 = 0 \quad (II)$$

From equation (I),

$$g_1 = a, f_1 = b, c_1 = b^2$$

From equation (II),

$$g_2 = c, f_2 = d, c_2 = c^2$$

If circle are orthogonal then,

$$2[g_1g_2 + f_1f_2] = c_1 + c_2$$

$$2(ac + bd) = b^2 + c^2$$

$$2ac + 2bd = b^2 + c^2$$

$$2ac - c^2 = b^2 - 2bd$$

Further simplify the above,

$$c(2a - c) = b(b - 2d)$$

$$b(b - 2d) = c(2a - c)$$

53. The equation of the circle passes through $x^2 + y^2 - 8x = 0$ and

$x^2 + y^2 - 9 = 0$ is,

$$(x^2 + y^2 - 8x) + \lambda(x^2 + y^2 - 9) = 0$$

$$(1 + \lambda)x^2 + (1 + \lambda)y^2 - 8x - 9\lambda = 0$$

$$x^2 + y^2 - \frac{8}{1 + \lambda}x - 9\frac{\lambda}{1 + \lambda} = 0$$

So, center is $C\left(\frac{4}{1 + \lambda}, 0\right)$

Now, equation of common chord is,

$$32 = 9 + 9\lambda$$

$$\lambda = \frac{23}{9}$$

$$1 + \lambda = \frac{32}{9}$$

So, the equation of circle is,

$$\frac{32}{9}x^2 + \frac{32}{9}y^2 - 8x - 23 = 0$$

$$32x^2 + 32y^2 - 72x - 207 = 0$$

54. Consider the equation of the parabola,

$$y^2 - 8x - 4y - 12 = 0$$

Simplify the above,

$$y^2 - 4y = 8x + 12$$

$$(y - 2)^2 = 8(x + 2)$$

$$4a = 8$$

$$a = 2$$

So, the parametric equations are,

$$x + 2 = at^2, y - 2 = 2at$$

$$x = -2 + 2t^2, y = 2 + 4t$$

55. Consider the equation of tangent to the parabola,

$$mx - y + (10 + m^2) = 0$$

This implies,

$$m^2 + mx + (10 - y) = 0$$

Since above line is tangent so, $D = 0$,

$$b^2 - 4ac = 0$$

$$x^2 - 4(1)(10 - y) = 0$$

$$x^2 = 4(10 - y)$$

$$x^2 = -4(y - 10)$$

56. The slope of tangent of the given parabola at intersection

$$(at^2, 2at) \text{ is } m_1 = \frac{1}{t}.$$

The slope of tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at intersection

$$\text{point } (at^2, 2at) \text{ is } m_2 = -\frac{b^2}{a^2} \left(\frac{t}{2} \right)$$

Now, since $m_1 m_2 = -1$ so,

$$\left(\frac{1}{t}\right)\left(-\frac{b^2}{a^2}\left(\frac{t}{2}\right)\right) = -1$$

$$\frac{b^2}{a^2} = 2$$

Then,

$$e^2 = 1 - \frac{a^2}{b^2}$$

$$= 1 - \frac{1}{2}$$

$$2e^2 = 1$$

57. The equation of ellipse is

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

Let the point $P(5\cos\theta, 4\sin\theta)$ on ellipse then the equation of tangent to ellipse is,

$$x\left(\frac{\cos\theta}{5}\right) + y\left(\frac{\sin\theta}{4}\right) = 1 \quad (I)$$

So, the point Q is $Q\left(\frac{5}{\cos\theta}, 0\right)$ and point R is $R(0, 5\sec\theta)$

The equation of the circle with QR as diameter is,

$$x(x - 5\sec\theta) + y(y - 5\sec\theta) = 0$$

$$x^2 + y^2 - (5\sec\theta)x - (5\sec\theta)y = 0$$

The above circle passes through the origin.

58. The equation of normal to hyperbola $xy = 4$ at point $\left(2t, \frac{2}{t}\right)$ is,

$$2t^4 - xt^3 + yt - 2 = 0 \quad (I)$$

Above equation passes through (a, b) so,

$$2t^4 - at^3 + bt - 2 = 0$$

Have roots $\frac{\alpha_1}{2}, \frac{\alpha_2}{2}, \frac{\alpha_3}{2}, \frac{\alpha_4}{2}$ so,

$$\frac{1}{2}(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) = \frac{a}{2}$$

$$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = a$$

$$\sum \left(\frac{\alpha_1 \alpha_2}{4} \right) = 0$$

$$\sum \left(\frac{\alpha_1 \alpha_2 \alpha_3}{8} \right) = \frac{-b}{2} \text{ and } \frac{\alpha_1 \alpha_2 \alpha_3 \alpha_4}{16} = -1$$

This implies,

$$\begin{aligned}
 \beta_1 + \beta_2 + \beta_3 + \beta_4 &= \frac{4}{\alpha_1} + \frac{4}{\alpha_2} + \frac{4}{\alpha_3} + \frac{4}{\alpha_4} \\
 &= 4 \frac{\sum \alpha_1 \alpha_2 \alpha_3}{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \\
 &= 4 \frac{-\frac{b}{2} \times 8}{-16} \\
 &= b
 \end{aligned}$$

This implies,

$$\begin{aligned}
 \frac{(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4)}{\beta_1 + \beta_2 + \beta_3 + \beta_4} &= (\alpha_1 \alpha_2 \alpha_3 \alpha_4) \\
 &= \frac{a(-16)}{b} \\
 &= -16 \frac{a}{b}
 \end{aligned}$$

59. Consider the given points,

$$A = (1, 2, 3)$$

$$B = (3, -1, 5)$$

$$C = (4, 0, -3)$$

The direction of $\overrightarrow{AB}$ is $(2, -3, 2)$.

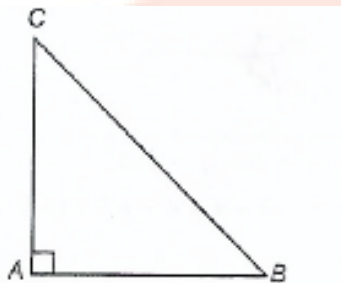
The direction of $\overrightarrow{BC}$ is $(1, 1, -8)$.

The direction of $\overrightarrow{AC}$ is $(3, -2, -6)$.

It can be noticed that $\overrightarrow{AB} \perp \overrightarrow{AC}$

So, $\angle A = 90^\circ$

So, triangle is,



The orthocenter of triangle ABC is vertex A so,

$$H = (1, 2, 3)$$

The circumcenter of ABC is equal to the mid point of BC so,

$$S = \left(\frac{7}{2}, \frac{-1}{2}, 1 \right)$$

Now, the required distance is,

$$(HS) = \sqrt{\frac{33}{2}}$$

60. Consider given points, the equation of plane passing through ABC is,

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

$$\begin{vmatrix} x - 2 & y + 1 & z - 5 \\ -1 & -2 & -1 \\ 3 & 3 & -4 \end{vmatrix} = 0$$

$$11x - 7y + 3z - 26 = 0$$

So, the required directions are,

$$\frac{11}{\sqrt{121+49+9}}, \frac{-7}{\sqrt{121+49+9}}, \frac{3}{\sqrt{121+49+9}} = \left(\frac{11}{\sqrt{179}}, \frac{-7}{\sqrt{179}}, \frac{3}{\sqrt{179}} \right)$$

61. From the given plane equations,

$$\pi_1 + \lambda \pi_2 = 0$$

$$(x + 3y - 6) + \lambda(3x - y + 4z) = 0$$

$$(1 + 3\lambda)x + (3 - \lambda)y + 4\lambda z = 0 \quad \text{(I)}$$

The perpendicular distance from $(0, 0, 0)$ to above plane is 1 so,

$$\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}} = 1$$

$$\frac{|-6|}{\sqrt{(1+3\lambda)^2 + (3-\lambda)^2 + (4\lambda)^2}} = 1$$

$$1 + 9\lambda^2 + 6\lambda + 9 + \lambda^2 - 6\lambda + 16\lambda^2 = 16$$

$$26\lambda^2 = 26$$

$$\lambda = \pm 1$$

Then equation (I) gives,

$$4x + 2y + 4z - 6 = 0$$

$$2x + y + 2z - 3 = 0$$

62. Consider the given expression,

$$l_1 = \lim_{x \rightarrow 2^+} (x^2 + [x])$$

$$= \lim_{x \rightarrow 2^+} (x^2 + 2)$$

$$= 4 + 2$$

$$= 6$$

And,

$$\begin{aligned}l_2 &= \lim_{x \rightarrow 3^-} (2x - [x]) \\&= \lim_{x \rightarrow 3^-} (2x - 2) \\&= 2(3) - 2 \\&= 4\end{aligned}$$

And,

$$l_3 = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{x - \frac{\pi}{2}} \right)$$

Take $x - \frac{\pi}{2} = y$ then,

$$\begin{aligned}l_3 &= \lim_{y \rightarrow 0} \left(\frac{\cos\left(\frac{\pi}{2} + y\right)}{y} \right) \\&= \lim_{y \rightarrow 0} \left(-\frac{\sin y}{y} \right) \\&= -1\end{aligned}$$

This implies,

$$l_3 < l_2 < l_1$$

63. Consider the expression,

$$\lim_{x \rightarrow 0} \frac{((a-n)nx - \tan x) \sin nx}{x^2} = 0, n \neq 0$$

Then,

$$\lim_{x \rightarrow 0} \left(\frac{((a-n)nx)}{x} - \frac{\tan x}{x} \right) \frac{\sin(nx)}{x} = 0 \quad \left[\because \lim_{x \rightarrow 0} \frac{\sin nx}{nx} = 1 \right]$$

$$n(an - n^2 - 1) = 0$$

$$a = n + \frac{1}{n}$$

This implies,

$$\frac{n + \frac{1}{n}}{2} \geq \sqrt{n \cdot \frac{1}{n}}$$

$$\frac{a}{2} \geq 1$$

$$a \geq 2$$

So, minimum possible value of a is 2.

64. Consider the function,

$$f(x) = \begin{cases} x-1, & -\infty < x < 1 \\ 0, & x = 1 \\ x^3-1, & 1 < x < \infty \end{cases}$$

Now, LHL at $x = 1$ is,

$$\lim_{x \rightarrow 1} (x - 1) = 1 - 1 \\ = 0$$

And, RHL at $x = 1$ is,

$$\lim_{x \rightarrow 1} (x^3 - 1) = 1^3 - 1 \\ = 0$$

So, $\text{LHL} = \text{RHL} = f(1)$

Therefore, $f(x)$ is continuous at $x = 1$,

Now, LHD at $x = 1$ is,

$$\lim_{x \rightarrow 1} \frac{(x - 1) - 0}{x - 1} = 1$$

RHD at $x = 1$ is,

$$\lim_{x \rightarrow 1} (x^2 + x + 1) = 3$$

So, $\text{LHL} \neq \text{RHL}$

Therefore, $f(x)$ is not differentiable at $x = 1$.

65. Consider the function,

$$\cos(f(x)) = \frac{1 - x^2}{1 + x^2}$$

Put $x = \tan \theta$ then,

$$\begin{aligned}\cos(f(x)) &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \\ &= \cos 2\theta\end{aligned}$$

$$\begin{aligned}f(x) &= 2\theta \\ &= 2 \tan^{-1} x\end{aligned}$$

And,

$$\tan(g(x)) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

Put $x = \tan 3\theta$ then,

$$\begin{aligned}\tan(g(x)) &= \tan 3\theta \\ g(x) &= 3\theta \\ &= 3 \tan^{-1} x\end{aligned}$$

Then,

$$\begin{aligned}\frac{f'(x)}{g'(x)} &= \frac{2}{3} \\ \frac{df}{dg} &= \frac{2}{3}\end{aligned}$$

66. Consider the equation,

$$x^4 + y^4 = t^2 + \frac{1}{t^2}$$

And,

$$x^2 + y^2 = t - \frac{1}{t}$$

Square on both sides then,

$$x^4 + y^4 + 2x^2y^2 = t^2 + \frac{1}{t^2} - 2$$

$$t^2 + \frac{1}{t^2} + 2x^2y^2 = t^2 + \frac{1}{t^2} - 2$$

$$x^2y^2 = -1$$

$$y^2 = \frac{-1}{x^2}$$

Differentiate w.r.t x,

$$2yy' = \frac{2}{x^3}$$

$$y' = \frac{1}{x^3y}$$

67. It is given that,

$$x = \sin \theta \text{ and } y = \cos p\theta$$

Then,

$$\frac{dy}{d\theta} = -p \sin p\theta \text{ and } \frac{dx}{d\theta} = \cos \theta$$

This implies,

$$\begin{aligned}\frac{dy}{dx} &= -\frac{p \sin p\theta}{\cos \theta} \\ &= -p \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}\end{aligned}$$

$$\sqrt{1-x^2} \left(\frac{dy}{dx} \right)^2 = p^2 (1-y^2)$$

Square both sides,

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = p^2 (1-y^2)$$

Differentiate above,

$$-2x \left(\frac{dy}{dx} \right)^2 + (1-x^2) 2 \frac{dy}{dx} \left(\frac{d^2y}{dx^2} \right) = -p^2 2y \frac{dy}{dx}$$

$$-x \frac{dy}{dx} + (1-x^2) \frac{d^2y}{dx^2} = -p^2 y$$

$$(1-x^2) y_2 = x y_1 - p^2 y$$

68. Consider the curve equation,

$$3y^2 = 4x^2$$

Let point P(h,k) on the curve then,

$$\left(\frac{dy}{dx} \right)_{(h,k)} = \frac{2h^2}{k}$$

Now the length of sub tangent is,

$$\begin{aligned} T &= \left| k \frac{k}{2h^2} \right| \\ &= \frac{1}{2} \frac{k^2}{h^2} \end{aligned}$$

The length of sub normal is,

$$\begin{aligned} N &= \left| k \frac{2h^2}{k} \right| \\ &= 2h^2 \end{aligned}$$

Then,

$$\begin{aligned} (3T)^2 &= \frac{9}{4} \frac{k^4}{h^4} \\ &= \frac{16h^6}{4h^4} \\ &= 4h^2 \\ &= 2N \end{aligned}$$

69. Consider the function,

$$f(x) = \frac{\log(7+x)}{\log(3+x)}, x > 0$$

The above function is decreasing so, $f'(x) < 0$ so,

$$\frac{\log(3+x) \frac{1}{7+x} - \frac{1}{3+x} \log(7+x)}{[\log(3+x)]^2} < 0$$

$$(3+x)^{(3+x)} < (7+x)^{(7+x)}$$

Above is true for every value of x so,

$$x \in (0, \infty)$$

70. By Lagrange mean theorem,

$$\frac{f(7) - f(2)}{7 - 2} = f'(x) \leq 5$$

$$f(7) - f(2) \leq 25$$

$$f(7) \leq 28$$

71. Consider the function,

$$f(x) = 2x^3 - 3x^2 - 12x + 5$$

So,

$$f'(x) = 6x^2 - 6x - 12 = 0$$

Now,

$$x^2 - x - 2 = 0$$

$$(x+1)(x-2) = 0$$

$$x = -1, 2 \in [-2, 4]$$

Therefore,

$$f(-2) = -16 - 12 + 24 + 5 = 1$$

$$f(-1) = -2 - 3 + 12 + 5 = 12$$

$$f(2) = 16 - 12 - 24 + 5 = -15$$

$$f(4) = 128 - 48 - 48 + 5 = 37$$

So, the absolute maxima occurs at $x = 4$.

72. Consider the function,

$$\int \frac{\cos 4x + 1}{\cot x - \tan x} dx$$

Simplify the above,

$$\begin{aligned} \int \frac{\cos 4x + 1}{\cot x - \tan x} dx &= \int \frac{2 \cos^2 x}{\cos^2 x - \sin^2 x} \sin x \cos x dx \\ &= \int \frac{\cos^2 2x}{\cos 2x} \sin 2x dx \\ &= \int \sin 2x \cos 2x dx \\ &= \frac{1}{2} \int \sin 4x dx \end{aligned}$$

Simplify the above,

$$I_1 = -\frac{1}{8}\cos 4x + c$$

$$\text{So, } k = -\frac{1}{8}$$

73. Consider the integral,

$$\begin{aligned} I(x) &= \int e^{2x} (\cos(3x+4) + 5x^2) dx \\ &= \int e^{2x} (\cos(3x+4)) dx + 5 \int e^{2x} x^2 dx \\ &= e^{2x} \left[\frac{2}{13} \cos(3x+4) + \frac{3}{13} \sin(3x+4) \right] + \frac{5x^2}{2} e^{2x} - 5 \int x e^{2x} dx \\ &= e^{2x} \left(\frac{2}{13} \cos(3x+4) + \frac{3}{13} \sin(3x+4) \right) + \frac{5x^2}{2} e^{2x} - \frac{5}{2} x e^{2x} + \frac{5}{2} \int e^{2x} dx \end{aligned}$$

Further simplify the above,

$$\begin{aligned} I(x) &= e^{2x} \left[\left(\frac{2}{13} \cos(3x+4) + \frac{3}{13} \sin(3x+4) \right) \right] + \frac{5x^2}{2} e^{2x} - \frac{5}{2} x e^{2x} + \frac{5}{4} e^{2x} \\ &= e^{2x} \left[\frac{2}{13} \cos(3x+4) + \frac{3}{13} \sin(3x+4) + \frac{5x^2}{2} - \frac{5}{2} x + \frac{5}{4} \right] \end{aligned}$$

74. Consider the integral,

$$\int \frac{5 \cot x + 1}{(\cot x - 1)(\cot x - 2) \sin^2 x} dx$$

Take $\cos x = t$ then,

$$-\operatorname{cosec}^2 x dx = dt$$

This implies,

$$\int \frac{5t + 1}{(t - 1)(t - 2)} dt = \frac{A}{t - 1} + \frac{B}{t - 2}$$

Then,

$$5t + 1 = (A + B)t - (2A + B)$$

So, $A + B = 5$ and $2A + B = -1$

This gives, $A = -6, B = 11$

Then,

$$\begin{aligned} I &= 6 \int \frac{dt}{t - 1} - 11 \int \frac{dt}{t - 2} \\ &= 6 \log |t - 1| - 11 \log |t - 2| + c \\ &= 6 \log |\cot x - 1| + 11 \log |(\cot x - 2)^{-1}| + c \end{aligned}$$

On comparison,

$$(f(x), g(x)) = ((\cot x - 1), (\cot x - 2)^{-1})$$

75. Consider the integral,

$$I_{m,n} = \int e^{mx} \cdot x^n dx$$

Simplify the above,

$$\begin{aligned} & \frac{1}{m} x^n e^{mx} - \int (nx^{n-1}) \left(\frac{e^{mx}}{m} \right) dx \\ &= \frac{x^n e^{mx}}{m} - \frac{n}{m} \int e^{mx} x^{n-1} dx \\ &= \frac{x^n e^{mx}}{m} - \frac{n}{m} I_{m,n-1} + c \\ I_{m,n} + \frac{n}{m} I_{m,n-1} &= \frac{x^n e^{mx}}{m} + c \end{aligned}$$

76. Consider the integral,

$$P = \lim_{n \rightarrow \infty} \frac{1}{n} [(n+1)(n+2) \dots (2n)]^{\frac{1}{n}}$$

Apply log on both sides,

$$\begin{aligned}
 \log P &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \left(\frac{n+1}{n} \right) \left(\frac{n+2}{n} \right) \dots \left(\frac{n+n}{n} \right) \right] \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \frac{r}{n} \right) \\
 &= \int_0^1 \log(1+x) \\
 &= \left[x \log(1+x) \right]_0^1 - \int_0^1 \frac{x}{1+x} dx
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 \log P &= \log 2 - \left(x - \log(1+x) \right)_0^1 \\
 &= \log 2 - 1 + \log 2 \\
 &= \log 4 - 1 \\
 &= \log \left(\frac{4}{e} \right)
 \end{aligned}$$

This implies,

$$P = \frac{4}{e}$$

77. Consider the integral,

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx \quad (I)$$

Apply the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ then,

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^2 x}{\cos x + \sin x} dx \quad (II)$$

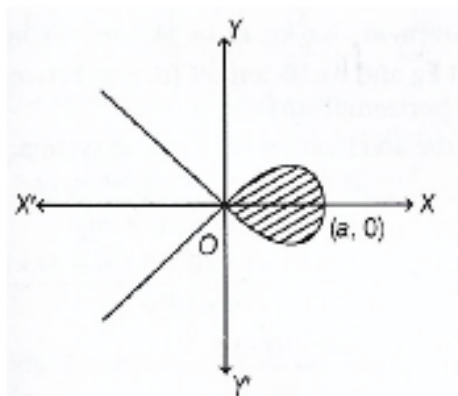
Add equation (I) and (II),

$$\begin{aligned} 2I &= \int_0^{\frac{\pi}{2}} \frac{\sin^2 x + \cos^2 x}{\sin x + \cos x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{dx}{\sin x + \cos x} \\ &= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \operatorname{cosec} \left(\frac{\pi}{4} + x \right) dx \\ &= \frac{1}{\sqrt{2}} \left[\log \left| \operatorname{cosec} \left(\frac{\pi}{4} + x \right) - \cot \left(\frac{\pi}{4} + x \right) \right| \right]_0^{\frac{\pi}{2}} \end{aligned}$$

Further simplify the above,

$$\begin{aligned} I &= \frac{1}{2\sqrt{2}} \left[\log(\sqrt{2} + 1) - \log(\sqrt{2} - 1) \right] \\ &= \frac{1}{2\sqrt{2}} \left[2\log(\sqrt{2} + 1) \right] \\ &= \frac{1}{\sqrt{2}} \left[\log(\sqrt{2} + 1) \right] \end{aligned}$$

78. Consider the diagram,



The required area is,

$$2 \int_0^a x \sqrt{\frac{a-x}{a}} dx = \frac{2}{\sqrt{a}} \int_0^a [x \sqrt{a-x}] dx$$

Put $a - x = t^2$ then,

$$-dx = 2t dt$$

So, required area is,

$$\begin{aligned}\int_{\sqrt{a}}^0 \frac{4}{\sqrt{a}} t^2 (a - t^2) dt &= \frac{4}{\sqrt{a}} \int_{\sqrt{a}}^0 (at^2 - t^4) dt \\&= \frac{4}{\sqrt{a}} \left[\frac{at^3}{3} - \frac{t^5}{5} \right]_0^{\sqrt{a}} \\&= \frac{4}{\sqrt{a}} \left(\frac{a^2 \sqrt{a}}{3} - \frac{a^2 \sqrt{a}}{5} \right) \\&= \frac{8}{15} a^2\end{aligned}$$

79. Consider the equation,

$$y = e^x (A \cos x + B \sin x)$$

So,

$$y' = e^x (-A \sin x + B \cos x) + e^x (A \cos x + B \sin x)$$

$$y' - y = e^x (-A \sin x + B \cos x)$$

$$\begin{aligned}y'' - y' &= e^x (-A \cos x - B \sin x) + e^x (-A \sin x + B \cos x) \\&= -y + y' - y\end{aligned}$$

This implies,

$$y'' - 2y' + 2y = 0$$

80. Consider the expression,

$$x \frac{dy}{dx} = y - x \tan \frac{y}{x}$$
$$\frac{dy}{dx} = \frac{y}{x} - \tan \left(\frac{y}{x} \right)$$

Let $y = vx$ then,

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

So,

$$v + x \frac{dv}{dx} = v - \tan v$$

$$\frac{dv}{\tan v} = -\frac{dx}{x}$$

$$\sin v = \frac{k}{x}$$

$$v = \sin^{-1} \left(\frac{k}{x} \right)$$

Further simplify the above,

$$y = x \sin^{-1} \left(\frac{k}{x} \right)$$

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Solutions – Physics

81. In a decimal number less than 1, the zero on the right of decimal point but left of the first non-zero digit are not significant. So, the number 0.00764 has three significant digits.

Thus, option (1) is correct.

82. The expression of the acceleration is,

$$a = \frac{dv}{dt}$$

$$-2.5\sqrt{v} = \frac{dv}{dt}$$

$$-2.5dt = \frac{dv}{\sqrt{v}}$$

Integrate above equation and substitute values to find the time taken to come to rest.

$$-2.5 \int_0^t dt = \int_{6.25}^0 \frac{dv}{\sqrt{v}}$$

$$-2.5t = \left[2\sqrt{v} \right]_{6.25}^0$$

$$t = 2 \text{ s}$$

83. According to the given condition,

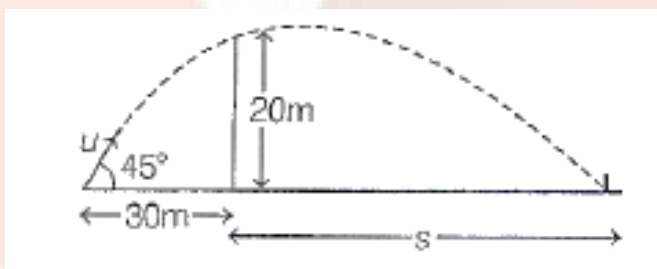
$$R = \frac{R_{\max}}{2}$$

$$\frac{u^2 \sin 2\theta}{g} = \frac{u^2/g}{2}$$

$$\sin 2\theta = \frac{1}{2}$$

$$\theta = 15^\circ$$

84. Consider the projectile path of the body as shown below.



If t is the time taken by projectile to cross the pole, then

$$30 = (u \cos 45^\circ)t$$

$$ut = 30\sqrt{2} \quad \dots\dots (I)$$

And,

$$20 = (u \sin 45^\circ)t - \frac{10}{2}t^2$$

$$20 = 30\sqrt{2} \times \frac{1}{\sqrt{2}} - 5t^2$$

$$t = \sqrt{2} \text{ s}$$

Now, from equation (I),

$$u\sqrt{2} = 30\sqrt{2}$$

$$u = 30 \text{ m/s}$$

The range is calculated as,

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R = \frac{(30)^2 \sin(2 \times 45^\circ)}{10}$$

$$R = 90 \text{ m}$$

The distance between pole and the point at which body strike on the ground is,

$$s = (90 - 30) \text{ m}$$

$$s = 60 \text{ m}$$

85. According to the conservation of momentum,

$$P_i = P_f$$

$$0 = P_1 + P_2 + P_3$$

$$P_3 = -(P_1 + P_2)$$

The magnitude of P_3 is calculated as,

$$|P_3| = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$m_3 v_3 = \sqrt{12^2 + 16^2 + 2(12)(16)\cos 90^\circ}$$

Substitute values,

$$m_3(4) = 20$$

$$m_3 = 5 \text{ kg}$$

The total mass of the rock is,

$$m = m_1 + m_2 + m_3$$

$$m = 1 + 2 + 5$$

$$m = 8 \text{ kg}$$

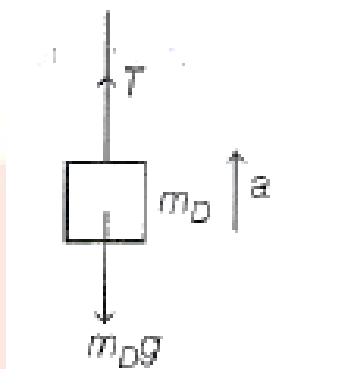
86. The acceleration of the complete system is calculated as,

$$a = \frac{m_C g - m_D g - \mu(m_A + m_D)g}{m_A + m_B + m_C + m_D}$$

$$a = \frac{(6)(10) - (1)(10) - 0.2(6+3)(10)}{6+3+6+1}$$

$$a = 2 \text{ ms}^{-2}$$

The free body diagram of the block D is shown below.



From equilibrium equation,

$$m_D a = T - m_D g$$

$$(1)(2) = T - (1)(10)$$

$$T = 12 \text{ N}$$

87. According to work energy theorem,

$$W = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$Pt = \frac{1}{2} m (v_f^2 - v_i^2)$$

Substitute values.

$$7t = \frac{1}{2} (15) (5^2 - 3^2)$$

$$t = 17.14 \text{ s}$$

The acceleration is calculated as,

$$a = \frac{V_f - V_i}{t}$$

$$a = \frac{5 - 3}{17.14}$$

$$a = 0.116 \text{ m/s}^2$$

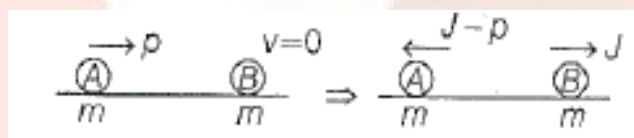
The distance travelled is calculated by,

$$S = \frac{v_f^2 - v_i^2}{2a}$$

$$S = \frac{5^2 - 3^2}{2(0.116)}$$

$$S \approx 70 \text{ m}$$

88. The collision between A and B is shown below,



The coefficient of restitution is,

$$e = \frac{\frac{J-p}{m} + \frac{J}{m}}{\frac{p}{m} - 0} \quad \left(\text{As, } v = \frac{mv}{m} = \frac{p}{m} \right)$$

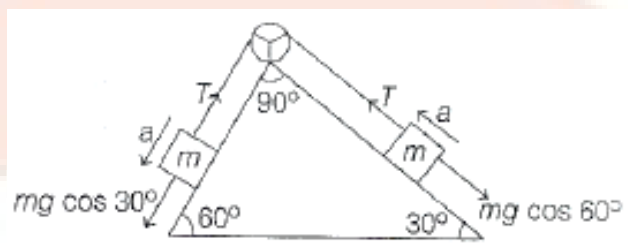
$$e = \frac{2J}{p} - 1$$

During impact, the impulse generated by B is,

$$J_B = p + J - p$$

$$J_B = J$$

89. Consider the figure shown below.



Write the force equation for both the blocks.

$$mg \cos 30^\circ - T = ma$$

$$T = m(g \cos 30^\circ - a) \quad \text{..... (I)}$$

$$T - mg \cos 60^\circ = ma \quad \text{..... (II)}$$

From equation (I) and (II), we get

$$ma = m(g \cos 30^\circ - a) - mg \cos 60^\circ$$

$$a = \frac{(\sqrt{3} - 1)}{4} g$$

The expression of the magnitude of the acceleration at centre of mass is,

$$a_{\text{cm}} = \left| \frac{ma_1 + ma_2}{m + m} \right|$$

$$a_{\text{cm}} = \left| \frac{a_1 + a_2}{2} \right|$$

Substitute values.

$$a_{cm} = \left| \frac{\left(\frac{\sqrt{3}-1}{4}\right)g\hat{i} + \left(\frac{\sqrt{3}-1}{4}\right)g\hat{j}}{2} \right|$$

$$a_{cm} = \frac{g}{2} \sqrt{\left(\frac{\sqrt{3}-1}{4}\right)^2 + \left(\frac{\sqrt{3}-1}{4}\right)^2}$$

$$a_{cm} = \left(\frac{\sqrt{3}-1}{4\sqrt{2}}\right)g$$

90. The torque is calculated as,

$$\tau = FR$$

$$\tau = mgR$$

$$\tau = (0.4)(10)(8 \times 10^{-2})$$

$$\tau = 0.32 \text{ Nm}$$

The angle by which wheel is rotated is,

$$\theta = \frac{\text{arc}}{\text{radius}}$$

$$\theta = \frac{1}{0.08}$$

$$\theta = 12.5 \text{ rad}$$

From the equation of rotational motion,

$$\theta = \omega_0 t + \frac{\alpha t^2}{2}$$

$$12.5 = (0)(10) + \frac{\alpha(10)^2}{2}$$

$$\alpha = 0.25 \text{ rad} \cdot \text{s}^{-2}$$

The moment of inertia of the wheel is,

$$I = \frac{\tau}{\alpha}$$

$$I = \frac{0.32}{0.25}$$

$$I = 1.28 \text{ kg} \cdot \text{m}^2$$

91. The mass of body after collision is,

$$M_{\text{body}} = M + m$$

$$M_{\text{body}} = 1 + 0.5$$

$$M_{\text{body}} = 1.5 \text{ kg}$$

From the conservation of momentum, the velocity of the body after collision is,

$$(M + m)v' = mv$$

$$(1 + 0.5)v' = (0.5)(3)$$

$$v' = 1 \text{ m/s}$$

The maximum amplitude of spring mass system is,

$$A = \frac{v'}{\omega} = \frac{v'}{2\pi f}$$

$$A = \frac{1}{2\pi \left(\frac{10}{\pi} \right)}$$

$$A = 0.05 \text{ m}$$

$$A = 5 \text{ cm}$$

The frequency of oscillation is calculated as,

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{M_{\text{body}}}}$$

$$\frac{10}{\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{1.5}}$$

$$k = \left(2\pi \times \frac{10}{\pi} \right)^2 (1.5)$$

$$k = 600 \text{ Nm}^{-1}$$

92. The gravitational potential is given by,

$$V = - \int_0^r \vec{E} \cdot d\vec{r}$$

$$V = - \int_0^r \left(E_x \hat{i} + E_y \hat{j} \right) \cdot \left(dr_x \hat{i} + dr_y \hat{j} \right)$$

Substituting the values, we get

$$V = - \left[\int (5\hat{i} + 12\hat{j}) \cdot (dr_x \hat{i} + dr_y \hat{j}) \right]$$

$$V = - \left[5 \int_0^{12} dr_x + 12 \int_0^{15} dr_y \right]$$

$$V = - [5(12 - 0) + 12(15 - 0)]$$

$$V = -240 \text{ J} \cdot \text{kg}^{-1}$$

The change in gravitational potential energy of the particle is,

$$E = mV$$

$$E = (2)(-240)$$

$$E = -480 \text{ J}$$

93. The Poisson's ratio is given by the expression,

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

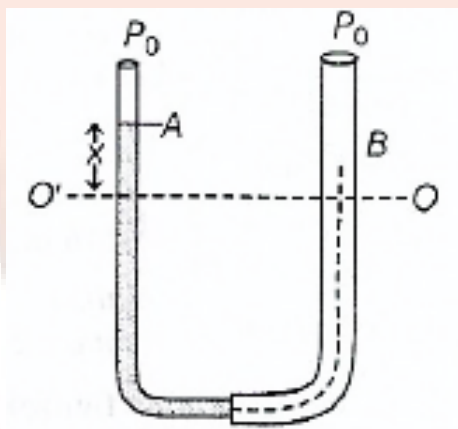
$$\sigma = \frac{\left(\frac{\Delta D}{D} \right)}{\left(\frac{\Delta L}{L} \right)} = \left(\frac{\Delta D}{D} \right) \left(\frac{L}{\Delta L} \right)$$

Substituting the values, we get

$$0.3 = \left(\frac{\Delta D}{0.6 \times 10^{-3}} \right) \left(\frac{10}{6 \times 10^{-3}} \right)$$

$$\Delta D = 10.8 \times 10^{-8} \text{ m}$$

94. The U shaped tube is shown below,



The tube is open at both ends. So, $P_A = P_B = P_0$

The pressure at point B is,

$$P_B = \frac{2T}{R_2}$$

The pressure at point A is,

$$P_A = \frac{2T}{R_1}$$

For the same height,

$$-\rho g x + \frac{2T}{R_1} = \frac{2T}{R_2}$$

$$x = \frac{2T}{\rho g} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Substituting the values, we get

$$x = \frac{(2)(7.3 \times 10^{-2})}{(10^3)(10)} \left[\frac{1}{2 \times 10^{-3}} - \frac{1}{4 \times 10^{-3}} \right]$$

$$x = 3.65 \text{ mm}$$

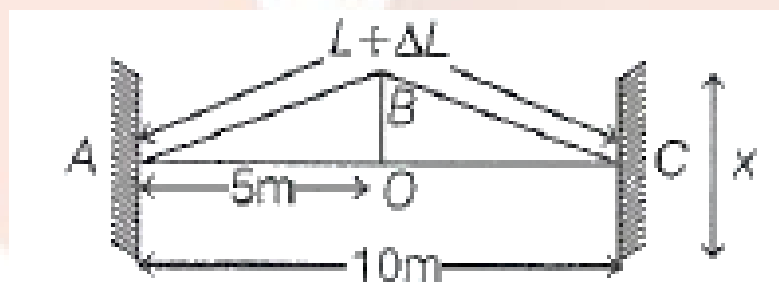
95. The change in length is calculated as,

$$\Delta L = L \alpha \Delta t$$

$$\Delta L = (10)(2.5 \times 10^{-6})(40)$$

$$\Delta L = 0.1 \text{ cm}$$

Consider the figure of system as shown below,



Let the displacement of midpoint of bar is x .

In $\triangle OBA$

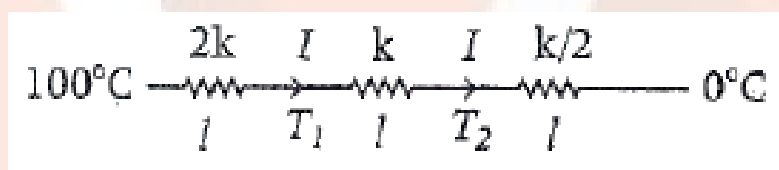
$$x = \sqrt{(AB)^2 - (OA)^2}$$

$$x = \sqrt{\left(\frac{10 + 0.001}{2}\right)^2 - (5)^2}$$

$$x = 0.223 \text{ m.}$$

$$x = 22.3 \text{ cm}$$

96. The conduction of heat through series combinations of rods is shown below.



The expression of the rate of heat flow for series conduction is,

$$\frac{dQ_1}{dt} = \frac{dQ_2}{dt} = \frac{dQ_3}{dt} = \frac{T_i - T_f}{k_{eq}}$$

$$\frac{k_1 A_1 (T_i - T_1)}{l_1} = \frac{k_2 A_2 (T_1 - T_2)}{l_2} = \frac{k_3 A_3 (T_2 - T_f)}{l_3}$$

Substitute $A_1 = A_2 = A_3$ and $l_1 = l_2 = l_3$ in above expression,

$$2k(100 - T_1) = k(T_1 - T_2) = 0.5(T_2 - 0) \quad \dots (I)$$

The equivalent coefficient of thermal conductivity is,

$$\frac{1}{k_{eq}} = \frac{1}{2k} + \frac{1}{k} + \frac{2}{k}$$

$$k_{eq} = \frac{2k}{7}$$

The heat current equation is,

$$\frac{dQ}{dt} = \frac{dQ_1}{dt}$$

$$\frac{k_{eq} A (T_i - T_f)}{l} = \frac{k_1 A_1 (T_i - T_1)}{l_1} \quad \dots (II)$$

The equivalent length is, $l = l_1 + l_2 + l_3 = 3l_1$

The equivalent area is, $A = A_1 + A_2 + A_3 = 3A_1$

Substitute values in equation (II),

$$\frac{2k}{7} \frac{3A_1 (100 - 0)}{3l_1} = \frac{2kA_1 (100 - T_1)}{l_1}$$

$$\frac{1}{7} \times 100 = 100 - T_1$$

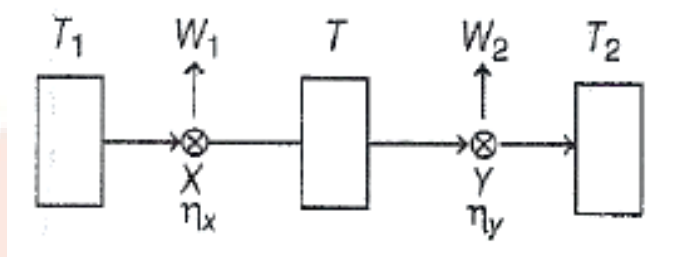
$$T_1 = \frac{600^\circ\text{C}}{7}$$

Substitute values in equation (I),

$$2k \left(100 - \frac{600}{7} \right) = 0.5kT_2$$

$$T_2 = \frac{400^\circ\text{C}}{7}$$

97. The series combination of two heat engines is shown below.



The expression of the efficiency of x heat engine is,

$$\eta_x = 1 - \frac{T}{T_1}$$

$$\eta_x = 1 - \frac{T}{900}$$

The expression of the efficiency of y heat engine is,

$$\eta_y = 1 - \frac{T_2}{T}$$

$$\eta_y = 1 - \frac{400}{T}$$

From the given condition, $\eta_x = \eta_y$

$$1 - \frac{T}{900} = 1 - \frac{400}{T}$$

$$T = 600 \text{ K}$$

98. The universal gas constant is given by,

$$R = C'_p - C'_v$$

$$R = mC_p - mC_v$$

$$R = m(C_p - C_v)$$

Here, C'_p and C'_v are the molar specific heat of gas at constant pressure and constant volume.

Substituting the values, we get

$$R = m(620 - 420)$$

$$m = \frac{R}{200}$$

Using the ideal gas equation,

$$pV = nRT$$

$$pm = \rho RT \quad \left(\text{As, } n = 1 \text{ and } \rho = \frac{m}{V} \right)$$

$$10^5 \times \frac{R}{200} = \rho R (273)$$

$$\rho = 1.85 \text{ Kg-m}^{-3}$$

99. The expression for the rms velocity of any gas is,

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$v_{\text{rms}} \propto \sqrt{\frac{T}{M}}$$

The above equation shows that the rms velocity of the gas depends on the temperature of gas and the molecular weight of the gas.

So, at constant temperature, the rms speed is independent of quantity of gas present in the gas mixture.

Thus, the rms speed of O_2 in vessel C will be v_1 .

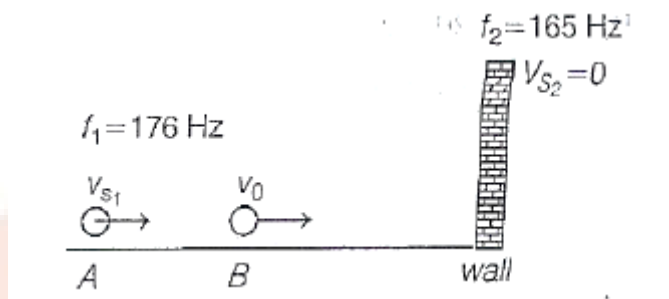
100. The vibrations are perpendicular to the direction of propagation in transverse wave.

The vibrations are perpendicular to the direction of propagation in longitudinal wave.

The beats are produced due to the superposition of the waves, travelling in the same direction.

The stationary waves are produced due to the superposition of the waves, travelling in opposite direction.

101. The figure below shows the given situation.



The apparent frequency of police car is,

$$f'_1 = \frac{v - v_0}{v - v_s} f_1$$

$$f'_1 = \frac{330 - v_0}{330 - 22} (176)$$

$$f'_1 = \frac{330 - v_0}{308} (176)$$

The apparent frequency of stationary siren is,

$$f'_2 = \frac{v + v_0}{v} f_2$$

$$f'_2 = \frac{330 + v_0}{330} (165)$$

The number of beats heard by the cyclist is zero. Therefore,

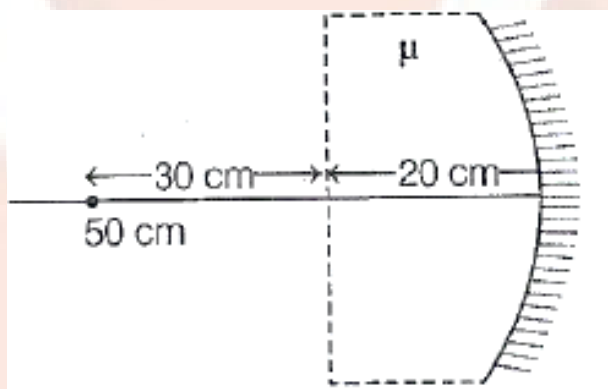
$$n = f'_1 - f'_2$$

$$0 = \frac{330 - v_0}{308}(176) - \frac{330 + v_0}{330}(165)$$

$$(330 - v_0) \frac{176}{308} = (330 + v_0) \frac{165}{330}$$

$$v_0 = 22 \text{ ms}^{-1}$$

102. The position of object and concave mirror is shown below.



Given, $v_{\text{real}} = 50 \text{ cm}$ and $v_{\text{apparent}} = 40 \text{ cm}$

The distance outside the transparent medium is,

$$D_{\text{real}} = v_{\text{real}} - \text{thickness of transparent medium}$$

$$D_{\text{real}} = 50 - 20$$

$$D_{\text{real}} = 30 \text{ cm}$$

The apparent distance is,

$$D_{\text{apparent}} = (40 - 20)$$

$$D_{\text{apparent}} = 20 \text{ cm}$$

The refractive index of the liquid is,

$$\mu = \frac{D_{\text{actual}}}{D_{\text{apparent}}}$$

$$\mu = \frac{30}{20}$$

$$\mu = \frac{3}{2}$$

103. The fringe width is given by,

$$\beta = \frac{\lambda D}{d}$$

Therefore,

$$\frac{\beta_1}{\beta_2} = \frac{\lambda_1 D_1}{d_1} \times \frac{d_2}{\lambda_2 D_2}$$

Given $\lambda_1 = \lambda_2$ and $D_1 = D_2$

Thus,

$$\frac{\beta_1}{\beta_2} = \frac{d_2}{d_1}$$

Substituting the values, we get

$$\frac{1.2 \times 10^{-3}}{\beta_2} = \frac{\frac{3}{2}d}{d}$$
$$\beta_2 = 0.8 \text{ mm}$$

104. The electrostatic force between charges is,

$$F_1 = \frac{kQ_1Q_2}{R^2}$$

$$F_1 = \frac{k}{R^2}(3.72 \times 1.86)10^{-12} \text{ N}$$

$$F_1 = \frac{k}{R^2}(6.9192 \times 10^{-12})$$

If 20% of Q_1 is given to Q_2 then,

$$Q'_2 = Q_2 + \frac{20}{100}Q_1$$

$$Q'_2 = 1.86 + \frac{20}{100}(3.72)$$

$$Q'_2 = 2.604 \mu\text{C}$$

And

$$Q'_1 = \frac{80}{100}Q_1$$

$$Q'_1 = 2.976 \mu\text{C}$$

The new electrostatic force between charges is,

$$F_2 = \frac{kQ'_1Q'_2}{R^2}$$

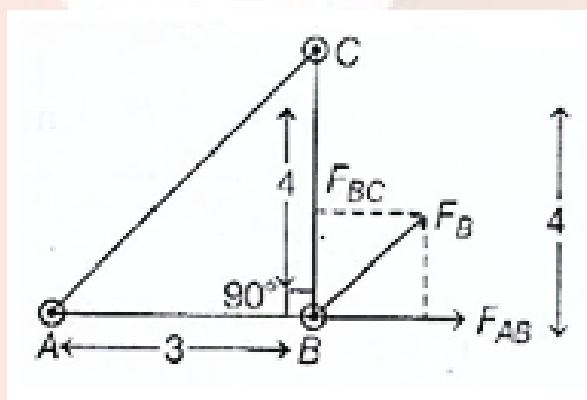
$$F_2 = \frac{k}{R^2}(2.976 \times 2.604)10^{-12} \text{ N}$$

$$F_2 = \frac{k}{R^2}(7.7495)$$

The % increases in electrostatic force between charges is,

$$\begin{aligned} \frac{F_2 - F_1}{F_1} \times 100 &= \frac{k}{R^2} \frac{(7.7495 - 6.9192) \times 10^{-12}}{\frac{k}{R^2} \times 6.9192 \times 10^{-12}} \times 100 \\ &= 12\% \end{aligned}$$

105. The position of charges is shown below.



The electrostatic force F_{AB} is calculated as,

$$F_{AB} = \frac{kQ_A Q_B}{r_{AB}^2}$$

$$F_{AB} = \frac{k(15 \times 10^{-6})(12 \times 10^{-6})}{9 \times 10^{-4}}$$

$$F_{AB} = k(20 \times 10^{-8}) \text{ N}$$

Similarly, the electrostatic force F_{BC} is calculated as,

$$F_{BC} = \frac{k(20 \times 10^{-6})(12 \times 10^{-6})}{16 \times 10^{-4}}$$

$$F_{BC} = k(15 \times 10^{-8}) \text{ N}$$

The resultant electrostatic force at B is,

$$F_B = \sqrt{F_{AB}^2 + F_{BC}^2} \quad (\text{As, } \theta = 90^\circ)$$

$$F_B = (9 \times 10^9) \sqrt{(20 \times 10^{-8})^2 + (15 \times 10^{-8})^2}$$

$$F_B = 2250 \text{ N}$$

106. The capacitance in first case is,

$$C_1 = 4\pi\epsilon_0 \left[\frac{R_1 R_2}{R_2 - R_1} + R_2 \right]$$

$$C_1 = 4\pi\epsilon_0 \left[\frac{(2)(5)}{5-2} + 5 \right]$$

$$C_1 = 4\pi\epsilon_0 \left(\frac{25}{3} \right) \quad \text{..... (I)}$$

The capacitance in second case is,

$$C_2 = 4\pi\epsilon_0 \left[\frac{R_1 R_2}{R_2 - R_1} \right]$$

$$C_2 = 4\pi\epsilon_0 \left[\frac{(2)(5)}{5-2} \right]$$

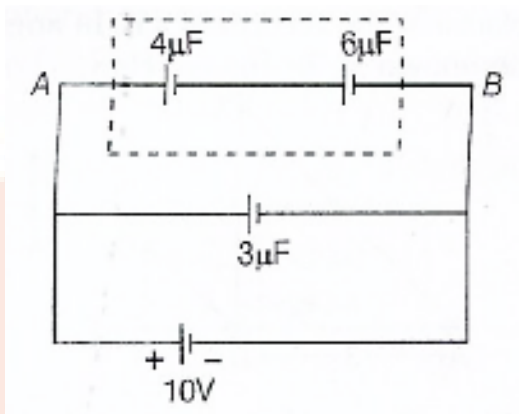
$$C_2 = 4\pi\epsilon_0 \left(\frac{10}{3} \right) \quad \text{..... (II)}$$

Divide equation (I) by (II).

$$\frac{C_1}{C_2} = \frac{25}{10}$$

$$\frac{C_1}{C_2} = \frac{5}{2}$$

107. The circuit is shown below,



The equivalent capacitance of $4\mu\text{F}$ and $6\mu\text{F}$ capacitors is,

$$C_{\text{eq}} = \frac{4 \times 6}{4 + 6}$$

$$C_{\text{eq}} = 2.4\mu\text{F}$$

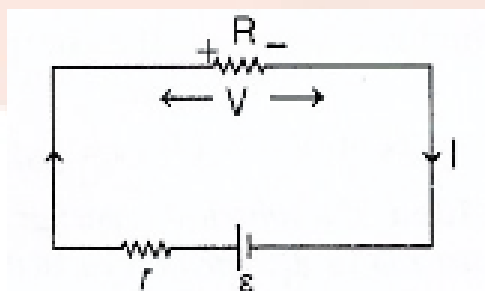
The charge stored by $4\mu\text{C}$ is calculated as,

$$Q = C_{\text{eq}} V$$

$$Q = (2.4 \times 10^{-6}) \times 10$$

$$Q = 24\mu\text{C}$$

108. The figure below shows the circuit diagram.



From the circuit,

$$Ir = E - V$$

$$\frac{V}{r} = E - V$$

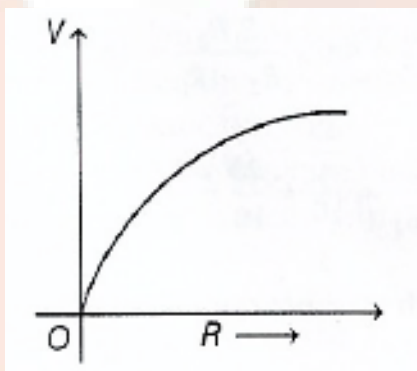
$$V \left[1 + \frac{r}{R} \right] = E$$

$$V = \frac{E}{R + r} R$$

The general equation of a straight line passing through the origin is $y = mx$

Here, $y = V$, $m = \frac{E}{R + r}$ and $x = R$

The line is passing through the origin and has a slope which reduces as R increases. So the graph is,



109. The equilibrium equation for meter bridge is,

$$\frac{R_1}{R_2} = \frac{l_1}{L - l_1}$$

$$\frac{R_1}{3x} = \frac{l_1}{l_1} \quad (\text{As, } l_1 = L - l_1)$$

Substitute values.

$$\frac{2}{3x} = 2$$

$$x = 6 \Omega$$

110. The current in the circular loop is given by,

$$B = \frac{\mu_0 I}{2R}$$

$$I = \frac{2BR}{\mu_0}$$

The magnetic moment of the loop is given by,

$$M = IA$$

$$M = \frac{2BR}{\mu_0} A \quad \dots\dots (I)$$

The radius is given by,

$$A = \pi R^2$$

$$R = \sqrt{\frac{A}{\pi}} \quad \dots\dots (II)$$

From equation (I) and (II), we get

$$M = \frac{2B\sqrt{A/\pi}}{\mu_0} A$$

$$M = \frac{2BA\sqrt{A}}{\mu_0\sqrt{\pi}}$$

111. The expression of the work done in turning a loop from θ_1 and θ_2 is calculated as,

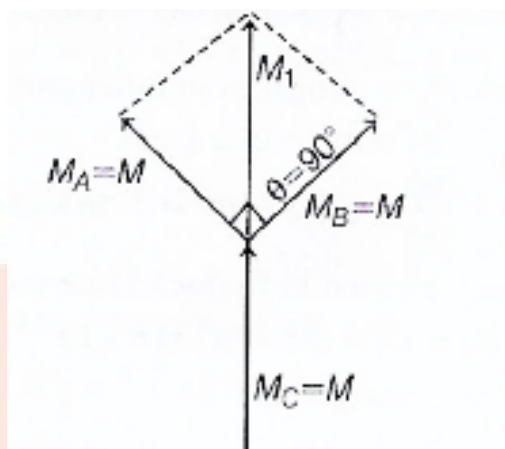
$$W = MB(\cos\theta_1 - \cos\theta_2)$$

$$W = NIAB(\cos\theta_1 - \cos\theta_2)$$

$$W = (100)(0.5)(\pi \times 100 \times 10^{-4})(2)(\cos 0^\circ - \cos 180^\circ)$$

$$W = 2\pi \text{ J}$$

112. The arrangement of three magnetic dipoles is shown below.



The magnetic moment M_1 is calculated as,

$$M_1 = \sqrt{M_A^2 + M_B^2 + 2M_A M_B \cos \theta}$$

$$M_1 = \sqrt{M^2 + M^2 + 2MM \cos 90^\circ}$$

$$M_1 = M\sqrt{2}$$

The resultant magnetic moment is,

$$\begin{aligned} M_{\text{resultant}} &= M_1 + M_C \\ &= M\sqrt{2} + M \\ &= (1 + \sqrt{2})M \end{aligned}$$

113. The magnetic field inside a solenoid of infinite length is calculated as,

$$\Delta B = \mu_0 n (i_f - i_i)$$

$$\Delta B = (4\pi \times 10^{-7})(2000)(5.5 - 1.5)$$

$$\Delta B = 32\pi \times 10^{-4}$$

So, the induced emf in a loop is,

$$e = -\frac{d\phi}{dt}$$

$$e = -\frac{\Delta BA}{\Delta t} \cos \theta$$

$$e = -\frac{(32\pi \times 10^{-4})\pi(3 \times 10^{-2})^2}{\frac{\pi^2}{100}} \cos 0^\circ$$

$$e = 0.288 \text{ mV}$$

114. The expression of the impedance of series LCR circuit is,

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Substitute values.

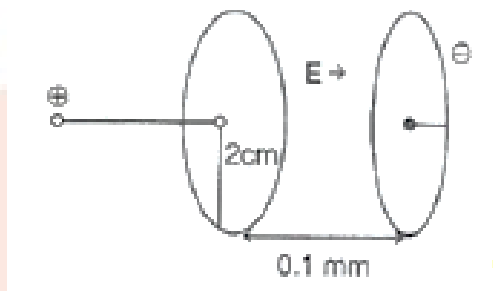
$$Z = \sqrt{10^2 + \left[(70 \times 10^3)(10 \times 10^{-6}) - \frac{1}{(70 \times 10^3)(1 \times 10^{-6})} \right]^2}$$

$$Z = \sqrt{100 + (-13.58)^2}$$

Here, negative sign shows that the circuit properties are more likely to be capacitive than the inductive i.e., $X_C > X_L$

Thus, the circuit behaves like series RC circuit.

115. The parallel plate capacitor is shown below.



The displacement current in the capacitor is given by,

$$I_d = \epsilon_0 \frac{d\phi}{dt}$$

$$I_d = \epsilon_0 \frac{dE}{dt} A$$

$$I_d = \epsilon_0 \frac{A}{d} \frac{dV}{dt}$$

Substitute values.

$$I_d = \frac{(8.85 \times 10^{-12}) \pi (4 \times 10^{-4}) (5 \times 10^6)}{1 \times 10^{-4}}$$

$$I_d = 556.28 \times 10^{-6} \text{ A}$$

$$I_d = 0.556 \text{ mA}$$

116. The energy of photon is calculated as,

$$E = \frac{hc}{\lambda}$$

$$E = \frac{1250}{488} \text{ eV}$$

$$E = 2.56 \text{ eV}$$

The maximum kinetic energy of a photo electron is,

$$KE_{\text{max}} = eV_0$$

$$KE_{\text{max}} = 0.38 \text{ eV}$$

Using Einstein's photoelectric equation, the work function is

$$E = KE_{\text{max}} + W_0$$

$$2.56 = 0.38 + W_0$$

$$W_0 = 2.18 \text{ eV}$$

$$W_0 \approx 2.16 \text{ eV}$$

117. Let the energy of hydrogen like atom in ground state be $-E$.

For, 1st excitation,

$$E = E_{2\text{nd}} - E_{1\text{st}}$$

$$15 \text{ eV} = \left(\frac{-E}{2^2} \right) - \left(\frac{E}{1^2} \right)$$

$$\frac{3E}{4} = 15 \text{ eV}$$

$$E = 20 \text{ eV}$$

The 3rd excitation energy is,

$$E_3 = E_{4\text{th}} - E_{1\text{st}}$$

$$E_3 = \left(\frac{-E}{4^2} \right) - \left(\frac{E}{1^2} \right)$$

$$E_3 = \frac{15}{16} E$$

$$E_3 = \frac{15}{16} (20)$$

$$E_3 = \frac{75}{4} \text{ eV}$$

118. The number of nucleus in 100 gm of ${}_{92}\text{U}^{235}$ is calculated as,

$$N = \frac{100}{M} \times N_A$$

$$N = \frac{100}{235} \times (6.022 \times 10^{23})$$

$$N = 2.56 \times 10^{23} \text{ nucleus}$$

The energy released in the fission of one nucleus is,

$$E'_0 = E_0$$

$$E'_0 = 1.88 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

$$E'_0 = 300.8 \times 10^{-13} \text{ J}$$

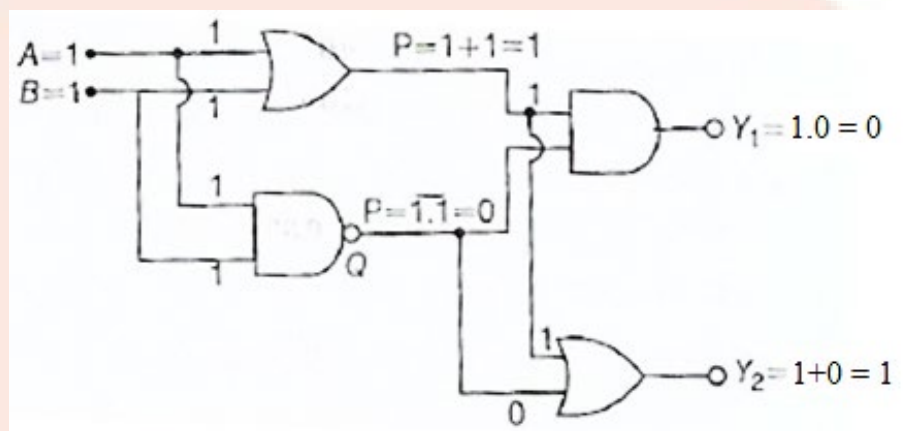
The energy released in N nucleus fission is calculated as,

$$E = E'_0 N$$

$$E = 300.8 \times 10^{-13} \times 2.56 \times 10^{23}$$

$$E = 7.71 \times 10^{12} \text{ J}$$

119. The logical circuit is shown below,



So, $Y_1 = 0$ and $Y_2 = 1$

120. For 100% modulation, $\mu = 1$

Using the expression of the modulation index, we get

$$\mu = \frac{E_m}{E_c}$$

$$1 = \frac{E_m}{E_c}$$

$$E_m = E_c$$

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Solutions – Chemistry

121. The expression for work function for Li is,

$$\nu_0 = \frac{W_0}{h}$$

Here, the threshold frequency is given as ν_0 and the work function is W_0 .

Substitute the given value in above expression.

$$\nu_0 = \frac{2.42 \times 1.602 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = 0.584 \times 10^{15} \text{ s}^{-1}$$

The expression for work function for K is,

$$\nu_0 = \frac{W_0}{h}$$

Substitute the given value in above expression.

$$\nu_0 = \frac{2.25 \times 1.602 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = 0.54 \times 10^{15} \text{ s}^{-1}$$

The expression for work function for Mg is,

$$\nu_0 = \frac{W_0}{h}$$

Substitute the given value in above expression.

$$\nu_0 = \frac{3.70 \times 1.602 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = 0.89 \times 10^{15} \text{ s}^{-1}$$

The expression for work function for Ag is,

$$\nu_0 = \frac{W_0}{h}$$

Substitute the given value in above expression.

$$\nu_0 = \frac{4.30 \times 1.602 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = 1.03 \times 10^{15} \text{ s}^{-1}$$

The expression for work function for Cu is,

$$\nu_0 = \frac{W_0}{h}$$

Substitute the given value in above expression.

$$\nu_0 = \frac{4.80 \times 1.602 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = 1.16 \times 10^{15} \text{ s}^{-1}$$

The frequency of light is given as,

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{540 \times 10^{-9}} = 0.56 \times 10^{15} \text{ s}^{-1}$$

The condition at which the photoelectric effect is shown is

$\nu > \nu_0$, which is only observed in case of Potassium.

122. The mass of the particle is calculated using de-Broglie relation:

$$\lambda = \frac{h}{mv}$$

$$m = \frac{h}{\lambda \times v}$$

Here, the wavelength of particle is λ and the speed of moving particle is v .

Substitute the given value in above expression.

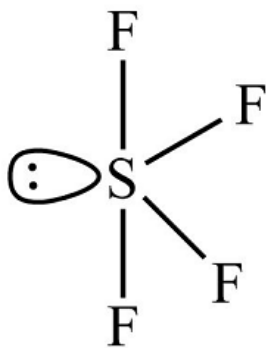
$$m = \frac{6.626 \times 10^{-34} \text{ Js}}{3.313 \times 2 \times 10^8 \times 10^{-10}} = 1 \times 10^{-32} \text{ kg}$$

123. It is given that the electronic configuration of M^{3+} is $[\text{Xe}]4f^3$.

and, the electronic configuration of Nd^{3+} is $[\text{Xe}]4f^3$.

Thus, the element is Nd^{3+} .

124. The option ii is only the correct choice; the shape of SF_4 is shown below,



See-saw shape

The option i and iii is incorrect as ClF_3 has 3 bond pairs and 2 lone pairs so, it is categorized under AB_3E_2 type and SO_2 has 2 bond pairs and 1 lone pair shows AB_2E type.

125. The electronic configuration of the particles are,

$$\text{C}_2 = \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2$$

$$\text{N}_2 = \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^2$$

$$\text{O}_2 = \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \sigma 2p_x^2 < \pi 2p_z^2 < \pi^* 2p_x^1 = \pi 2p_y^1$$

$$\text{F}_2 = \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \sigma 2p_x^2 < \pi 2p_z^2 < \pi^* 2p_x^2 = \pi 2p_y^2$$

The molecule acquires equal number of bonding and anti bonding molecular orbitals.

126. The expression for rate of diffusion for two particles is given as,

$$\frac{R_A}{R_B} = \sqrt{\frac{M_B}{M_A}}$$

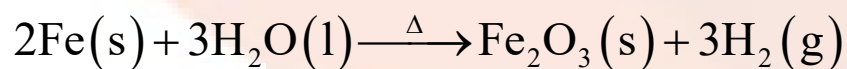
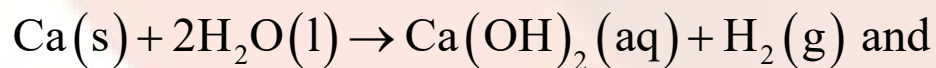
Here, the molecular weight of the particle is M .

Substitute the given value in above expression.

$$\frac{1}{0.707} = \sqrt{\frac{32}{M_A}}$$

$$M_A \approx 16$$

127. In non-metal displacement reaction, either a metal or a non-metal will displace another non-metal of a compound present in the reaction. Usually, the non metal undergoing displacement is hydrogen. The reaction in which non-metal displacement reaction occurs is,



128. In an isothermal reversible expansion process,

$$\Delta U = C_v, \Delta T = 0$$

In case of adiabatic wall $q = 0$

According to 1st law of thermodynamics,

$$\Delta U = q + W$$

$$\Delta U = W_{\text{adiabatic}}$$

In case of thermal conducting walls

$$\Delta U = -q$$

The first law of thermodynamics for close system is,

$$\Delta U = q - W$$

Thus, the correct match is II-A, V-B, III-C, I-D

129. The pressure of the gas N_2O_4 using an ideal gas equation is,

$$p = \frac{nRT}{V}$$

Substitute the given value in above expression.

$$\begin{aligned} p &= \frac{18.4 \times 0.083 \times 400}{1 \times 92} \\ &= 6.64 \text{ bar} \end{aligned}$$

At equilibrium condition, the pressure of N_2O_4 is $p - p_i$ and the pressure of NO_2 is $2p_i$.

The total pressure at an equilibrium condition is given as,

$$p_T = p - p_i + 2p_i$$

$$p_T = p + p_i$$

$$\begin{aligned} p_i &= 10.64 - 6.64 \\ &= 4 \text{ bar} \end{aligned}$$

The value of equilibrium constant is,

$$K_p = \frac{[p_{\text{NO}_2}]^2}{[p_{\text{N}_2\text{O}_4}]} = \frac{(2 \times 4)^2 \text{ bar}}{(6.64 - 4) \text{ bar}} = 24.29$$

130. The expression of pH value using Henderson-Hasselbalch equation is,

$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Here, the pH value of buffer is taken as 10.5.

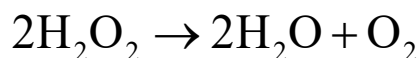
Substitute the value is above expression.

$$10.5 = \text{pK}_a + \log \frac{0.01}{0.1}$$

$$\text{pK}_a = 11.5$$

131. For vessel A,

The reaction of hydrogen peroxide is shown below,



The 10 mL O₂ is obtained by,

$$w = \frac{2 \text{ mol of H}_2\text{O}_2 \times V_{\text{H}_2\text{O}_2}}{V_{\text{at NTP}}}$$

Substitute the value in above we get,

$$w = \frac{2 \times 34 \times 10}{22400} = 0.03 \text{ g H}_2\text{O}_2$$

For vessel B,

The solution of 30 mL O₂ is obtained by,

$$w = \frac{2 \text{ mol of H}_2\text{O}_2 \times V_{\text{H}_2\text{O}_2}}{V_{\text{at NTP}}}$$

Substitute the value in above we get,

$$w = \frac{2 \times 34 \times 30}{22400} = 0.09 \text{ g H}_2\text{O}_2$$

For vessel B,

The molarity is give as,

$$M = \frac{n_{\text{H}_2\text{O}_2}}{V_{\text{sol}}}$$

Here, the number of moles of H₂O₂ is $n_{\text{H}_2\text{O}_2}$ and the volume off the solution is V_{sol}

Substitute the value in above expression.

$$2 = \frac{x / 34}{250 / 1000}$$

$$x = 17 \text{ g}$$

Thus, the order of weight of H_2O_2 in vessel is $\text{C} > \text{B} > \text{A}$.

132. The statement i, ii and iii are correct as,

- i. Beryllium oxide is actually an amphoteric oxide which reacts with both acid and base.
- ii. The blue colour of the solution is due to ammoniated electron.
- iii. The hydration energy depends upon size of cation, which increases down the group.

133. The statement i is correct as in aqueous solution, the molecule H_3BO_3 reacts with a molecule of water to form a hydrated solution. The compound NaBH_4 is a selective reducing agent for aldehydes, ketones and ester.

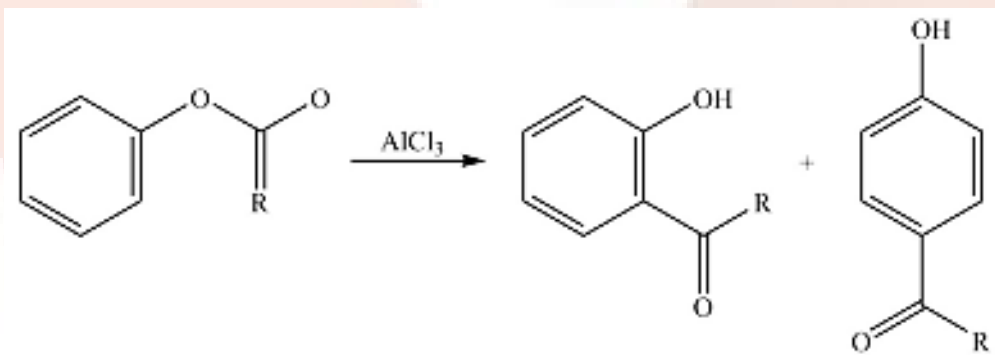
Thus, the statements i and iii are correct.

134. The statements i, ii, iv and v are correct as, in C_{60} all atoms of carbon are sp^2 hybridized. It contains 12 rings of 5 carbon each

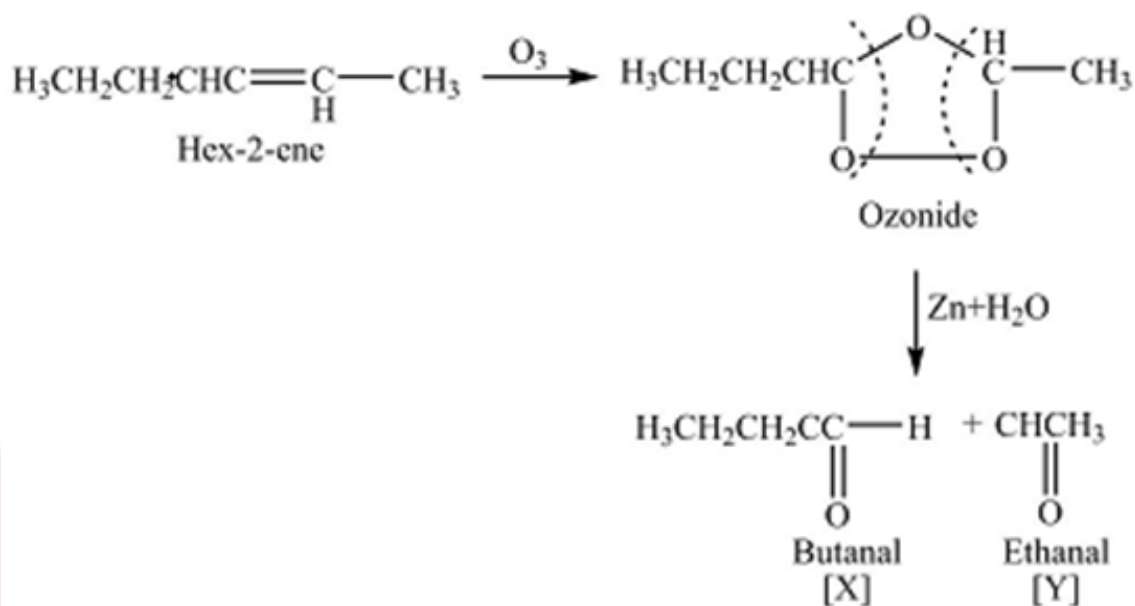
and 20 rings of 6 carbon each. It is anti aromatic with spherically surrounded by pi-electrons.

135. The statement ii is incorrect as the EPM has set maximum allowable concentration of fluorine in water at 2 ppm to prevent enamel fluorosis and 4 ppm to prevent skeletal fluorosis.

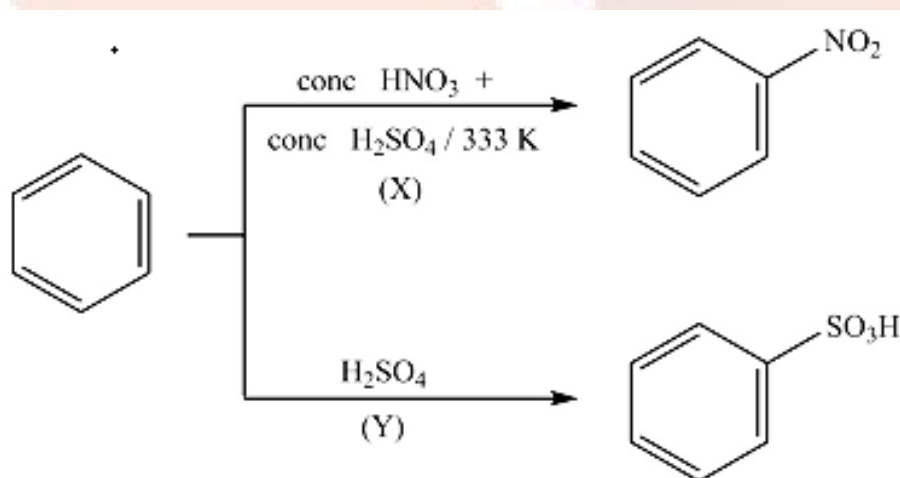
136. The Fries rearrangement reaction is an organic name reaction which involves the conversion of phenolic esters into hydroxyaryl ketones on heating in the presence of a catalyst. The reaction is given below.



137. The sequence of reaction of Hex-2-ene is shown below.



138. The reaction on benzene in presence of concentrated HNO_3 and H_2SO_4 to form product as nitrobenzene. Whereas, when benzene reacts with sulphuric acid the product obtained is called benzene sulphonic acid. The reaction is shown below.



139. The statements B and C are incorrect as silicon or germanium belongs to group 14, so they can be used as semiconductors. P and Ar belong to group 15 element, it contains 5 valence electrons, 4 is used for covalent bonding and one electron is extra and become delocalized.

140. The number of moles of solute (n_1) will be $7.5/M$ if we take the molar mass of the solute to be M g/mol.

The moles of solvent (n_2) will be $90/18 = 5$ moles

The partial pressure comes out to be,

$$\begin{aligned}
 p &= \frac{P_0 - P_a}{P_0} \\
 &= \frac{P_0 - 2.8}{P_0} \\
 &= \frac{7.5/M}{5} \\
 \frac{P_0 - 2.8}{P_0} &= \frac{1.5}{M_2} \quad \dots\dots (i)
 \end{aligned}$$

Now, when 18 gram of water is added then

$$\frac{P_0 - 2.81}{P_0} = \frac{7.5 / M}{108 / 18} \dots\dots (ii)$$

$$= \frac{1.25}{M}$$

On solving equation (i) and (ii) we get,

$$M = 71.5 \text{ g/mol}$$

141. The expression for total pressure is given as,

$$p_{\text{total}} = p_A + p_B$$

$$p_{\text{total}} = \chi_A p_A^0 + \chi_B p_B^0$$

Substitute the values in above expression.

$$p_{\text{total}} = \frac{2}{5} \times 100 \text{ mm} + \frac{3}{5} \times 160 \text{ mm}$$

$$p_{\text{total}} = 136 \text{ mm}$$

The pressure of liquid A is given as,

$$p_A = y_A p_{\text{total}}$$

$$y_A = \frac{p}{p_{\text{total}}}$$

Substitute the value in above expression.

$$y_A = \frac{40 \text{ mm}}{136 \text{ mm}} = 0.294$$

The mole fraction of liquid B is,

$$\begin{aligned}y_B &= 1 - y_A \\&= 1 - 0.294 \\&= 0.706\end{aligned}$$

142. Leclanche cell is a primary cell, handy for sporadic use, with positive anode of zinc encompassed by a mixture of manganese dioxide and powdered carbon in a pot, which is porous.

143. The rate of the reaction is given as,

$$R = k[A][B]^2$$

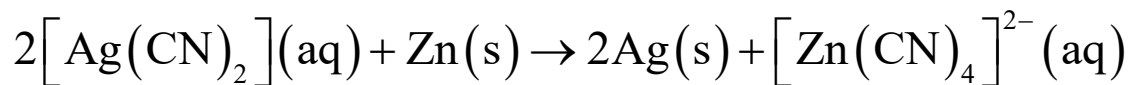
Substitute the given value in above expression.

$$R = 50 \times 10^{-6} [0.05][0.1]^2$$

$$R = 2.5 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$$

144. The statement 3 is incorrect as the layer of positive or negative charge acquired by selective adsorption of ion on the surface of a colloidal particle is known as adsorption layer.

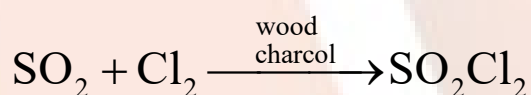
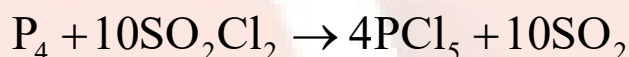
145. The Zn is used as a reducing agent in the extraction process of Ag and Au. The reaction is given as,



The statement ii is also correct as the impure zinc can be refined by the process of distillation.

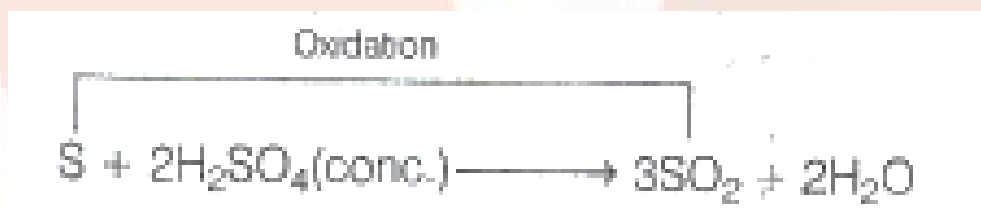
Thus, the option 4 is correct.

146. The reaction of white phosphorous with sulphuryl chloride is given as,



Thus, the compound X is SO_2 and Y is SO_2Cl_2 .

147. The reaction iv is given as,

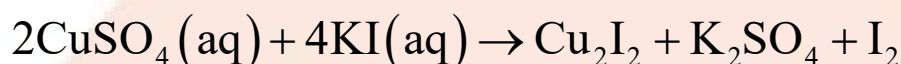


The sulphur gets oxidized to sulphur dioxide.

148. The helium has lowest boiling point due to weak dispersion force of attraction between helium atoms.

Thus, both assertion and reason is correct and reason is the correct explanation.

149. The reaction CuSO_4 with KI is incorrect, the correct reaction is given as,



150. The overall stability constant is given as,

$$\beta_4 = K_1 \times K_2 \times K_3 \times K_4$$

Here, the rate of the reaction is given as K .

Substitute the value in above expression.

$$\beta_4 = 1 \times 10^4 \times 1 \times 10^3 \times 1 \times 10^3 \times 1 \times 10^2$$

$$\beta_4 = 1 \times 10^{12}$$

151. The expression for number average molecular mass of polymer is give as,

$$M_n = \frac{\sum N_i M_i}{\sum N_i}$$

Substitute the given value in above expression.

$$M_n = \frac{15 \times 8000 + 15 \times 80000}{15 + 15}$$

$$M_n = 44000$$

152. For the statements A, B and D, the correct explanation is,

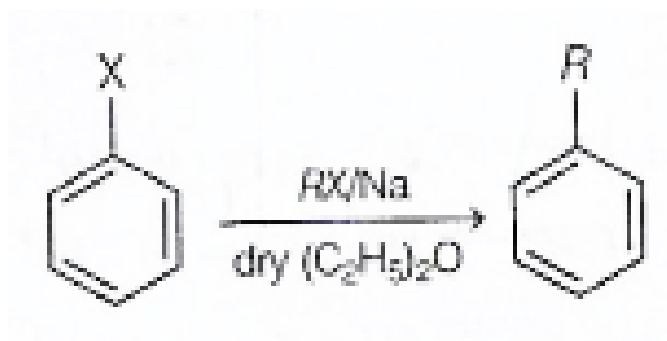
A. A tripeptide contains of 3 amino acid linked by two peptide linkage.

B. A pentapeptide contains 5 amino acids linked by four peptide linkages.

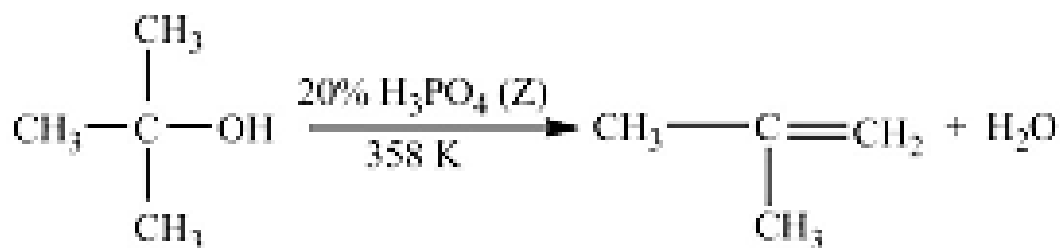
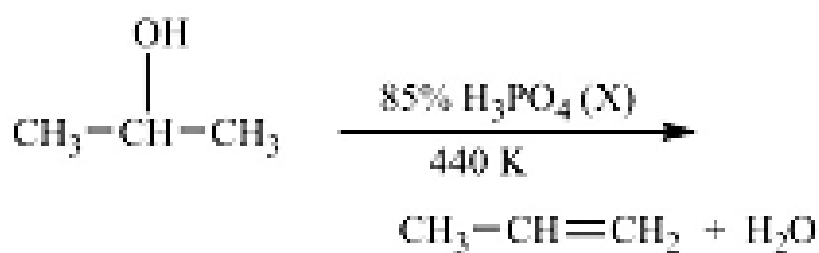
D. In case of cellulose, β -D-glucose units are joined by the glycosidic linkage between C-1 of one glucose and C-4 of the next glucose unit.

153. Anti-histamines are drugs which treat allergic rhinitis and other allergies. Among the following given drugs, dimetane and seldane are anti-histamine.

154. The reaction is called as Wurtz-Fittig reaction. In this alkyl halide is reacted with aryl halide with sodium in presence of dry ether. The reaction is shown below.



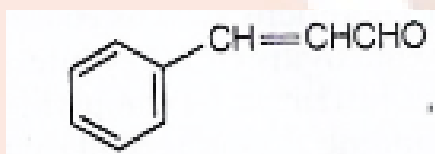
155. The reaction is known as dehydration reactions with H_3PO_4 and H_2SO_4 as a dehydrating agent given as,



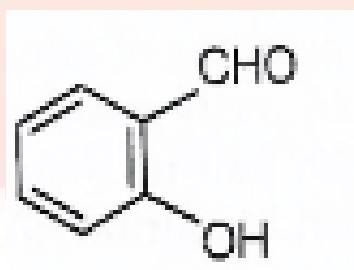
156. The reaction is known as Williamson's synthesis reaction which is shown below.



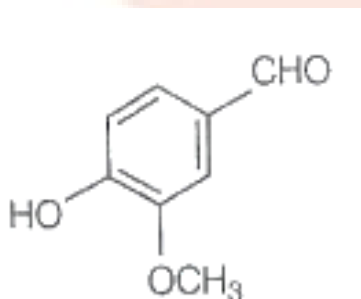
157. The structure of cinnamaldehyde (I) is,



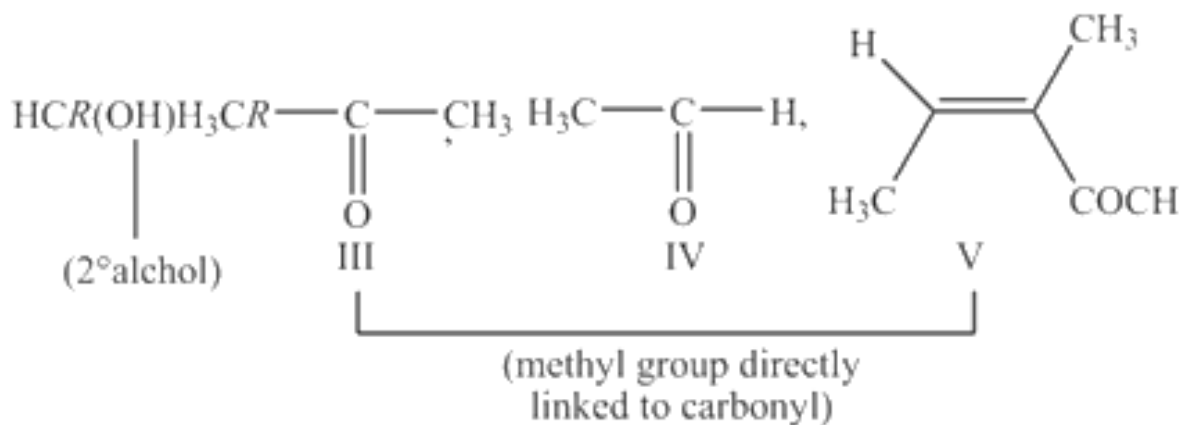
The structure of salicylaldehyde (II) is,



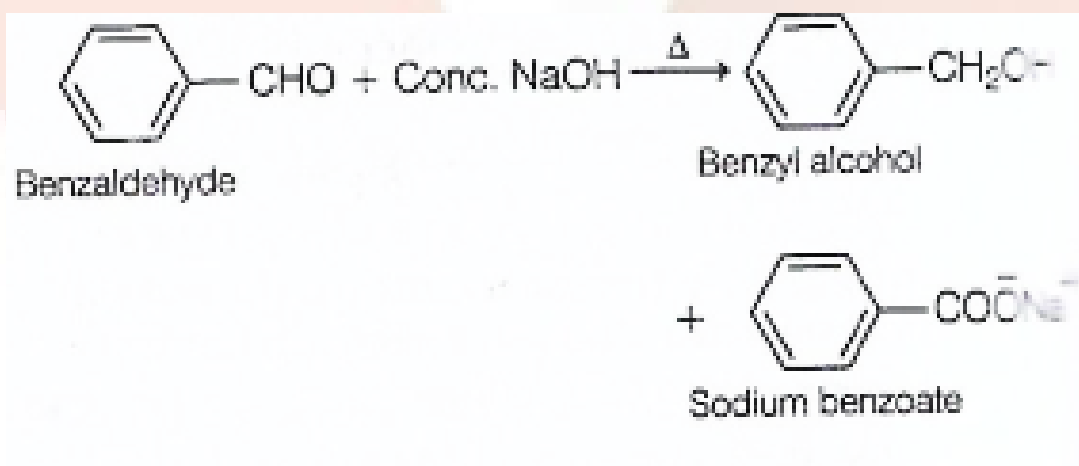
The structure of vanillin (III) is,



158. The aldehydes and ketone group attached with methyl group gets oxidized by NaOCl. The compounds which gets oxidized by NaOCl is,



159. The reaction of benzaldehyde with concentrated NaOH is shown below.



160. The given reaction is called as Gattermann reaction where diazonium salt reacts with copper metal in presence of HCl to give chlorobenzene.

